



**GREEN
CLIMATE
FUND**

Meeting of the Board
30 June – 3 July 2025
Port Moresby, Papua New Guinea
Provisional agenda item 11

GCF/B.42/02/Add.08

9 June 2025

Consideration of funding proposals – Addendum VIII

Funding proposal package for FP266

Summary

This addendum contains the following seven parts:

- a) A funding proposal titled "Strengthening the resilience of ecosystems and populations in four regional hubs in northern Mauritania";
- b) No-objection letter issued by the national designated authority(ies) or focal point(s);
- c) Environmental and social report(s) disclosure;
- d) Secretariat's assessment;
- e) Independent Technical Advisory Panel's assessment;
- f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
- g) Gender documentation.

Table of Contents

Funding proposal submitted by the accredited entity	3
No-objection letter issued by the national designated authority(ies) or focal point(s)	72
Environmental and social report(s) disclosure	73
Secretariat's assessment	76
Independent Technical Advisory Panel's assessment	89
Response from the accredited entity to the independent Technical Advisory Panel's assessment	102
Gender documentation	104

Disclaimer:

The designations and the presentation of the materials used in this document, including their respective citations, maps and references, have been included by the relevant Accredited Entity and do not imply the expression of any opinion whatsoever on the part of the Green Climate Fund concerning the legal status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries. Also, the boundaries and names shown, and the designations used in this document have been included by the relevant Accredited Entity and do not imply official endorsement or acceptance by the Green Climate Fund.

Funding Proposal

Project/Programme title:	<u>Strengthening the resilience of ecosystems and populations in four regional hubs in northern Mauritania</u>
Country(ies):	<u>Islamic Republic of Mauritania</u>
Accredited Entity:	<u>United Nations Environment Programme</u>
Date of first submission:	<u>[YYYY/MM/DD]</u>
Date of current submission	<u>[YYYY/MM/DD]</u>
Version number	<u>[V.000]</u>



GREEN
CLIMATE
FUND

Contents

Section A	PROJECT / PROGRAMME SUMMARY
Section B	PROJECT / PROGRAMME INFORMATION
Section C	FINANCING INFORMATION
Section D	EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA
Section E	LOGICAL FRAMEWORK
Section F	RISK ASSESSMENT AND MANAGEMENT
Section G	GCF POLICIES AND STANDARDS
Section H	ANNEXES

Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 10.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

"FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]"

PROJECT/PROGRAMME SUMMARY				
A.1. Project or programme	Project	A.2. Public or private sector	Public	
A.3. Request for Proposals (RFP)	If the funding proposal is being submitted in response to a specific GCF Request for Proposals, indicate which RFP it is targeted for. Please note that there is a separate template for the Simplified Approval Process and REDD+. <u>Not applicable</u>			
A.4. Result area(s)	Check the applicable GCF result area(s) that the overall proposed project/programme targets below. For each checked result area(s), indicate the estimated percentage of GCF and Co-financers' contribution devoted to it. The total of the percentages when summed should be 100% for GCF and Co-financers' contribution respectively.			
		GCF contribution	Co-financers' contribution¹	
	Mitigation total	<u>Enter number</u> %	<u>Enter number</u> %	
	☐ Energy generation and access	<u>Enter number</u> %	<u>Enter number</u> %	
	☐ Low-emission transport	<u>Enter number</u> %	<u>Enter number</u> %	
	☐ Buildings, cities, industries and appliances	<u>Enter number</u> %	<u>Enter number</u> %	
	☐ Forestry and land use	<u>Enter number</u> %	<u>Enter number</u> %	
	Adaptation total	<u>Enter number</u> %	<u>Enter number</u> %	
	☒ Most vulnerable people and communities	35 %	35 %	
	☒ Health and well-being, and food and water security	15 %	15 %	
	☒ Infrastructure and built environment	15 %	15 %	
	☒ Ecosystems and ecosystem services	35%	35 %	
A.5. Expected mitigation outcome (Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)	Indicate greenhouse gas (GHG) emission reductions or removals in tCO ₂ eq over total lifespan of the project/programme ²	A.6. Expected adaptation outcome (Core indicator 2: direct and indirect beneficiaries reached)	Total beneficiaries 230,237 (107,396 men; 122,841 women)	
			Direct beneficiaries 85,229 (40,877 men; 44,352 women)	Indirect beneficiaries 145,008 (66,519 men; 78,489 women)
			1.78% of the country's total population	3.02 % of the country's total population
A.7. Total financing (GCF + co-finance³)	33,560,345 USD	A.9. Project size	Small (Upto USD 50 million)	
A.8. Total GCF funding requested	30,902,775 USD <i>For multi-country proposals, please fill out annex 17.</i>			

¹ Co-financer's contribution means the financial resources required, whether Public Finance or Private Finance, in addition to the GCF contribution (i.e. GCF financial resources requested by the Accredited Entity) to implement the project or programme described in the funding proposal.

A.10. Financial instrument(s) requested for the GCF funding	☒ Grant <u>30,902,775 USD</u> ☐ Equity <u>Enter number</u> ☐ Loan <u>Enter number</u> ☐ Results-based payment <u>Enter number</u> ☐ Guarantee <u>Enter number</u>		
A.11. Implementation period	72 months	A.12. Total lifespan	20 years
A.13. Expected date of AE internal approval	11/27/2023	A.14. ESS category	B
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☒ No ☐
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?	Yes ☒ No ☐
A.19. Complementarity and coherence	<i>Does the project/programme complement other climate finance funding (e.g. GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1.</i> Yes ☒ No ☐		
A.20. Executing Entity information	Executing Entity: Ministry of Environment and Sustainable Development The EE also serves as the NDA.		
A.21. Executive summary (max. 750 words, approximately 1.5 pages)			
Prolonged droughts across the country have accelerated vegetation loss in desert environments, reducing ecosystem services related to sediment stabilisation and water infiltration. As a result, the rate of dune advancement along the Sahara-Sahel boundary has increased and greater sand encroachment has been observed in oases, with subsequent impacts on water resources, agricultural productivity and income generation across northern Mauritania. Siltation of dams, boreholes, wells and rivers has reduced the accessibility and availability of already scarce water resources in the region. Moreover, reduced rainfall and water infiltration resulting from vegetation loss have increased the depth of the water table. This has led to many pastoral wells drying up, impacting pastoralists' abilities to maintain nomadic lifestyles. Consequently, many nomadic pastoralists are adopting sedentary lifestyles around rural oases, increasing pressure on limited natural resources such as the demand for food, water and income to support rural-urban communities growth. Sand inundation is threatening important socio-economic infrastructure such as roads, schools and hospitals located within the Sahel-Sahara region. Indeed, in extreme cases, sand encroachment has resulted in the displacement of entire villages and communities. These processes will continue to worsen, with the Sahara Desert expected to spread southwards into the country's most fertile lands in the Sahelian region and near the Senegal River. To slow the rate of desertification and enhance water security in Mauritania, the restoration and conservation of ecosystems along the Sahel-Saharan boundary is required to enhance ecosystem services relating to dune stabilisation, water infiltration and the provision of natural resources. Four regional hubs, where desertification processes are most severe and adaptation needs are greatest, have been identified in northern Mauritania. These hubs include: i) Aoujeft, in the Adrar province; ii) Rachid, in the Tagant province; iii) Tamcheket, in the Hodh El Gharbi province; and iv) Nema, in the Hodh El Chargui province.			

² The total lifespan of the project/programme is defined as the maximum number of years over which the outcomes of the investment are expected to be effective. This is different from the project/programme implementation period.

³ Refer to the Policy of Co-financing of the GCF.

PROJECT/PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

Climate change is profoundly impacting Mauritanian communities, particularly those living along the Sahel-Saharan boundary that extends from the northwest to the southeast of the country. A noticeable impact of climate change-induced desertification is the accelerated advance of sand dunes. Dams, boreholes, wells and rivers are becoming increasingly silted as a result of sand inundation, reducing access to and the availability of already-strained water resources (see Figure 60 in Annex 2). Arable land, rangeland and pastures are being inundated, severely compromising the agricultural (see Figure 27 in Annex 2) and livestock practices in the oases. In addition, sand inundation threatens critical socio-economic infrastructure, including, *inter alia*, roads, school buildings and hospitals. In extreme cases, sand dunes have displaced entire villages, in some cases more than once (see Figure 15 in Annex 2). The ongoing decline in the water table has also dried up wells commonly used by pastoralist herders, impacting their ability to practice transhumance⁴. Combined with the climate change-induced loss of rangelands and pastures, water scarcity has contributed to previously nomadic herders becoming more sedentary around rural oases where resource access is reliable. However, sedentarisation increases pressure on limited resources, creating a source of conflict — particularly between farmers and herders.

Four regional hubs have been identified along the boundary where desertification processes are most severe and the need for adaptation is greatest. These hubs are: i) Aoujeft, in the wilayah of Adrar; ii) Rachid, in the wilyah of Tagant; iii) Tamcheket, in the wilayah of Hodh El Gharbi; and iv) Nema, in the wilayah of Hodh El Chargui. Each hub covers an area of 80–85 km², spans several communes and comprises an urban centre surrounded by rural hinterland. The locations of the four hubs — Aoujeft, Rachid, Tamcheket and Nema. Aoujeft hub comprises six communes with a total population of ~35,600; Rachid hub spans three communes and sustains ~25,600 people; Tamcheket hub covers five communes and houses ~43,300 inhabitants; and Nema hub encompasses ten communes with a total population of ~99,600⁵.

Climate science basis⁶

Observed climate change

Observations of historical climate trends show that the mean annual temperature in Mauritania has increased by 0.3°C per decade over the last 70 years — from 26.10°C in 1950 to 28.46°C in 2020⁷. Although warming has been observed across all major climatic zones in the country, temperature increases have been greatest over the Sahara Desert and lowest over the Senegal River Headwaters (**Error! Reference source not found.**)⁸. Moreover, the latest available 30-year weather station data (1981–2010) indicate that, in general, temperatures have also increased within the focal hubs of Aoujeft, Rachid, Tamcheket and Nema (see Table 27 in Annex 2 for observed temperature changes in the four focal hubs)⁹.

In addition to shifting temperatures, climate change in Mauritania has induced: i) an increase in the frequency of extreme rainfall events and associated flash floods; and ii) a general decline in low-intensity, long-duration precipitation events (see Table 32 in Annex 2 for observed precipitation trends in the four focal hubs). Consequently, while heavy precipitation events have become more intense, rainfall is increasingly concentrated over shorter time periods. This shift has resulted in increased rainfall irregularity and prolonged dry spells across the region¹⁰. Precipitation has decreased during the wet season and over the entire year in Nema, Tamcheket and Rachid. In contrast, the Aoujeft hub has undergone a small but statistically significant increase in its overall precipitation. Despite the overall decline in average precipitation, an increase in extreme daily rainfall, which often results in flash floods, has been observed in all four hubs over both time frames.

⁴ The action or practice of moving livestock from one grazing ground to another in a seasonal cycle, typically to lowlands in winter and highlands in summer.

⁵ Annex 2 — Appendix 4: Climate Change Risk Assessment

⁶ Detailed primary and secondary climate analyses for the proposed project are presented in Annex 2: Feasibility Study, in Section 3 and Appendices 3 and 4, respectively. The information presented below is a summary of what is presented in the more detailed analyses. While detailed climate analyses in the Feasibility Study were conducted using CMIP 5 data, the findings presented here are also consistent with more recent CMIP 6 model findings.

⁷ World Bank. 2022. Climate change knowledge portal: Mauritania. Available at: <https://climateknowledgeportal.worldbank.org/country/mauritania/trends-variability-historical>.

⁸ AfDB. 2018. National climate change profile. Available at: <https://www.afdb.org/en/documents/mauritania-national-climate-change-profile>.

⁹ Annex 2 - Appendix 2: Trend Analysis Report for Mauritania.

¹⁰ AfDB. 2018. National climate change profile. Available at: <https://www.afdb.org/en/documents/mauritania-national-climate-change-profile>.

Observed potential evapotranspiration (ETP) values are high in all four hubs (1,700–2,000 mm/year), with the highest values observed in Nema and Tamcheket. In addition, each hub has negative differences between precipitation and evapotranspiration (P-ETP), indicating that water loss via ETP exceeds water gained via precipitation.

Sand movements are driven by the prevalence of strong winds from the north of Mauritania including: i) the north-eastern trade winds; and ii) the *harmattan* — a dry wind strongest from late November to mid-March. Together, these winds carry large amounts of dust and sand across the Mauritanian territory.

Projected climate change

All four target hubs are projected to experience an increase in both minimum and maximum temperatures under future RCP4.5 (intermediate) and RCP8.5 (high) greenhouse gas emissions scenarios, with larger temperature increases projected under RCP8.5 than RCP4.5. These projected changes are reflected in Table 31, Annex 2. In line with increasing temperatures, the frequency of heat waves is predicted to increase across all four target hubs by 2050 under both the RCP4.5 and RCP8.5 emissions scenarios¹¹. Projected changes in the number of days with heatwaves per decade are presented in Table 42 in Annex 2.

The projected evolution of precipitation trends varies among hubs. For example, no significant change in precipitation is projected in Aoujeft by 2050 under the RCP4.5 GHG emissions scenario. In contrast, annual precipitation is expected to significantly decrease in Rachid, Tamcheket and Nema, relative to the reference period (1981–2010). Similarly, under the RCP8.5 emissions scenario, mean annual precipitation in Aoujeft is not projected to change significantly relative to the reference period. However, by 2050, mean annual precipitation is expected to decrease in Rachid, Tamcheket and Nema. Projected changes in precipitation within the four target hubs are presented in Table 35 in Annex 2.

In line with increasing hub-level temperature trends and decreasing precipitation trends, potential evapotranspiration (ETP) is projected to increase across all four hubs by 2050 under both the RCP4.5 and RCP8.5 emissions scenarios (Table 40 in Annex 2). As a result of increased ETP and limited rainfall, the water balance deficit (P-ETP) in all four hubs is also projected to increase significantly ($p < 0.1$) by 2050, as shown in Table 41, Annex 2¹².

The combination of projected temperature rise, ETP and declining rainfall will increase the extent and severity of droughts at national and local levels. By 2040, the median number of consecutive dry days (CDD) for Mauritania is predicted to be between 219 and 230, compared to the historical value of 216 days in 1995. Moreover, by 2100, annual CCD is expected to reach between 225 and 236 days, representing a change of between 9 and 20 days relative to 1995. Another indicator of future drought conditions — the Standardised Precipitation-Evapotranspiration Index (SPEI)¹³ — displays a clear trend toward worsening drought conditions in Mauritania. All Shared Socioeconomic Pathways (SSP)¹⁴ scenarios predict that SPEI values will fall below zero in the future, indicating an increase in dry years relative to the historical average. Moreover, by 2100, 90th percentile values are expected to fall below negative two, signifying an increase in extremely dry years.

Although data regarding projected changes in wind regimes across the country are limited, a recent global review indicates that near-surface wind speeds are projected to decrease under future global warming scenarios (1.5–4°C above pre-industrial levels) over the parts of the Northern Hemisphere, including Mauritania¹⁵. At the hub level, models projecting changes in wind speed and direction under future climate change scenarios are currently unavailable. Consequently, the assumption that wind conditions will remain constant under future climate scenarios has been applied to all four hubs. The combination of constant windy conditions and a generally warmer and drier climate is likely to facilitate the southward expansion of the Sahara into the Sahelian region of the country. This trend will further impact the communities of the project's focal hubs.

B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

¹¹ Annex 2 — Appendix 2: Trend Analysis Report for Mauritania.

¹² Annex 2 — Appendix 4: Climate Change Risk Assessment for Mauritania.

¹³ The Standardised Precipitation-Evapotranspiration Index (SPEI) is an indicator of drought severity that accounts for precipitation and potential evapotranspiration. Values +2 or higher indicate extremely wet years, while values -2 or less indicate extremely dry years.

¹⁴ The IPCC AR6 identifies five potential scenarios based on emission levels and Shared Socioeconomic Pathways (SSPs) that are used to predict future climate change trends. SSP1-2.6 describes a 'green road' scenario whereby the world shifts towards a sustainable path, with CO₂ emissions reaching net zero by 2075. SSP2-4.5 describes a 'middle of the road' scenario whereby emissions remain around current levels before falling around the mid-century as socioeconomic factors follow historic trends, with slow progress towards sustainability. SSP5-8.5 describes a worst-case scenario whereby CO₂ emissions double by 2050, with rapid economic growth dependent on fossil fuels and energy-intensive lifestyles.

¹⁵ Zha J, Shen C, Li Z, Wu J, Zhao D, Wenxuan F, Sun M, Azorin-Molina C & Deng K. 2021. Projected changes in global terrestrial near-surface wind speed in 1.5–4.0°C global warming levels. *Environmental Research Letters*. DOI: 10.1088/1748-9326/ac2fdd.

Problem statement

Climate change will continue to exacerbate the drivers of desertification in Mauritania, threatening the structural integrity of the fragile oases that underpin the natural resource-based livelihoods of rural communities. Drivers of desertification include changes in the spatial and temporal patterns of rainfall and increases in temperatures and ETP that, when combined, lead to: i) more frequent, severe and prolonged drought events¹⁶; ii) reduced water security; iii) more intense desertification processes; and iv) siltation of dams, boreholes, wells and rivers, which reduces the accessibility and availability of already strained water resources v) the drying up of wells commonly used by pastoralist herders. Of these impacts, the increasing frequency, severity and time span of droughts are particularly damaging, as they already contribute to vegetation loss, promoting dune advancement and accelerated sand encroachment in fragile oases (see Figure 140 in Annex 2).

Without urgent intervention, these processes will worsen in the future. The Sahara Desert is expected to spread south into the country's most fertile lands in the Sahelian region. Rural communities in Mauritania have limited financial and coordination capacity to slow the rate of desertification along the Sahel-Saharan boundary and improve water security.

Preferred adaptation solution

An ecosystem-based adaptation approach within the target hubs is the preferred option for responding to the impacts of climate change-induced desertification within the four target hubs because it is cost-effective, easily implemented and has the potential to provide valuable ecosystem goods and services to climate-vulnerable communities dependent on the oases (for an example, see Figure 145 in Annex 2). This approach includes establishing 2,123 ha of green-grey dune fixation infrastructure and 120 km of protective fence line to facilitate the rehabilitation and maintenance of degraded landscapes and enhance ecosystem services related to dune stabilisation and the supply of natural resources.

To support the adaptive capacity of the communities in the target hubs, the proposed project will develop or strengthen sustainable agriculture-based livelihoods. By providing an alternative source of income to current extraction-based livelihoods, these livelihood options will improve the sustainability of green-grey infrastructure approaches by mitigating the risk of communities resorting to maladaptive coping mechanisms.

Barriers to the preferred solution

Barrier 1: Limited coordination and technical capacity to implement adaptation interventions at the community level

Implementing entities, as well as government and non-governmental stakeholders in Mauritania, have limited institutional capacities and incentive to coordinate efforts to address climate risks, which has restricted CCA efforts in the four target hubs. Government officials at national-, regional- and local levels, as well as staff from crucial sectors at the local level, have limited technical capacity to develop and implement adaptation initiatives. Local authorities and stakeholders at the forefront of managing climate change impacts, such as community leaders, often have inadequate training on incorporating climate change considerations into decision-making or designing and implementing climate-resilient natural resource management and land-use planning.

Barrier 2: Limited financial capacity to invest in climate-resilient land and natural resources management at the community level

Most urban centres in Mauritania are characterised by predominantly informal trade, limited infrastructure and public-service coverage, high levels of self-employment and low human capital. These characteristics generally discourage private-sector investment in sustainable development and CCA¹⁷. Although the Government of the Islamic Republic of Mauritania (GoM) has introduced a number of regulatory and legislative reforms in an attempt to boost the private sector by 2025 — for example, through the establishment of the Higher Council for Investment in 2020 — several obstacles persist, impeding local and foreign investment in sustainable development. These obstacles include: i) difficulty in accessing microfinance for local business development; ii) limited market access within rural and semi-urban areas; iii) low levels of technical and professional training; iv) limited governance within the financial services sector; and v) limited private-sector organisation¹⁸. Consequently, past projects have been unable to mobilise substantial private-sector investment in CCA. For this reason, the proposed project will leverage public-sector funding to co-finance adaptation interventions.

¹⁶ AfDB. 2018. National climate change profile. Available at: <https://www.afdb.org/en/documents/mauritania-national-climate-change-profile.v>

¹⁷ Centre for Affordable Housing Finance in Africa. 2018. Africa Housing Finance Yearbook 2018 — Mauritania. Available at: <http://housingfinanceafrica.org/app/uploads/Mauritania-2018-Final.pdf>

¹⁸ Arab NGO Network for Development (ANND). 2022. The Private Sector and Sustainable Development in Mauritania. Available at: https://annd.org/uploads/publications/Arab_Watch_Report_-_The_Private_Sector_and_Sustainable_Development_in_Mauritania_-_En.pdf

Barrier 3: Limited knowledge of the climate change impacts, risks, vulnerabilities and adaptation options across the four target hubs

Communities in rural areas — particularly farmers and pastoralists — are not well informed about current and expected climate change impacts, risks and vulnerabilities related to water availability, agricultural production and other land-use practices. Moreover, the target communities have limited technical knowledge on adapting to these changes and increasing climate resilience. To date, attempts to stabilise dunes in the four target hubs have rarely been consistent, strategic or integrated into comprehensive plans. The NAP Knowledge Management Platform is still under development, as a result, national-level stakeholders seldom pass on information to regional or local levels. There is limited inter-sectoral collaboration and knowledge-sharing between ministries, departments and stakeholders.

Barrier 4: Limited integration of climate change adaptation into land-use planning and natural-resource management strategies at the regional and local levels

Although urban plans developed for Nema and Rachid incorporate some measures for climate-risk reduction, mainstreaming adaptation and mitigation strategies into plans at the regional and local levels has largely been overlooked. In the past, failure to recognise the importance of a participatory approach has resulted in weak coordination between climate change-related projects and associated activities within the target region^{19,20}. While some basic water-management plans exist within the target region, pasture- and rangeland-management plans that consider climate change are notably absent in the target hubs²¹.

Project strategy for overcoming barriers to adaptation and achieving a paradigm shift in Mauritania

The goal of the proposed project is to enhance the resilience of communities and ecosystems in northern Mauritania against the impacts of climate change. This will be achieved by investing in land rehabilitation and sand-dune fixation, water resources management, climate-resilient agriculture and livelihood diversification in four rural-urban hubs: Aoujeft (in the Adrar wilayah), Rachid (in the Tagant wilayah), Tamcheket (in the Hodh El Gharbi wilayah) and Nema (in the Hodh El Chargui wilayah). To ensure the success of this approach, the project will focus on two main outcomes:

- **Outcome 1.** Increased implementation and upscaling of Ecosystem-based Adaptation (EbA) measures across the Sahara-Sahel boundary in Mauritania.; and
- **Outcome 2.** Communities in four rural-urban hubs along the Sahara-Sahel boundary in Mauritania have increased livelihood and water security as a result of investment in EbA measures, water access, and sustainable land- and natural resources management practices.

Output 1.1 will address all four of the identified barriers to adaptation by establishing coordination platforms to improve technical capacity, facilitate knowledge sharing on natural resource management and sustainable land-use planning and the implementation of EbA activities at the regional and commune level. By creating and disseminating knowledge on EbA, Output 1.2 will contribute to addressing barrier 1 and barrier 3.

Under Output 2.1, ~2,123 ha of green-grey infrastructure will be established across the four target hubs to facilitate the rehabilitation and maintenance of degraded landscapes and enhance ecosystem services related to dune stabilisation and the supply of natural resources. Co-developing commune-level rehabilitation plans will support the integration of climate change adaptation into land-use planning and natural-resource management strategies at the regional and local levels. Investment in climate-resilient land management will contribute to overcoming barriers 2 and 3.

Under Output 2.2, community-managed Water User Groups (WUGs) will be established. Commune-level water monitoring and regulation plans will be developed, and physical water management infrastructure and water access points will be established along historical transhumance routes. This will contribute to overcoming barriers 2, 3 and 4.

Output 2.3 will facilitate the adoption of climate-smart agricultural practices and sustainable diversified livelihoods for farmers and community members within the target communes. Farmers will be trained by MEDD and MDR extension officers and representatives from commune-level technical committees (CTCs) to practice climate-resilient crop agriculture and to use improved agricultural technologies. These extension officers will also develop seed banks and provide climate-resilient seed varieties to newly establishing nurseries — for distribution to 500 farmers and sale

¹⁹ Food and Agriculture Organisation. 2010. Fighting sand encroachment: Lessons from Mauritania. Available at: <https://www.fao.org/3/i1488e/i1488e.pdf>

²⁰ World Bank. 1994. Mauritania Country Environmental Strategy Paper. Available at: <https://documents1.worldbank.org/curated/en/537131468756600019/pdf/multi-page.pdf>

²¹ The Adaptation Partnership. 2011. Review of current and planned adaptation action — Mauritania. Available at: <https://www.preventionweb.net/publication/mauritania-review-current-and-planned-adaptation-action>

thereafter — and training to nursery operators, to support land rehabilitation, dune fixation and horticulture activities. As some nurseries already exist within the project's target regions, there is an opportunity for upscaling as well as the establishment of new nurseries and market gardens, encouraging livelihood diversification and empowering women in agriculture. In addition, an on-granting mechanism will promote and support community-led initiatives, fostering the autonomous uptake of EbA practices by community members, thereby increasing community ownership and the sustainability of project activities beyond the project implementation timeline.

The theory of change (ToC) states that **If** an ecosystem-based adaptation (EbA) approach is used to restore ecosystems, fix dunes in place and sustainably manage natural resources along the Sahara-Sahel boundary in Mauritania, supported by transformational change in policies, planning, knowledge management and capacity-building, **then** the resilience of communities and ecosystems to the impacts of climate change-induced desertification — including sand inundation, water scarcity, rangeland degradation and reduced agricultural productivity — will be strengthened **because** the provision of ecosystem services will be enhanced, communities in rural-urban hubs will have strengthened livelihood security and sub-national decision-makers will have strengthened capacities to invest in and implement EbA measures. This resilience will be enhanced particularly if supported by transformational change in policies, planning, knowledge management and capacity-building. As ecosystem services are enhanced, livelihood security for communities in rural-urban hubs will be enhanced. Moreover, sub-national decision-makers' capacity to invest in and implement EbA measures will be strengthened. This, in turn, will support a paradigm shift towards effective CCA in Mauritania.

The assumptions underpinning the success of the proposed project are that: i) target communities and the government will continue to engage in the development and maintenance of adaptation interventions; ii) climate change knowledge generated will be used to inform effective management practices in the long term; iii) sustainable livelihoods will provide sufficient incentive to prevent the continuation of unsustainable land-management practices; and iv) improving market access will increase adoption of sustainable agriculture practices.

The ToC is provided in Figure 1 below.

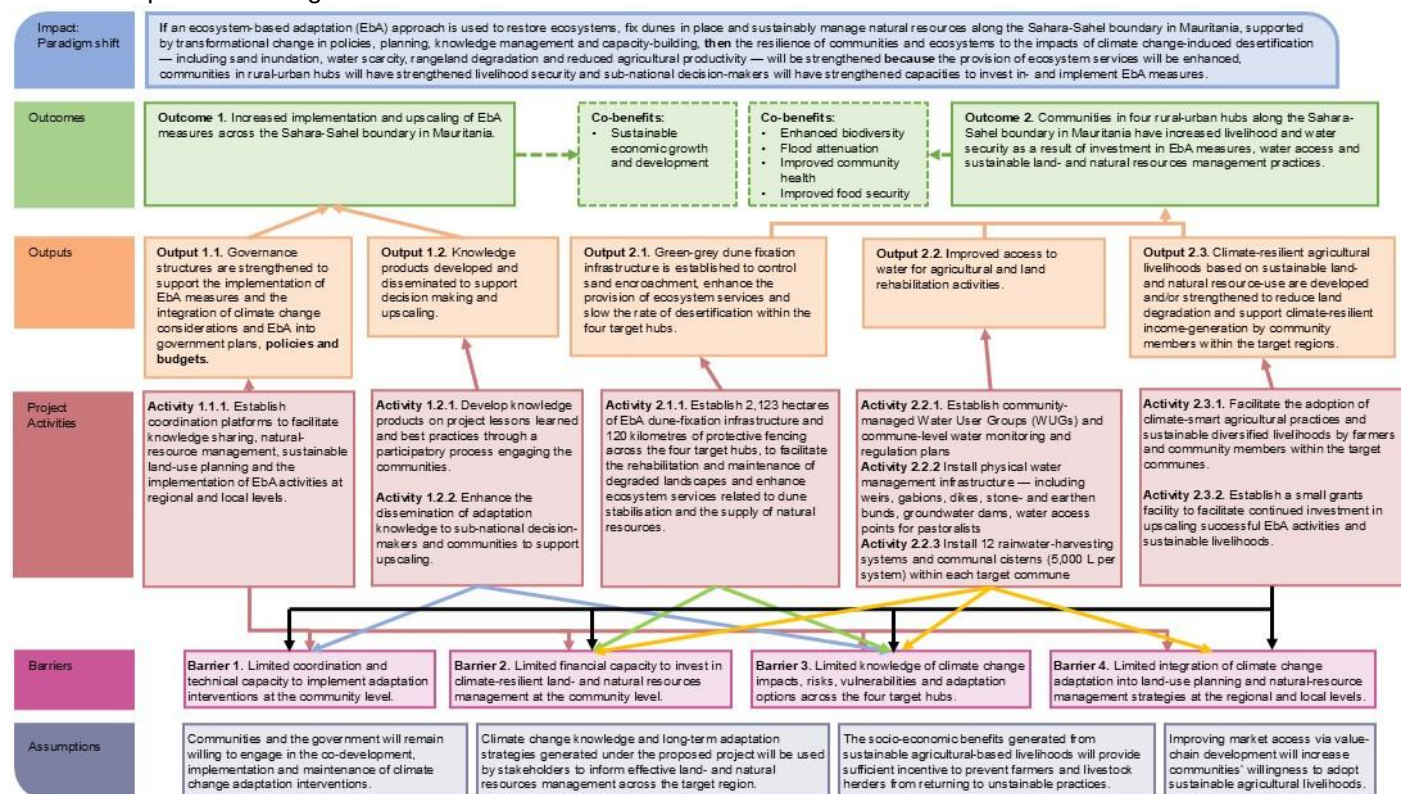


Figure 1. Theory of change for the GCF Funding Proposal.

B.2 (b). Outcome mapping to GCF results areas and co-benefit categorisation

Outcome number	GCF Mitigation Results Area (MRA 1-4)				GCF Adaptation Results Area (ARA 1-4)			
	MRA 1 Energy generation and access	MRA 2 Low-emission transport	MRA 3 Building, cities, industries, appliances	MRA 4 Forestry and land use	ARA 1 Most vulnerable people and communities	ARA 2 Health, well-being, food and water security	ARA 3 Infrastructure and built environment	ARA 4 Ecosystems and ecosystem services
Outcome 1	☐	☐	☐	☐	☒	☐	☐	☒
Outcome 2	☐	☐	☐	☐	☒	☒	☒	☒

The selection of ARA1, ARA2, ARA3 and ARA4 is based on the direct and measurable contribution of the project outcomes and outputs to enhancing community and ecosystem resilience in the target hubs. The proposed project adopts an integrated ecosystem-based adaptation (EbA) approach that addresses environmental and socio-economic vulnerabilities caused by desertification and climate change.

Outcome 1

- **ARA1:** Under Outcome 1, the project will strengthen governance and technical capacity for climate adaptation by establishing local coordination structures and integrating climate change considerations into sub-national plans and budgets. The involvement of community representatives in these structures will enhance community ownership of adaptation measures, ensuring that vulnerable groups have a platform to influence decision-making and benefit directly from resilience-building interventions.
- **ARA4:** The strengthened governance and knowledge-sharing mechanisms established under Outcome 1 will improve the planning and management of ecosystem restoration activities. By mainstreaming ecosystem-based adaptation into local plans and enhancing coordination among stakeholders, the project will promote sustainable land-use practices and strengthen the long-term resilience of ecosystems.

Outcome 2

- **ARA1:** Under Outcome 2, the project will improve the resilience of vulnerable communities by reducing exposure to desertification and sand encroachment through the establishment of green-grey infrastructure. Additionally, training and technical support for climate-resilient agriculture, sustainable livestock management and alternative livelihoods will enable communities to diversify income sources and reduce dependency on fragile ecosystems.
- **ARA2:** Under Outcome 2, the project will enhance water and food security by establishing water management infrastructure, installing improved irrigation systems and promoting sustainable water management practices. Increased access to water for agricultural and household use will improve community health and well-being while climate-resilient agricultural practices will strengthen food security.
- **ARA3:** Under Outcome 2, the project will increase the resilience of infrastructure by supporting the establishment of green-grey dune fixation infrastructure and protective greenbelts. These measures will protect houses, roads and other critical infrastructure from sand encroachment, thereby reducing long-term maintenance costs, increasing the longevity of built infrastructure and mitigating displacement resulting from sand encroachment on infrastructure.
- **ARA4:** Under Outcome 2, the project will restore and protect ecosystems by establishing green-grey dune fixation infrastructure and improving natural resource management. Increased ecosystem health will enhance the provision of ecosystem services such as soil stabilisation, groundwater recharge and improved biodiversity, strengthening the overall resilience of both ecosystems and communities.

This alignment demonstrates how the project's integrated approach will generate social, economic and environmental adaptation benefits, directly supporting the GCF's strategic objectives. Further information regarding the proposed project's contribution to the GCF results areas is provided in Section D.1.

Co-benefit number	Co-benefit					
	Environmental	Social	Economic	Gender	Adaptation	Mitigation
Co-benefit 1: Improved health of target communities	☒	☒	☐	☒	☒	☐
Co-benefit 2: Enhanced biodiversity in the target communes	☒	☒	☐	☐	☒	☒
Co-benefit 3: Improved flood attenuation	☒	☒	☒	☐	☒	☒
Co-benefit 4: Sustainable economic growth and development	☒	☒	☒	☒	☐	☐
Co-benefit 5: Improved food security	☐	☒	☒	☒	☒	☐

B.3. Project/programme description (max. 2500 words, approximately 5 pages)

The objective of the proposed project is to enhance the resilience of communities and ecosystems²² against climate change-induced desertification in the Sahara-Sahel boundary of Mauritania. The project will focus on four rural-urban hubs: Aoujeft (in the Adrar wilayah), Rachid (in the Tagant wilayah), Tamcheket (in the Hodh El Gharbi wilayah) and Nema (in the Hodh El Chargui wilayah). For a more detailed description of the proposed project's Outcome, Outputs and activities, please refer to Section 11 of Annex 2: Feasibility Study.

Outcome 1. Increased implementation and upscaling of Ecosystem-based Adaptation (EbA) measures across the Sahara-Sahel boundary in Mauritania.

Outcome 1 will contribute to creating an enabling environment for climate change and gender-responsive land-use planning and natural resource management at the sub-national level in Mauritania. This will enhance the long-term sustainability and scalability of climate resilience-building interventions across the Sahara-Sahel boundary.

Output 1.1 will strengthen governance structures to support the implementation, maintenance and upscaling of EbA measures across the four target hubs and the integration of climate change considerations and Ecosystem-based Adaptation (EbA) into government plans, policies and budgets. This will be achieved by establishing and training commune-level technical committees (CTCs) in 12 priority communes across Aoujeft, Rachid, Tamcheket and Nema (process for selection of priority communes is detailed in Section 6 of Annex 2: Feasibility Study), to facilitate natural resource management, sustainable land-use planning, EbA implementation and knowledge sharing at regional and local levels. Each CTC will be chaired by the respective commune's mayor, with members comprising DREDD representatives; 1-2 members of the Communal Council, representatives of Associations for Natural Resources (AGLCs) and of Association of Oasis Producers' Groups (AOPGs); civil society representatives; and local community members. The CTCs will be trained by extension officers from the MEDD and DREDDs to supervise, manage and upscale CCA measures introduced under Output 2.1, Output 2.2 and Output 2.3 of the proposed project. The CTCs will also coordinate knowledge-sharing between local community members, commune-level authorities (mayors) and regional authorities (DREDDs and walis).

Output 1.2 will focus on developing and disseminating knowledge products that support informed decision-making for climate change adaptation and the upscaling of the project approach and best practices. Knowledge products generated under the project will include training manuals, implementation guides, lessons learnt and best practices identified by both experts and local stakeholders, and monitoring and evaluation reports. These knowledge products will be disseminated to stakeholders across the four wilayahs (including non-target communes) using the NAP Knowledge Management Platform, which is currently under development by the NAP readiness project. The project will also support the NAP Knowledge Management Platform to better reach local-level end users, by packaging knowledge in formats that are accessible to sub-national decision-makers and communities.

²² Table 74 in Annex 2 shows how the project aligns with the GCF Ecosystem and Ecosystem Services Sectoral Guide (2022).

Output 1.1. Governance structures are strengthened to support the implementation of EbA measures and the integration of climate change considerations and EbA into government plans, policies and budgets.

Activity 1.1.1. Establish coordination platforms and train participants to facilitate knowledge sharing, natural-resource management, sustainable land-use planning and the implementation of EbA activities at regional and local levels.

Under Activity 1.1.1, the proposed GCF project will:

- establish 12 commune-level technical committees (CTCs) across the four target hubs;
- enhance capacity for coordination and knowledge-sharing between local and regional stakeholders within the target hubs; and
- support the integration of climate change considerations and EbA into sub-national plans, policies and budgets.

To maximise the impact potential project activities, given the finite nature of financial resources and human capital²³, CTCs will be established within the 12 priority communes targeted by the project (i.e., most vulnerable and exposed), with the view to upscale project interventions to additional communes in the future.

Table 1. The 12 priority communes for the proposed project

	Hub			
	Aoujeft	Rachid	Tamcheket	Nema
Target communes	• Ain Savra • El Medah • Maeden • Aoujeft	• Tidjikja • El Wahat	• El Mabrouk • Gueate Teidoume	• Oualata • N'Beiket Lahwach • Noual • Jreif

Representatives of the CTCs will coordinate with DREDD directorates in each target wilayah to oversee the following mandates:

- enhance cross-sectoral coordination for CCA between national, regional and local stakeholders;
- ensure that target communities' needs and concerns are considered when developing and implementing natural-resource management and sustainable land-use plans and policies;
- facilitate two-way information- and knowledge-sharing between local-level stakeholders and regional authorities²⁴;
- support the training of community members to implement and maintain adaptation interventions introduced under proposed the project;
- support the integration of climate change considerations into regional- and commune-level policies, plans and budgets for natural-resources management and land-use planning; and
- develop the technical and institutional capacities of relevant stakeholders to drive scalable and sustainable climate action.

In accordance with Mauritanian law, the 12 CTCs will be formally established via announcement; however, over the course of the project, their legitimacy in the relevant legislature will be pursued via registration with the Ministry of the Interior. Formalisation of these project structures will maintain their medium- to long-term sustainability and facilitate community ownership of project activities beyond the project implementation period.

Each CTC will be chaired by the mayor of its respective commune and its members will include, *inter alia*, representatives from the regional environmental delegation (DREDD), 1-2 members of the Communal Council, civil society representatives, and community members. Existing community structures and systems — such as already established Associations for Natural Resources (AGLCs) — will be leveraged, where applicable, to accelerate the establishment of the CTCs and support their sustainability.

Sub-activities implemented to deliver this activity are:

- **Sub-activity 1.1.1.1.** Establish commune-level technical committees (CTCs) within each priority commune in the four target hubs.
- **Sub-activity 1.1.1.2.** Deliver training workshops to CTCs, enhancing members' capacities to implement and manage project activities, support the integration of climate change in regional- and commune-level policies, plans and budgets, and facilitate knowledge sharing between regional- and local-level stakeholders.

²³ Human capital describes the skills, knowledge, and experience possessed by individuals or a population, viewed in terms of their value or cost to an organisation or country.

²⁴ CTCs will provide a platform for regular engagement between stakeholders at all administrative levels. This will facilitate the two-way flow of information between representatives at the village, commune, moughataa, wilayah and national levels, as shown in Table 10.

- **Sub-activity 1.1.1.3.** Train CTC members and government staff to use the NAP Knowledge Management Platform and facilitate the collection and dissemination of climate information and adaptation best practices by CTC members.
- **Sub-activity 1.1.1.4.** Conduct a review of existing wilayah-, moughataa- and commune-level development plans, policies and budgets and prepare policy briefs for the integration of climate considerations and gender-responsiveness into these documents.
- **Sub-activity 1.1.1.5.** Convene training workshops for Regional and Communal Councils and relevant sectors to support the integration of climate change in regional- and commune-level policies, plans and budgets, including through the presentation of the policy briefs prepared under Sub-activity 1.1.1.4.

Output 1.2. Knowledge products developed and disseminated to support decision making and upscaling.

Activity 1.2.1. Develop knowledge products on project lessons learned and best practices through a participatory process engaging the communities.

To ensure that project interventions are responsive to local adaptation needs and generate lasting benefits, Activity 1.2.1 will focus on systematically capturing community perspectives, lessons learned and best practices that emerge throughout project implementation. By integrating participatory approaches, the project will facilitate knowledge exchange between local stakeholders and the Project Management Unit (PMU), enabling adaptive management and the dissemination of successful adaptation strategies that can be referenced when upscaling or replicating climate change adaptation strategies across Mauritania, or when developing new climate change adaptation projects.

Under this activity, bi-annual gender-inclusive discussions will be held between the PMU, the CTCs and communities in the target hubs. These discussions will provide a structured forum for community members to share their experiences, voice concerns and highlight successes and challenges related to project interventions. Ensuring gender inclusivity in these engagements will help capture diverse perspectives, particularly from marginalised groups that may otherwise be excluded from decision-making processes. The insights gained from these discussions will be synthesised into community engagement reports, which will serve as a record of local experiences and provide valuable feedback for refining project implementation strategies.

In parallel, the project will undertake a systematic identification of best practices and lessons learned emerging from project interventions. These knowledge products will be documented in lessons learned and best practice reports, as well as in regular M&E reports, capturing successful approaches, lessons learned and strategies that have proven effective in enhancing climate resilience at the community level. These reports will be made accessible to the public via the NAP knowledge management platform and other channels appropriate to the local context (Activity 1.2.2).

By generating and sharing knowledge products that reflect community experiences and successful adaptation practices, this activity will contribute to evidence-based decision-making and capacity building at local, national and regional levels.

Sub-activities contributing to the delivery of Activity 1.2.1. are:

- **Sub-activity 1.2.1.1.** Hold bi-annual gender-inclusive discussions between representatives from the PMU, CTCs and communities in the target hubs on project intervention successes and challenges, and develop these discussions into community engagement reports.
- **Sub-activity 1.2.1.2.** Identify lessons learned and best practices used in project interventions, and develop these into implementation guides and best practice reports.

Activity 1.2.2. Enhance the dissemination of adaptation knowledge to sub-national decision-makers and communities to support upscaling.

To ensure that knowledge generated through the project is widely accessible and supports long-term climate change adaptation efforts, the proposed project will facilitate the dissemination of knowledge products through both national and international platforms. The primary mechanism for this will be the NAP knowledge management platform, which is currently under development in Mauritania, where monitoring reports, GCF evaluation reports, community engagement reports, records of the validation workshop, final policy briefs and best practices generated under Activity 1.2.1 will be uploaded and stored. This centralised repository will enhance access to critical adaptation information for policymakers, practitioners and community-level stakeholders.

To extend the reach of these knowledge products beyond digital platforms, the project will strengthen the dissemination of stored materials through community-based communication channels. A communications expert will be consulted to design knowledge-management campaigns that are adapted to the local context. These campaigns will include targeted knowledge-sharing efforts via local radio stations, TV programmes and printed materials such as flyers distributed by DREDD officers. These efforts will ensure that information is accessible to local beneficiaries, including vulnerable populations with limited internet access, thereby supporting enhanced awareness, informed decision-making and the replication of best practices at the community level.

Sub-activities contributing to the delivery of Activity 1.2.2. are:

- **Sub-activity 1.2.2.1.** Upload knowledge products (e.g. implementation guides, monitoring and evaluation reports, community engagement reports, policy briefs, lessons learned and best practice reports) onto the NAP knowledge management platform.
- **Sub-activity 1.2.2.2.** Package knowledge in the NAP knowledge management platform into formats that are accessible at local level (e.g. brochures, TV and radio programmes, awareness-raising materials).
- **Sub-activity 1.2.2.3.** Disseminate locally-accessible knowledge products in target and non-target communes across the four project wilayahs to catalyze upscaling, with support from the DREDDs and CTCs.

Outcome 2. Communities in four rural-urban hubs along the Sahara-Sahel boundary in Mauritania have increased livelihood and water security as a result of investment in EbA measures, water access, and sustainable land- and natural resources management practices.

Outcome 2 will enhance livelihood security for communities in four rural-urban hubs along the Sahara-Sahel boundary in Mauritania by promoting investment in Ecosystem-based Adaptation (EbA) measures and sustainable land and natural resource-management practices.

The project will stabilise dunes and reduce sand encroachment by utilising EbA dune-fixation measures (green infrastructure) within twelve priority communes across the Aoujeft, Rachid, Tamcheket and Nema hubs. EbA dune stabilisation techniques will include: i) establishing community-managed buffer zones, or protective greenbelts, along priority communes' borders; and ii) utilising both biological and mechanical dune-fixation techniques (described below) to stabilise dunes in vulnerable areas identified by target communities. These EbA measures will be supplemented by fencing (grey infrastructure), which will be installed around newly established dune-stabilisation sites to protect newly planted vegetation from livestock.

Climate-resilient livestock-management practices will include rotational grazing, supplementary feeding and 'livestock collar re-seeding'. By incorporating these climate-resilient livestock management practices, the project will enhance the resilience of pastoral livelihoods and reduce pressure on natural resources. This approach will complement the establishment of green-grey infrastructure to control sand encroachment and slow the rate of desertification.

Output 2.1. Green-grey dune fixation infrastructure is established to control sand encroachment, enhance the provision of ecosystem services and slow the rate of desertification within the four target hubs.

Activity 2.1.1. Establish 2,123 hectares of EbA dune-fixation infrastructure and 120 kilometres of protective fencing across the four target hubs, to facilitate the rehabilitation and maintenance of degraded landscapes and enhance ecosystem services related to dune stabilisation and the supply of natural resources.

Under this activity, ~2,123 hectares (ha) of EbA dune-fixation measures will be implemented across the four target hubs. These measures will include:

- ~1,138 ha of protective green belts (buffer zones); and
- ~985 ha of mechanical and biological dune-fixation infrastructure.

Green belts in strategic locations will slow sand encroachment and subsequent desertification under future climate change conditions, with natural vegetation serving as a barrier to approaching sand dunes (as shown in Figure 142 in Annex 2). This will involve restoring degraded ecosystems and actively planting protective tree lines along the boundaries of priority communes, to create buffer zones between vulnerable communities and mobile sand dunes (for an example of protected tree lines, see Figure 143 in Annex 2). Each green belt will be ~500 m wide, with restored ecosystems and planted treelines forming a continuous barrier against southward-shifting sand dunes. The strategic positioning of these buffer zones along the north-eastern boundaries of Aoujeft, Rachid and Tamcheket will, in turn, shield south-western communes against the impacts of sand encroachment in the long-term. Using GIS software, the number of hectares of EbA dune-fixation measures needed to establish proposed buffer zones in each hub was calculated (Table 2).

Table 2. Total area of EbA dune fixation measures needed to establish proposed buffer zones in each target hub.

Hub	Area of proposed buffer zone (ha)
Aoujeft	325
Rachid	296
Tamcheket	333
Nema	185
Total	1,139

Mechanical dune-fixation will involve the construction of windbreak hedges via wattling (Figure 145 in Annex 2 **Error! Reference source not found.**), to form ‘stop dunes’, or ‘check dunes’ (see Figure 146 in Annex 2). These structures form a physical barrier between advancing sand dunes and target communities, while the checkerboard pattern assists with capturing sand as it is carried by prevailing winds. Green-grey dune-fixation infrastructure will be installed along the perimeter of each vulnerable town, to ensure that all critical infrastructure is loosely enclosed by treelines, wattling and check dunes. Each of the five towns in Figure 147 in Annex 2 has an area of ~74 ha (740,000 m²); therefore, a total of ~370 ha of green-grey dune-fixation infrastructure (fences, wattling and check dunes) are needed to protect critical infrastructure against sand encroachment at these sites.

Community members in the target hubs of Aoujeft, Rachid, Tamcheket and Nema identified several important sites requiring biological and mechanical dune-fixation during stakeholder consultations- outlined in Table 3 below. Details regarding why these sites were selected are provided in Section 6 of Annex 2: Feasibility Study. The total area of mechanical- and biological dune fixation infrastructure required to protect vulnerable sites identified via GIS software and stakeholder consultation is equivalent to 985 ha.

Table 3. Number of hectares (ha) of green-grey dune-fixation infrastructure required to protect each of the vulnerable sites identified by stakeholders within the four target hubs.

Hub	Site	Area of green-grey dune-fixation infrastructure (ha)	Proposed dimensions (m)	Perimeter (m)
Aoujeft	Toueïguidit	125	2,500 x 500	6,000
	Ain Savra	70	7,000 x 100	14,200
Rachid	The road on the Hssey Lgara– Touajil axis	350	35,000 x 100	70,200
Tamcheket	Tamcheket town	20	2,000 x 100	4,200
Nema	City of Noual	50	5,000 x 100	10,200
Total		615		104,800

Fencing will be installed around green-grey dune-fixation sites, to protect vegetation against foraging and trampling by free-roaming animals. Accordingly, fences will be erected around each of the dune-fixation sites identified in Table 3 above. The planting of treelines for biological dune fixation in buffer zones will follow the best practices described in Section 6 of Annex 2: Feasibility Study and all tree seedlings used during planting exercises will be sourced from local nurseries established under Activity 2.2.1, to support livelihood diversification efforts undertaken by the project. Treelines will comprise a mixture of fast- and slow-growing, climate-resilient species, in accordance with the best practices and lessons learned outlined in Section 6 of Annex 2: Feasibility Study.

Regional livestock cooperatives and breeders’ associations exist within all of the proposed project’s target wilayah; therefore, the proposed project will collaborate with representatives from these organisations — via the commune-level CTCs — to facilitate the adoption of climate-resilient livestock-management practices. Measures will include replacing local livestock breeds with climate-resilient breeds, growing climate-resilient fodder varieties for supplementary feeding during the lean season²⁵, and establishing ‘set-aside’ rotational grazing systems to rehabilitate degraded ecosystems and rangelands. Additionally, pastoral water points and livestock corridors will be established to reverse sedentarisation trends among historically transhumant pastoralist communities, reducing anthropogenic pressure on natural ecosystems and minimising conflict between land users competing for natural resources. By employing these strategies, livestock farmers’ resilience to climate change-induced desertification within the four target hubs will be enhanced.

Sub-activities contributing to the delivery of Activity 2.1.1. are:

²⁵ The period between successive wet seasons and harvests

- **Sub-activity 2.1.1.1.** Support CTCs to co-develop commune-level land rehabilitation plans in collaboration with village-level stakeholders, members of project management teams and DREDD representatives.
- **Sub-activity 2.1.1.2.** Install ~2,123 ha of dune-stabilisation infrastructure (1,138 ha of green belts; 985 ha of biological and mechanical dune-fixation infrastructure) at strategic sites across the four target hubs — to protect critical areas against the impacts of sand inundation.
- **Sub-activity 2.1.1.3.** Install ~120 km of fence lines around dune-fixation sites established under sub-activity 2.1.1.2, to protect biological dune-fixation infrastructure against damage from livestock and unsustainable use of natural resources.
- **Sub-activity 2.1.1.4.** Train livestock herders within the target communes to implement climate-resilient livestock management practices, such as rotational grazing, transhumance, supplementary feeding, agro-silvopasture, collective herding and 'livestock collar re-seeding'.

Output 2.2. Improved access to water for land rehabilitation and agricultural activities.

Activity 2.2.1. Establish community-managed Water User Groups (WUGs) and commune-level water monitoring and regulation plans.

Commune-level water user groups (WUGs) will be formed in the 12 priority communes. Each WUG will be led by elected community leaders from the respective priority communes and will report to the CTCs. WUG members will comprise local community members and representatives from livestock cooperatives, farmer-based organisations and women's groups in each target commune. WUGs will be trained to monitor water availability and usage within their respective communes and to oversee and manage the construction, operation and maintenance of infrastructure for capturing surface water, managing runoff flow, increasing groundwater recharge and reducing flood risks.

A team of national experts — comprising two hydrogeologists from the Ministry of Water and Sanitation (MHA²⁶) and two meteorologists from the Meteorological Service of Mauritania (ONM) — will conduct hydrogeological studies to quantify and map groundwater and surface water resources within each of the target hubs. In particular, this team will be mandated to identify flood-risk areas and determine the interconnectivity of aquifers, informing the selection of strategic sites to install: i) groundwater dams, for improving aquifer recharge; ii) gabions, dikes and stone bunds for reducing flood risks; and iii) reservoirs, boreholes, well points and rainwater-harvesting systems to increase rainwater capture and storage. The selection of target sites for water management infrastructure will be summarised in the form of commune-level water management plans.

Sub-activities contributing to the delivery of Activity 2.2.1. are:

- **Sub-activity 2.2.1.1.** Establish community-managed Water User Groups (WUGs) within each target commune and train members to implement and maintain water-related activities introduced under Activity 2.2.2 of the project.
- **Sub-activity 2.2.1.2.** Support CTCs to raise awareness about sustainable water usage and co-develop commune-level water monitoring and regulation plans in collaboration with WUGs.
- **Sub-activity 2.2.1.3.** Conduct hydrogeological studies (or consult existing hydrogeological maps, where applicable) and engage with WUG members to identify priority sites to install water-management infrastructure (Sub-activities 2.2.2.1, 2.2.2.2 and 2.2.3.1) in each target commune. Summarise the finding of site selection into a commune-level water management plan for each priority commune.

Activity 2.2.2. Install physical water management infrastructure — including weirs, gabions, dikes, stone- and earthen bunds, groundwater dams and water access points for pastoralists.

Under Activity 2.2.2, water security within the proposed project's target hubs will be strengthened by installing flood management infrastructure — including weirs, gabions, dikes, stone bunds, earthen bunds and groundwater dams — in each priority commune. These measures will reduce surface water runoff, increase groundwater infiltration and recharge rates, and enhance the capture of rainfall for use in supporting sustainable agriculture and climate-resilient livelihoods.

Additionally, pastoral water access points — that is, boreholes equipped with solar pumps — will be established in *graras* and along historical transhumance routes, to support year-round agricultural production and transhumance and reverse sedentarisation trends among nomadic pastoralists. These physical measures will: i) improve the availability of fresh water for use in agriculture and livestock farming; ii) reduce unsustainable abstraction from aquifers; and iii) ensure the long-term sustainability of climate-resilient livestock management, agricultural practices and livelihoods introduced under Output 2.3.

²⁶ Ministère de l'Hydraulique et de l'assainissement

The extent of physical water-management infrastructure installed within each target hub should not exceed: i) four groundwater dams (each with a storage capacity of ~10,000 m³); ii) 4 km of clay dikes; iii) 4 km of stone gabions; and iv) 80 ha of stone bunds arranged in 50 x 50 m squares. Details regarding site selection for these interventions are provided in Section 6 and 7 of Annex 2: Feasibility Study.

Sub-activities contributing to the delivery of Activity 2.2.2. are:

- **Sub-activity 2.2.2.1.** Install physical water management infrastructure — including weirs, gabions, dikes, stone- and earthen bunds, groundwater dams and solar-powered pumps — at strategic sites within each target commune, to improve water access and availability, increase groundwater recharge rates and reduce flood risks in the target hubs.
- **Sub-activity 2.2.2.2.** Establish water access points along historical transhumance routes and in *graras*, to improve nomadic pastoralists' access to water and reduce sedentarisation among livestock herders.

Activity 2.2.3. Install 12 rainwater-harvesting systems and communal cisterns (5,000 L per system) within each target commune

Under Activity 2.2.3 of the proposed project, the installation of rainwater harvesting (RWH) systems and communal cisterns will increase the capture of surface water runoff during extreme rainfall events and increase target communities' capacities to store water for use during prolonged periods of drought. To promote fair access, sites located near communal buildings (town halls, schools and health posts) and agricultural areas (including market gardens and nurseries) will be given preference, to ensure water is used for agricultural activities and sustainable livelihoods introduced under the proposed project.

To establish the RWH systems and communal cisterns, local labour will be employed and supervised by trained WUG members. Twelve RWH systems and communal cisterns, with a capacity of 5,000 litres each, will be distributed to each of the 12 priority communes, totalling 144 RWH systems and cisterns across the four target hubs. Details regarding site selection for these interventions are provided in Section 6 and 7 of Annex 2: Feasibility Study.

Sub-activities contributing to the delivery of Activity 2.2.2. are:

- **Sub-activity 2.2.3.1.** Install 12 rainwater-harvesting systems and communal cisterns (5000L per system) within each target commune across the four hubs, to improve access to water for agricultural livelihood activities.

Output 2.3. Climate-resilient agricultural livelihoods based on sustainable land- and natural resource-use are developed and/or strengthened to reduce land degradation and support climate-resilient income-generation by community members within the target regions.

Activity 2.3.1. Facilitate the adoption of climate-smart agricultural practices and sustainable, diversified livelihoods by farmers within the target communes

The project will facilitate the adoption of climate-resilient practices (Activity 2.3.1). The project will encourage livelihood diversification by supporting the uptake of sustainable income-generating activities such as beekeeping, poultry farming and value-added processing, reducing dependence on traditional agricultural activities and building resilience against climate-related risks. To further improve rural farmers' resilience to the impacts of climate-change desertification, MEDD and the Ministry of Rural Development (MDR) extension officers will supply local nurseries with an initial stock of climate-resilient²⁷ seed varieties for sale and distribution to local farmers and market gardeners within each of the target hubs and train nursery operators to maintain sustainable stocks (or seed banks).

Horticultural centres established in each priority commune will be supplied with the initial materials needed to commence business operations. These may include beehives for apiculture, climate-resilient seed varieties for seed banks and irrigation kits for market gardens. Access to markets to sell horticultural produce will be improved by: i) organising weekly markets in each priority commune — arranged by CTCs — to facilitate trade between local community members; and ii) using 'community carts'²⁸ to transport goods to larger, urban markets within the target hubs and surrounding areas. Please refer to Table 82 in Annex 2 for a summary of the types of climate-smart agriculture (CSA) practices currently implemented and Table 84 in Annex 2 for a summary of potential alternative livelihoods.

²⁷ Crop varieties with drought- and heat-resistant genotypes

²⁸ 'Community carts' (horse-, ox-, or donkey-drawn carts) provide a means of transport in rural areas — for example, for transporting trees to be planted in protected areas, water for agriculture and nurseries, and stones for the construction of stone bunds. This transport can be used by individuals for a small fee, to generate income and maintain the condition of animals

Sub-activities contributing to the delivery of Activity 2.3.1 are:

- **Sub-activity 2.3.1.1.** Establish nurseries and seed banks in each target commune to supply activities related to land rehabilitation and dune fixation (Activity 2.1.1), CSA practices (Sub-activity 2.3.1.3) and horticultural activities such as market-gardening (Sub-activity 2.3.1.4).
- **Sub-activity 2.3.1.2.** Collect cuttings and seeds from agricultural crop species, as well as indigenous grass and tree species, to serve as stock material for nurseries and seed banks established under Sub-activity 2.3.1.1.
- **Sub-activity 2.3.1.3.** Train farmers within the target communes to practice climate-resilient crop agriculture and use improved agricultural technologies, including drip irrigation kits, solar powered pumps, integrated pest management strategies, zai pits and half-moons.
- **Sub-activity 2.3.1.4.** Conduct site visits and provide technical support to facilitate the uptake of sustainable livelihood activities — including horticulture (market-gardening), apiculture, poultry farming, livestock feed production and the collection and sale of non-timber forest products — by community members within the target communes.
- **Sub-activity 2.3.1.5.** Supply farmers and horticulturalists with water-efficient irrigation equipment and climate-resilient crop varieties to support the uptake of agricultural activities adopted under Sub-activities 2.3.1.3 and 2.3.1.4.
- **Sub-activity 2.3.1.6.** Improve access to urban markets and develop value chains for offloading agricultural produce within each target commune, to enhance income generated from sustainable agricultural livelihoods.

Activity 2.3.2. Establish a small grants facility to facilitate continued investment in upscaling successful EbA activities and sustainable livelihoods.

A project Small Grants Facility (SGF) will be established to enhance local communities' access to finance to upscale successful EbA interventions and sustainable, agriculture-based livelihoods introduced under the project. The on-granting mechanism will promote and support community-led initiatives, fostering the autonomous uptake of EbA practices by community members. GCF Proceeds will be used to finance the sub-grants. The SGF will be housed in MEDD, and sub-grant agreements will be executed between MEDD and each sub-grant recipient.

An SGF Steering Committee will be instituted to guide and oversee the establishment and operations of the SGF. It will be chaired by UNEP as the project AE, and will include representation from MEDD and other relevant Ministries (the Ministry of Economy and Finance; the Ministry of Agriculture; the Ministry of Livestock Farming; the Ministry of Housing, Urbanism and Regional Planning), as well as from UNDP (to ensure links and synergies with the Small Grants Programme it manages). The SGF Steering Committee will review the recommended funding proposals and jointly make the grant award decisions. UNEP as the chair of the SGF Steering Committee will report on its operations to the PSC. Table 81 in Annex 2 outlines the preliminary proposed design of the SGF, and the process for submitting and screening sub-grant proposals. The screening begins with CTCs (including representatives of vulnerable groups and CSOs) and is completed by the SGF team housed in MEDD (see below), before submission to the SGF Steering Committee.

Establishing and operating a project SGF requires procuring an SGF management team to be housed in MEDD, to: i) oversee the facility's operation and maintenance; and ii) manage the approval, disbursement and monitoring of small grants throughout the project implementation period. Prior to the SGF establishment, an international consultant (finance expert) will develop the SGF's policies, frameworks and guiding principles. These will be built on lessons learned and best practice from other projects in the region. For example, the experience with the project *Enhancing resilience of communities to the adverse effects of climate change on food security in Mauritania (PARSACC)* showed that building social and capital was a key success factor. The international consultant will also design a funding proposal template and accompanying selection criteria for the SGF, streamlining the small-grants approval process.

Capacity building support will be provided to MEDD to strengthen MEDD's institutional, operational and financial capacity for the design, execution and oversight of the SGF. This consultancy will also provide technical backstopping to MEDD on these aspects throughout the implementation of the SGF. As a first step, building on the findings of the HACT micro-assessment undertaken in 2023, specific capacity or process gaps will be identified that may impact the management of the SGF. In addition, experiences and lessons learnt from MEDD's ongoing small grants programme will be reviewed. This programme has been operational since 2022, with \$250,000 awarded annually to approximately 20 local NGOs, cooperatives and other CSOs for climate change adaptation and environmental initiatives.

Furthermore, to increase the capacity of potential recipients of the grants, a suitable service provider will be contracted to work with potential recipients to: i) strengthen their organisational and financial management capacity; ii) increase their capacity to develop and submit project concepts so as to access funds from the SGF; and iii) improve their capacity to successfully implement EbA measures funded through the SGF. It is anticipated that 5 projects per year in each of

the 12 target communes will receive grants (60 grants per year) over the final four years of the proposed project. Sub-grants will only be awarded to low risk projects, and will be screened to avoid duplication of other project activities.

Other indicative criteria for projects to be awarded sub-grants:

- Recipients should be legally registered entities, such as NGOs, CSO, cooperatives, associations or local enterprises.
- The proposed initiative to be financed by the sub-grant should support households or communities to adapt to specific climate change risks.
- The proposed initiative should support the uptake or strengthening of climate-resilient livelihoods, or EbA approaches that build the resilience of households or communities.
- The proposed initiative should include measures to ensure the long-term sustainability of the impacts to be achieved with the sub-grant financing.
- The proposed initiative should ensure the development of business skills and market linkages, where relevant. Level of co-financing contribution (cash or in-kind).
- The recipient entity should provide the required ratio of co-financing contribution (cash or in-kind) – to be set at SGF inception stage.
- Maximum request of ~\$25,000.

Indicative criteria for sub-grant recipients are included in Section 11 of Annex 2: Feasibility Study.

These criteria will be finalised during the initial stage of project implementation, in alignment with existing commune-level adaptation plans.

Sub-activities contributing the delivery of this activity are:

- **Sub-activity 2.3.2.1.** Establish and operationalise a Small Grants Facility (SGF) to facilitate continued investment in upscaling successful EbA activities and sustainable livelihoods introduced under the project.
- **Sub-activity 2.3.2.2.** Prepare budget briefs for directing regional government funds into the SGF established under Sub-activity 2.3.2.1. to promote government investment in CCA.
- **Sub-activity 2.3.2.3.** Develop monitoring and reporting mechanisms to ensure the traceability and risk management of funds between the SGF and local-level stakeholders.
- **Sub-activity 2.3.2.4.** Produce annual monitoring and evaluation reports for sub-projects funded via the on-granting mechanism established under Sub-activity 2.3.2.1.

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

Governance structures

The implementation arrangements and governance structure for the proposed project are illustrated in Figure1010Error! Reference source not found.

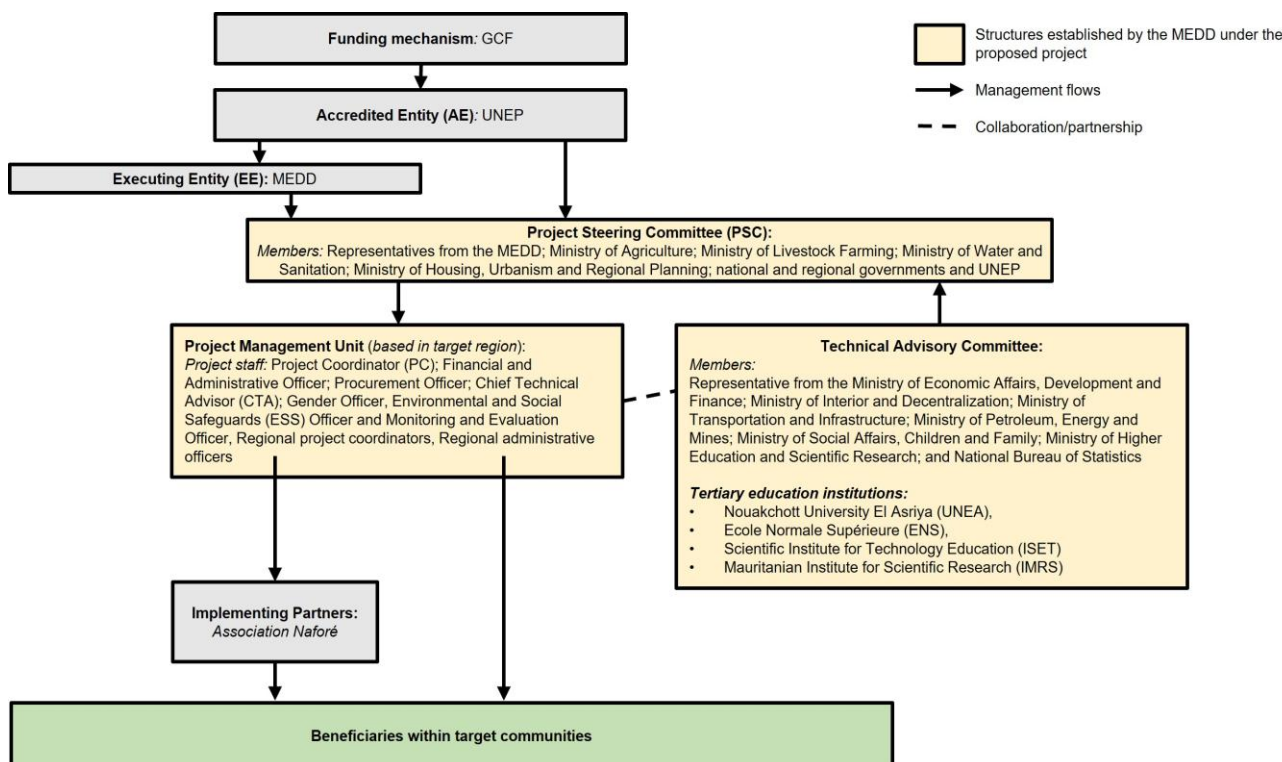


Figure10. Project implementation arrangements and governance structures

Accredited Entity

UNEP is the Accredited Entity (AE) for this project, as designated by the Government of the Islamic Republic of Mauritania (GoM). The AE will be responsible for the overall management of the project as well as for the financial monitoring and reporting aspects of the proposed project's activities. UNEP will be responsible for overseeing the implementation, monitoring and evaluation (M&E) of the project in coordination with the Project Steering Committee (PSC) and the Project Management Unit (PMU). A UNEP Programme Officer (PO) will be responsible for project supervision to ensure compliance with GCF's and UNEP's standard policies and operating procedures. The PO will be required to participate in annual Project Steering Committee (PSC) meetings. The PO will be supported by a UNEP Fund Management Officer (FMO) on financial monitoring and supervision.

Experience and track record of AE

UNEP has a broad global portfolio on climate change projects funded via the, *inter alia*, GCF, Least Developed Countries Fund (LDCF), Special Climate Change Fund (SCCF), Adaptation Fund (AF) and bilateral funds such as the European Commission. From these projects, UNEP has supported national governments and local communities to adapt to climate change. UNEP's knowledge-base is derived from completed and ongoing projects supporting: i) methods and tools for decision-making; ii) prioritising, designing and implementing adaptation interventions; iii) enhancing climate resilience by restoring vulnerable ecosystems that underpin community livelihoods; and iv) monitoring the socioeconomic and environmental benefits of adaptation interventions.

UNEP also has a longstanding relationship in Mauritania, having supported the country in GEF enabling activities, including the development of the National Adaptation Programme of Action (NAPA) as well as national communications since 1997. UNEP is also currently supporting the GCF National Adaptation Plan (NAP) Readiness Programme and has assisted the implementation of several CCA projects in Mauritania with

the support of international climate change funds, including the GEF-funded projects, EbA South and DIMS. As the GCF AE, UNEP will draw on previous experience and lessons learned to oversee the efficient and effective delivery of the project objectives during the implementation of the proposed project.

Executing Entity

The proposed project will be implemented over a six-year period from 2025 to 2030 and executed by Mauritania's Ministry of Environment and Sustainable Development (MEDD). The MEDD is tasked with developing and executing the National Sustainable Development Plan (SNEDD), implementing the National Environmental Action Plan (PANE II), formulating operational plans, and overseeing and securing financial and material support for program implementation. It has created sector-specific plans and programmes concerning desertification and climate change, biodiversity, wetland conservation, and coastal ordinance. These areas of expertise and context-specific knowledge allow the MEDD to provide the necessary oversight and guidance during the implementation of the project. The MEDD also serves as the country's focal point for the United Nations Framework Convention on Climate Change (UNFCCC). Moreover, the Coordination Cell for the National Climate Change Programme (*Cellule de Coordination du Programme National sur le Changement Climatique*, CCPNCC) operates within the MEDD to coordinate and implement climate change initiatives in the country, as well as handle reporting, such as the National Communications III. Both the MEDD and CCPNCC consult directly with several ministries and departments to promote these programs.

As the Executing Entity (EE), the MEDD will coordinate the implementation of the project at national level and will report to UNEP as the AE. To ensure that the project remains in line with national policies, MEDD and UNEP will establish a Project Steering Committee (discussed below).

Project Steering Committee

To ensure project compliance with national policies, MEDD and UNEP will establish a Project Steering Committee (PSC). This committee will be composed of at least one representative from the MEDD, the Ministry of Agriculture; the Ministry of Livestock Farming; the Ministry of Water and Sanitation; and the Ministry of Housing, Urbanism and Regional Planning, respectively. In addition, to ensure local representation, a representative of regional councils (rotating between project regions), a mayor representing communes (rotating between communes) and a CSO representative will be included in the PSC. A representative of UNEP will also form part of the PSC. This committee will have a decision-making capacity and will primarily serve to provide project oversight and advisory support, including: a) overseeing project implementation; and b) reviewing annual budget and workplans. The committee will meet at least twice a year (once per semester), with ad hoc meetings as required to discuss project results and performance indicators and provide strategic guidance.

Technical Committee

In addition to the PSC, a Technical Committee will be established under the chairmanship of the Ministry of Environment and Sustainable Development (MEDD) to provide technical guidance and ensure ownership, communication and reporting on the project with national adaptation communities. This Technical Committee will be composed of one representative from the Ministry of Economic Affairs, Development and Finance, the Ministry of Interior and Decentralization, the Ministry of Transportation and Infrastructure, the Ministry of Petroleum, Energy and Mines, the Ministry of Social Affairs, Children and Family, the Ministry of Higher Education and Scientific Research, and the National Bureau of Statistics, respectively. The Technical Committee will also request specific technical support from other relevant stakeholders, as required. These stakeholders will include tertiary education institutions in Mauritania, notably the Nouakchott University El Asriya (UNEA), Ecole Normale Supérieure (ENS), Scientific Institute for Technology Education (ISET) or the Mauritanian Institute for Scientific Research (IMRS). The Technical Committee will seek cooperation opportunities with potential project partners including, *inter alia*, UNDP Country Office, UNOPS, IFAD, FAO, World Bank, AfDB, AFD, GIZ, CIDA and IUCN, as well as national NGOs and private sector associations. The Technical Committee will report to the PSC and will meet at least twice a year (once per semester), with ad hoc meetings held when required.

Project Management Unit

The PMU will be embedded within the MEDD and will consist of: i) a Project Coordinator (PC); ii) a Financial and Administrative Officer; iii) a Procurement Officer; iv) a Chief Technical Advisor (CTA); v) a Communications Officer; vi) a Gender Officer; vii) an Environmental and Social Safeguards (ESS) Officer; and viii) a Monitoring and Evaluation Officer.

The PC will be a full-time, competitively recruited staff member, responsible for the daily implementation and management of project activities. The PC will: i) report directly to the PSC on project management-related matters; ii) manage the project in accordance with the specified workplans and allocated budget; iii) ensure that all project interventions are implemented according to GCF and UNEP guidelines; iv) manage and disburse financial resources allocated for the project during project implementation; v) work closely with national and local authorities to ensure that the project is managed effectively and that the needs of all beneficiary groups are considered; and vi) oversee the efficient and effective information and knowledge-transfer to relevant project partners. All PMU members will report to the PC. The CTA will provide technical guidance at all stages of project implementation, to ensure project interventions are implemented effectively and efficiently, in line with the latest best practices and lessons learned from around the globe. The CTA will be a UNEP National Officer. A part-time Communications Officer will be responsible for managing all awareness-raising and knowledge disseminations activities over the course of the proposed project.

Implementing Partners

A local NGO in Mauritania, Association Naforé²⁹, has been identified as an eligible implementing partner for the proposed project. Naforé will assist the MEDD to implement on-the-ground interventions (Outputs 2.1, 2.2 and 2.3) within the proposed project's four target hubs. Nafore will report directly to the MEDD and collaborate with the PMU to implement ecosystem restoration, livelihood and water infrastructure activities. Additional Implementation Partners may be onboarded during the project implementation stage.

Since its inception in 2009, Naforé has implemented several projects in the domains of environmental protection, health, hygiene and sanitation, showcasing its capacity through diverse projects and a wide range of expertise among its members. Naforé's main strengths are summarised below:

- **Organisational strength:** Naforé boasts a membership base of 25 individuals from various professional backgrounds, contributing to its rich pool of expertise and allowing it to respond adeptly to environmental, health, hygiene and sanitation challenges.
- **Mission and objectives:** The organisation is dedicated to supporting state policies focused on environmental protection and poverty alleviation using participatory approaches that involve all stakeholders. Its goals include contributing to state environmental protection policies, improving the living conditions of disadvantaged groups through income-generating activities that are compatible with natural resource conservation, and participating in national, sub-regional or international actions promoting childhood support and indigent care.
- **Environmental initiatives:** Naforé is actively engaged in conservation efforts that address challenges related to biodiversity, species protection, habitat protection, natural resource conservation, climate change mitigation, waste management, pollution control, sustainable agriculture and rural hygiene and sanitation. These initiatives reflect a comprehensive approach to environmental stewardship, with the aim of maintaining biodiversity, supporting sustainable land and natural resource use and enhancing community resilience to climate change and its effects.
- **Project execution:** Naforé has a proven track record of successfully managing projects funded by several national and international bodies, including inter alia UNDP, IUCN, UNICEF, FAO and GIZ. Projects such as promoting green jobs for rural youth, environmental awareness campaigns and improving hygiene and sanitation in healthcare centres demonstrate Naforé's ability to effectively utilise funds and achieve project goals.
- **Strategic partnerships:** The organisation has built strong partnerships with state agencies, international institutions and local communities, showcasing its collaborative approach and ability to work within the broader framework of Mauritania's development policies.
- **Expertise in environmental impact studies:** Naforé has been involved in significant environmental impact studies for several sectors, including oil exploration and infrastructure development. These activities highlight its expertise in assessing and mitigating environmental impacts in line with national and international standards.

Operational arrangements

- **Key contractual agreements:** The partnership between GCF, UNEP, the MEDD and other Implementing Partners is governed by a set of legal agreements outlining the scope, responsibilities, and financial arrangements of the project. These agreements define the operational framework and set the foundation for project execution.
 - A Funded Activity Agreement (FAA) will be established between the GCF and UNEP.

²⁹ Association Nafore. N.d. A propos de nous. Available at: <https://naforeorg.wordpress.com/>

- A Project Cooperation Agreement (PCA) will be established between UNEP and the MEDD (EE). This will serve as the Subsidiary agreement for the purposes of the FAA.
- Naforé's engagement will be formalised through contractual agreements with the MEDD, outlining project activities, budget allocations and reporting requirements. These agreements will ensure that Naforé's implementation efforts are fully integrated into the project's overall governance and reporting structures.
- **Disbursement and financial management:** Financial flows from GCF to UNEP and subsequently to the MEDD are regulated through contractual agreements, ensuring accountability and efficient use of funds. This arrangement includes provisions for financial auditing, reporting and oversight mechanisms to maintain financial integrity throughout the project lifecycle.

The flow of funds, legal instruments and reporting mechanisms for the project are shown in Figure 11.

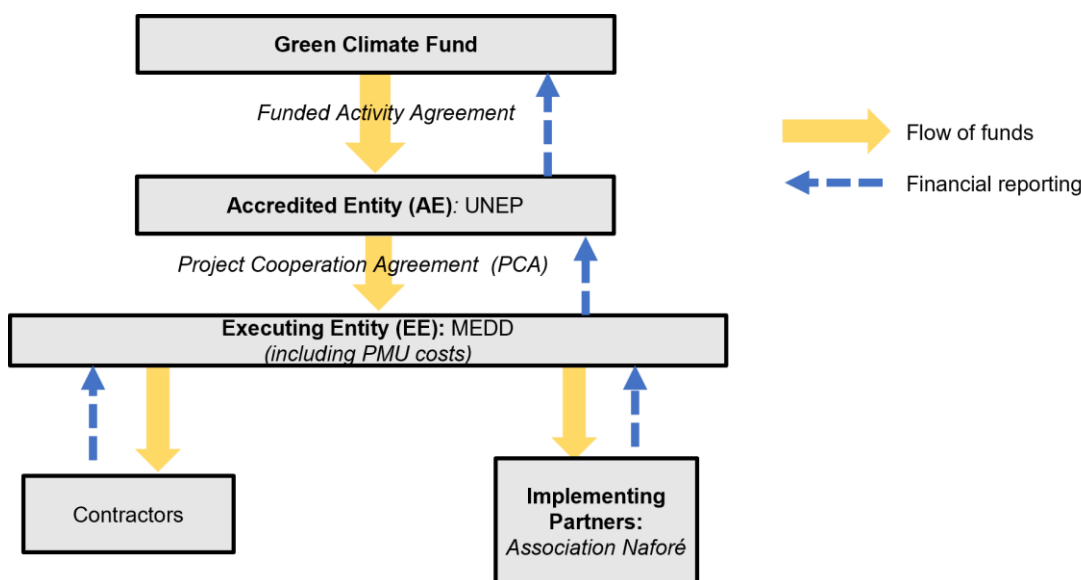


Figure 11. Flow of funds and reporting mechanisms for the proposed GCF project.

A separate flow of funds related explicitly to the on-granting mechanism established under Activity 2.3.2 is shown in Figure 159 of Annex 2, Section 11. **Error! Reference source not found.**

The project will be operated in accordance with UN regulations, rules and policies including the Anti-Fraud and Anti-Corruption Framework of the United Nations Secretariat and fiduciary management of the project will be managed by UNEP in its role as AE. The proposed project will be guided by UN Financial Regulations, Rules and Practices. UNEP will also closely monitor the financial management of the Project using the established Monitoring and Evaluation procedure and financial reporting mechanism, including an annual audit; and establish internal controls for the Project and project fund management.

Furthermore, the Project Cooperation Agreement (PCA) between UNEP and MEDD will include warranties and caveats by MEDD to inter alia ensure compliance with the Anti-Fraud and Anti-Corruption Framework of the United Nations Secretariat, as well as the Green Climate Fund Policy on Prohibited Practices.

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

Grant finance from the GCF is the most appropriate form of concessional grant finance for the proposed project. This is because there are limited financial resources for investment in climate-resilient sustainable development in Mauritania, which is a considerable barrier to the implementation of CCA interventions in the country. In addition, there is a high level of public and external debt in Mauritania, which limits the capacity of the GoM to repay loans. Moreover, the country is a least developed country (LDC), with low income and severe structural impediments to sustainable development. As a result, communities in Mauritania require support from the GCF to achieve a paradigm shift towards climate-resilient sustainable development.

Communities living along the Sahara-Sahel boundary of Mauritania are highly vulnerable to the impacts of climate change, particularly the increasing rates of desertification, dune advancement and water scarcity. In 2019, Mauritania was ranked the 20th most vulnerable country to climate change and other global challenges

on the Notre Dame Global Adaptation Initiative (ND-GAIN) country index. Notably, the country had the highest overall sensitivity score³⁰, largely because of prevailing insecurity within the food and water sectors³¹. Moreover, of the 182 countries on the ND-GAIN readiness index, Mauritania was ranked the 75th least ready to respond to climate change and other global challenges, with education and control of corruption comprising the worst-scoring readiness indicators in the country³². As a result, there is an urgent need for support to improve the adaptive capacity of rural communities in Mauritania. Despite this, the GoM has limited resources to implement CCA, as is further specified in the constraints summarised below.

Development challenges in targeted rural communities

The vulnerability of Mauritania's population is high because of elevated levels of poverty and a strong dependence on 'climate change sensitive' sectors, including agriculture, fisheries, mining and livestock³³. In addition, access to basic services and facilities is unequal between rural and urban areas and major obstacles are encountered by those attempting to transition out of poverty³⁴. As a result, approximately 6% of Mauritania's population live below the international poverty line (2011 purchasing power parity US\$1.9 per day)³⁵ and ~28% below the national poverty line³⁶. Moreover, 70% of the country's impoverished households live in rural areas and are mainly engaged in agriculture and livestock-rearing. This high prevalence of poverty and limited access to basic resources means that rural communities living within the four target hubs lack the adaptive and financial capacity to invest in climate-resilient sustainable development and are almost entirely dependent on external support from both government and development partners.

Public debt and fiscal constraints at the national level

Mauritania's Intended Nationally Determined Contribution (INDC) to the UNFCCC states that major investments totalling USD9.4 billion are required for the Mauritanian population to adequately adapt between 2015 and 2030 to the impacts of climate change. It is further stated that USD8.2 billion is required to reduce projected emissions in the country by 22% before 2030. However, the high level of public debt (99% of GDP in 2016) constrains public financing for CCA, making it difficult for the GoM to secure financing for adaptation projects. Moreover, the fiscal capacity of the GoM to meet the commitments of the NDC without external support is further reduced by recent volatility in global financial markets as a result of the Russia-Ukraine War. As a result, baseline development issues are often prioritised in the constrained budget of the country, resulting in limited climate change adaptation and mitigation action.

The GoM is actively working to mobilise funding towards the INDC and large investments have already been made in climate change adaptation. For example, the GoM spent USD335 million between 2011 and 2015 to provide support to the drought-affected population, and invested USD330 million in poverty alleviation between 2006 and 2010 — particularly focusing on land management in agropastoral areas. However, further investments to address climate change are almost entirely reliant on assistance from external multilateral development partners, including the GCF. Consequently, the proposed outcomes, outputs and activities of this project have been designed to be fully aligned with the objectives of the GCF, with technical support provided by UNEP (see B.4. section).

Justification for level of concessionality

The proposed interventions aim to improve subsistence-level agriculture, water access, and health outcomes in marginalized communities. Grant funding is proposed as much of the value generated accrues to households through local consumption instead of through formal value chains and trade, which would generate the sufficient cashflows needed to sustain alternative funding instruments. In addition, the i) high levels of interdimensional poverty in Mauritania; ii) the high debt-to-GDP ratio of the GoM; and iii) the low levels of economic development in the country adds to the financial-level risk in the project. Furthermore, there is limited incentive for the private sector to participate in the proposed project because: i) there is limited presence of private sector entities in the target areas; and ii) the project has a low potential for cost-recovery. As a result, the private sector is excluded as an option for alternative funding for the proposed project.

B.6. Exit strategy (max. 500 words, approximately 1 page)

³⁰ Reflects the extent to which the country depends on 'climate change sensitive' sectors.

³¹ WB Group. 2021. Climate change knowledge portal: Mauritania — vulnerability.

³² University of Notre Dame Global Adaptation Initiative. 2022. Country index — Mauritania.

³³ WB Group. 2021. Climate change knowledge portal: Mauritania — vulnerability.

³⁴ WB Group. 2020. Poverty and equity brief — Mauritania.

³⁵ WB Group. 2020. Poverty and equity brief — Mauritania.

³⁶ United Nations High Commissioner for Refugees (UNHCR) Mauritania. 2021. Protection and solutions strategy.

Sustainability through technical and institutional capacity building

The project will create a strengthened enabling environment that will support the long-term sustainability and scaling of CCA measures in Mauritania. This enabling environment will include: i) improved vertical (across national, regional and local levels) and horizontal (between government departments at the hub level) coordination mechanisms, including the commune-level technical committees (CTCs); ii) updated policies, plans and strategies that mainstream CCA and natural resource management in a gender-responsive manner; and iii) strengthened knowledge management systems aimed at collecting, tailoring and disseminating CCA best practices. This will empower relevant government institutions at the national and sub-national levels to dedicate more resources and budget to incorporating gender-responsive CCA initiatives into development plans, policies and budgets.

Equipment provided through the project will also provide recipients (GoM, CTCs, cooperatives, associations etc.) with the technical capacity to continue implementing CCA measures. To ensure that this remains functional, the GoM will assume responsibility for the continued operation and maintenance of this equipment after the project lifespan. A structured plan, with responsible entities, estimated costs and funding sources will be developed as part of the project's exit strategy. In cases where the equipment is provided to community user groups (e.g. water user groups or natural resource management associations), a management system will be established, with a fee structure that generates funds for the operation, maintenance and repair of the equipment. Where needed, maintenance costs will be integrated into government (MEDD) budgets, with a set schedule for inspections to identify maintenance needs. This will promote the sustainability of the CCA measures.

Social and environmental sustainability

The geographical isolation of the target hubs means that community participation and ownership of the proposed project's on-the-ground interventions are critical to their longevity during and past the project's lifespan. Government stakeholders have highlighted this participatory approach as a critical component to ensure the success and longevity of project interventions. Community and decision-makers participation will be secured through the CTCs, WUGs and other community level groups.

Financial sustainability

Several financing opportunities at the government and community levels will enable the continued O&M of interventions beyond the project's lifespan. At the government level, the prioritisation of CCA, gender considerations and natural resource management into policies, plans and strategies will promote the allocation of sufficient funds to CCA initiatives through national, regional and local government budgets.

A critical component of achieving financial sustainability will be establishment of an on-granting mechanism under Activity 2.3.2, designed to provide financial stability and facilitate continued investment in upscaling successful EbA activities and sustainable livelihoods. To ensure the longevity and effectiveness of this mechanism beyond the project lifespan, the on-granting mechanism will be housed in MEDD, which has the mandate and capacity to manage and scale CCA and EbA initiatives. This entity will be responsible for overseeing the on-granting mechanism, ensuring that it continues to support community-led initiatives and foster the autonomous adoption and upscaling of successful EbA measures and sustainable agricultural-based livelihoods within the target hubs.

To solidify this transition, the project will work closely with the designated government entity to build its capacity in fund management, monitoring and evaluation, as well as in the strategic allocation of resources towards high-impact EbA interventions. This collaboration will include the development of clear guidelines for fund disbursement, criteria for project selection, and mechanisms for tracking and reporting on the impact of funded projects. The on-granting mechanism will be embedded within an existing governmental framework to encourage directing a percentage of regional government development funds into the SGF. The long-term viability of this mechanism will be supported by the development of a monitoring and reporting system that will ensure the traceability and risk management of funds between the SGF and local-level stakeholders.

At the household level, the financial benefits incurred through improved: i) climate-resilient agricultural and alternative livelihoods; ii) access to water, food and wood fuel; iii) protection from permanent losses; and iv) built and natural capital through dune stabilisation are expected to incentivise individuals to maintain introduced practices, equipment and infrastructure after the project has concluded. In total, the project's economic benefit to cost ratio is estimated at 5.86 to 1 (Annex 3: FEA and Annex 3b). Additionally, technical support will be given to communities to facilitate market access for alternative livelihoods introduced by the project, which will assist in ensuring the long-term viability of these livelihoods.

It is assumed that the capacity built through the project within institutions and among individuals will be retained post-project, ensuring the continuity of skills and knowledge necessary for the ongoing application and scaling of CCA measures. This retention will be facilitated by: i) institutionalising training programmes; ii) embedding CCA and sustainable natural resource management practices into standard operating procedures; and iii) fostering a culture of continuous learning and adaptation within participating organisations.

FINANCING INFORMATION						
C.1. Total financing						
(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount			Currency		
	30,902,775			million USD (\$)		
GCF financial instrument	Amount	Tenor	Grace period	Pricing		
(i) Senior loans	<u>Enter amount</u>	<u>Enter years</u>	<u>Enter years</u>	<u>Enter %</u>		
(ii) Subordinated loans	<u>Enter amount</u>	<u>Enter years</u>	<u>Enter years</u>	<u>Enter %</u>		
(iii) Equity	<u>Enter amount</u>			<u>Enter % equity return</u>		
(iv) Guarantees	<u>Enter amount</u>	<u>Enter years</u>				
(v) Reimbursable grants	<u>Enter amount</u>					
(vi) Grants	30,902,775					
(vii) Results-based payments	<u>Enter amount</u>					
(b) Co-financing information	Total amount			Currency		
	2,657,570			million USD (\$)		
Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
GoM	<u>In kind</u>	<u>2,657,570</u>	<u>million USD (\$)</u>	<u>Enter years</u> <u>Enter years</u>	<u>Enter%</u>	<u>Options</u>
Click here to enter text.	<u>Options</u>	<u>Enter amount</u>	<u>Options</u>	<u>Enter years</u> <u>Enter years</u>	<u>Enter%</u>	<u>Options</u>
Click here to enter text.	<u>Options</u>	<u>Enter amount</u>	<u>Options</u>	<u>Enter years</u> <u>Enter years</u>	<u>Enter%</u>	<u>Options</u>
Total financing (c) = (a)+(b)	Amount			Currency		
	<u>33,560,345</u>			<u>million USD (\$)</u>		
(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)	The maximum contribution to parallel financing through projects and programmes already operating in the proposed project's target regions is estimated at USD41.3 million.					

C.2. Financing by component

Component	Output	Indicative cost million USD (\$)	GCF financing		Co-financing		
			Amount million USD (\$)	Financial Instrument	Amount million USD (\$)	Financial Instrument	Name of Institutions
<u>Outcome 1.</u>	<u>Output 1.1.</u> Governance structures are strengthened, sustainable financial mechanisms are established and policy briefs are prepared to support the integration of climate change considerations and EbA into plans, policies and budgets.	<u>USD 2.159 million</u>	<u>USD 2.149 million</u>	<u>Grants</u>	<u>USD 9,600</u>	<u>In kind</u>	<u>MEDD</u>
	<u>Output 1.2.</u> Knowledge products developed and disseminated to support decision making and upscaling.	<u>USD 0.417 million</u>	<u>USD 0.393 million</u>	<u>Grants</u>	<u>USD 24,000</u>	<u>In kind</u>	<u>MEDD</u>
<u>Outcome 2.</u>	<u>Output 2.1.</u> Green-grey dune fixation infrastructure is established to control sand encroachment, enhance the provision of ecosystem services and slow the rate of desertification within the four target hubs.	<u>USD 10.572 million</u>	<u>USD 8.453 million</u>	<u>Grants</u>	<u>USD 2.119 million</u>	<u>In kind</u>	<u>MEDD</u>
	<u>Output 2.2.</u> Improved access to water for agricultural and land rehabilitation activities.	<u>USD 6.838 million</u>	<u>USD 6.685 million</u>	<u>Grants</u>	<u>USD 152,800</u>	<u>In kind</u>	<u>MEDD</u>
	<u>Output 2.3.</u> Climate-resilient agricultural livelihoods based on sustainable land- and natural resource-use are developed and/or strengthened to reduce land degradation and support climate-resilient income-generation by community members within the target regions.	<u>USD 10.664 million</u>	<u>USD 10.522 million</u>	<u>Grants</u>	<u>USD 142,000</u>	<u>In kind</u>	<u>MEDD</u>
Project management cost (USD)		<u>USD 1.466 million</u>	<u>USD 1.256 million</u>	<u>Grants</u>	<u>USD 0.21 million</u>	<u>In kind</u>	<u>MEDD</u>
Project monitoring and evaluation cost (USD)		<u>USD 1.444 million</u>	<u>USD 1.444 million</u>	<u>Grants</u>			
Indicative total cost (USD)		<u>USD 33.560 million</u>	<u>USD 30.902 million</u>		<u>USD 2.657 million</u>		

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities? Yes ☒ No ☐

C.3.2. Does GCF funding finance technology development/transfer? Yes ☒ No ☐

Technology transfer

Outcome 1 of the proposed project will create a conducive environment for climate change and gender-responsive land-use planning and natural resource management in Mauritania, facilitating technology transfer through several key strategies. First, by strengthening the institutional capacity of decision-making bodies, Outcome 1 will enable the adoption and implementation of new technologies by national- and sub-national decision-makers. Knowledge sharing and dissemination are promoted under Outcome 1, by training stakeholders to leverage the NAP Knowledge Management Platform to facilitate the exchange of information and best practices. Overall, this outcome promotes a supportive environment that encourages capacity building, knowledge sharing, financial support, collaboration and long-term sustainability across all four target hubs, enabling beneficiaries to adopt and scale up innovative technologies to address climate change challenges and improve natural resource management in Mauritania.

Outcome 2 of the proposed project will facilitate technology transfer by implementing EbA interventions to restore the provisioning of ecosystems services towards water infiltration and control of sand encroachment in the four target hubs. These interventions will include establishing ~2,123 ha of green-grey dune-stabilisation infrastructure — using ecosystem restoration and biological- and mechanical dune fixation measures — at strategic sites within twelve priority communes across the four rural hubs. The installation of this infrastructure will contribute to ~65,700 individuals in the target hubs experiencing improved protection from the increasing impacts of sand inundation. Approximately 120km of fencing will be established around dune-fixation sites, to ensure the protection of these sites and the ecosystem services accrued within them in the long-term.

The resilience of target communities to the impacts of drought will also be strengthened via the installation of physical structures such as gabions, dikes, boreholes, solar pumps, groundwater dams for improved aquifer recharge and rainwater harvesting systems. These structures will enhance the water security of more than 85,000 individuals across the target hubs by increasing groundwater storage and reducing overland water flow during heavy rainfall events, thereby limiting the amount of water lost through evaporation.

Capacity-building interventions

The engagement of the Ministry of Environment and Sustainable Development (MEDD) as the Executing Entity for the project will contribute significantly to building the government's technical, operational and project management capacity. In addition to supporting MEDD in addressing any procurement, human resource and/or financial management capacity gaps identified in the HACT micro-assessment and procurement assessment, the project interventions include specific capacity building support to strengthen MEDD's institutional, operational and financial capacity for the design, execution and oversight of the Small Grants Facility.

Capacity-building activities under Outcome 1 include strengthening the legal and institutional framework of Mauritania for sustainable natural resource management and strategic land-use planning, to build the resilience of the rural Mauritanian population against the impacts of climate change-induced drought and desertification. Overall, the project will create a strengthened enabling environment that will support the long-term sustainability and scaling of climate change adaptation measures in Mauritania. This enabling environment will include: i) improved vertical and horizontal coordination mechanisms; ii) updated policies, plans and strategies that mainstream climate change adaptation and natural resource management; and iii) strengthened knowledge management systems aimed at collecting, tailoring and disseminating best practices. This will empower relevant government institutions at the national and sub-national levels to dedicate more resources and budget to incorporating gender-responsive climate change adaptation initiatives into development plans, policies and budgets.

Capacity-building initiatives implemented under Outcome 2 largely involve training community members on how to design, implement, manage EbA dune-stabilisation infrastructure, water resources management infrastructure and CSA interventions. The establishment and training of Water User Groups, for example, will strengthen the communities' capacity to monitor water availability and usage, and to oversee and manage the construction, operation and maintenance of water infrastructure. Additionally, capacity-building under Outcome 2 includes training local community

members to uptake sustainable, alternative livelihoods in the four target hubs. The total cost of all capacity-building initiatives to be implemented under the project is USD7,200,000. This estimate includes all costs associated with Outcome 1 and all training activities and workshops implemented under Outcome 2.

Under the proposed project, ~1,800 national and sub-national decision-makers will be trained on how to design, implement and manage CCA interventions within the target regions, including members of CTCs and WUGs; ~34 individuals will benefit from strengthened capacities to use the NAP Knowledge Management Platform; and ~720 individuals will be capacitated to adopt sustainable natural resource-based livelihoods.

EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's [Initial Investment Framework](#).

D.1. Impact potential (max. 500 words, approximately 1 page)

Direct beneficiaries: defined as the ~85,229 people (40,877 males and 44,352 females) living within the twelve priority communes across the four target hubs, who will directly experience improved protection against sand inundation, enhanced access to markets and income-generating opportunities, and increased access to clean water via communal water points. The total number of direct beneficiaries represents ~1.8% of Mauritania's total population and ~7% of the population in the four target wilayahs. Please refer to Table 1 in Annex 23 for a detailed breakdown of direct beneficiaries.

Indirect beneficiaries: Those who benefit from the project outcomes indirectly through improved environmental conditions, policy changes or enhanced institutional capacities. This project estimates ~145,008 indirect beneficiaries (66,519 males and 78,489 females), representing individuals in surrounding or non-priority communes who will experience knowledge sharing benefits, improved coordination at national and regional levels and access to the small grants facility for scaling climate change adaptation interventions. This estimate represents ~3% of the country's total population and ~12% of the population in all four target wilayahs³⁷. Please refer to Table 2 in Annex 23 for a detailed breakdown of indirect beneficiaries.

Contribution to GCF Results Areas

The proposed project will contribute to three GCF climate change resilience results areas, namely: i) ARA1 — increased resilience of the most vulnerable people and communities; ii) ARA2 — increased health, well-being, food and water security; and iii) ARA4 — increased resilience of ecosystems and ecosystem services. Specific contributions to these result areas are described below.

ARA1 — Increased resilience of the most vulnerable people and communities.

Under Output 2.1, all individuals in the target hubs will gain enhanced protection from the escalating impacts of sand inundation. This will be achieved through enhanced ecosystem services generated by EbA and measures for physical dune fixation, including natural ecosystem restoration and conservation. The ecosystem services include dune fixation, improved soil stability and reduced airborne dust by establishing of natural windbreaks. EbA solutions will be complemented by installing physical structures (fencing) around dune fixation sites established under Output 2.1. Approximately 240 local community members will be given training on installing and maintaining these nature-based and physical structures, fostering resilience through ownership of these solutions.

Output 2.2 includes several activities that will increase community livelihood resilience to the impacts of climate change while reducing the anthropogenic landscape degradation, which increases ecosystems' and communities' exposure to desertification. Under this output: i) 500 individuals will benefit from training on climate-resilient and sustainable livestock-management practices; ii) 500 individuals will receive training and inputs (such as equipment and crop seeds) for climate-resilient crop agriculture; and iii) 500 individuals will be given technical support in adopting alternative, natural resource-based livelihoods.

ARA2 — Increased health, well-being, food and water security.

Under Activity 2.2.2, target communities' resilience to the impacts of drought will be enhanced through the installation of physical structures such as gabions, percolation tanks and injection wells, which increase aquifer recharge. These structures will enhance water security for ~85,000 people across the target hubs by increasing groundwater storage and reducing overland water flow during heavy rainfall events, limiting the amount of water lost through evaporation. Additionally, surface water flow and access for community members and nomadic pastoralists will be managed through

³⁷ Ministère des Affaires Economiques et du Développement (MAED) & Office National de la Statistique. 2013. Recensement Général de la Population et de l'Habitat (RGPH 2013). Available at: http://www.dgct.mr/wp-content/uploads/2016/06/Populations-2013-brochure-RGPH_Final-4-aou.pdf

the construction of weirs and dikes. Installing rainwater harvesting structures and communal cisterns will further enhance water security, directly benefitting all 85,229 individuals across the target communes^{38,39}.

ARA4 — Increased resilience of ecosystems and ecosystem services.

The proposed project will improve the resilience to climate change impacts of ~2,123 ha in the target hubs through ecosystem restoration, conservation and the sustainable management of natural ecosystems. Improved resilience will be achieved by actively restoring degraded habitats under Activity 2.1.1. The increased health of these managed ecosystems' health will, in turn, generate ecosystem services that benefit surrounding communities. These ecosystem services include, *inter alia*: i) hydrological regulation; ii) increased biodiversity; iii) protection from sand inundation; iv) enhanced soil quality; and iv) increased availability of natural resources that support livelihood diversification.

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

The interventions contribute to four paradigm-shifting pathways, as identified in the GCF sectoral guidance, aimed at enhancing the climate resilience of ecosystems, and the agricultural and water sectors. These pathways include: i) transformational planning and programming; ii) catalysing climate innovation; iii) mobilising finance at scale; and iv) establishing coalitions and enhancing knowledge to scale up successful outcomes. Please refer to Table 75 in Annex 2 for a description of each pathway.

Across the four target wilayah covered by the project, the number of communes falling outside of the rural-urban hubs is as follows: 19 communes in Hodh El Chargui (outside of Nema hub); 22 communes in Hodh El Gharbi (outside of Tamcheket hub); 7 communes in Tagant (outside of Rachid hub); and lastly, 5 communes in Adrar wilayah (outside of Aoujeft hub). Collectively, these communes, which are not covered by the project target region, house ~802,600 individuals⁴⁰. Additionally, in the future, successful project interventions could be replicated in all 8 remaining wilayah in Mauritania, which collectively house a population of ~2.7 million⁴¹.

The mechanisms used for intra-wilayah scaling are equally applicable in scaling up project interventions beyond the four target wilayah. Spatial scaling will be achieved through mainstreaming climate change and gender considerations into regional plans, policies and budgets, with support from the DREDDs. Once formally recognised by the Ministry of the Interior, CTCs can be established in all remaining communes, to further strengthen coordination and knowledge-sharing, both within and across administrative levels. Newly established CTCs will then use lessons learned and best practices drawn from the proposed project — access via the NAP Knowledge Management Platform — to implement and manage successful CCA interventions in communes not covered by the target region. Please refer to Table 51 in Annex 2 for a detailed list of lessons learned.

Spatial upscaling of ecosystem rehabilitation within the target hubs will be supported by establishing nurseries. For physical dune fixation, interventions will utilise locally sourced, sustainably collected materials at minimal costs. By training target community members to install and maintain these interventions, the project will facilitate the scaling up of these measures across other hubs and areas in the country affected by sand inundation.

Establishing livestock cooperatives will support the expansion of sustainable livestock management. Capacity building for crop farmers, creating demonstration plots and providing training on climate-resilient seed varietal replication will help disseminate practices introduced by the project across the target hubs. Improved market access for goods produced through these livelihood interventions will further incentivise replication of these practices among adjacent communities, leveraging the benefits and traditional communication systems. Community WUGs will be capacitated by raising awareness about water supply and demand management practices in the target hubs. Such innovations — particularly water demand management — are cost-effective and easy to replicate, encouraging their uptake by neighbouring communities.

³⁸Based on the assumption that a single 5000L RWH system and cistern services 4-6 occupants (average of 5 occupants), with occasional irrigation of small-scale gardens.

³⁹RainWater Harvesting UK. 2021. Rainwater Harvesting: Frequently Asked Questions. Available at: <https://www.rainwaterharvesting.co.uk/wp-content/uploads/2021/11/rainwater-harvesting-information.pdf>

⁴⁰Ministère des Affaires Economiques et du Développement (MAED) & Office National de la Statistique. 2013. Recensement Général de la Population et de l'Habitat (RGPH 2013). Available at: http://www.dgct.mr/wp-content/uploads/2016/06/Populations-2013-brochure-RGPH_Final-4-aou.pdf

⁴¹Ministère des Affaires Economiques et du Développement (MAED) & Office National de la Statistique. 2013. Recensement Général de la Population et de l'Habitat (RGPH 2013). Available at: http://www.dgct.mr/wp-content/uploads/2016/06/Populations-2013-brochure-RGPH_Final-4-aou.pdf

Creating an enabling environment through transformational planning and programming

An enabling environment for mainstreaming CCA and natural resource management is essential at all institutional levels to achieve the above upscaling and replication strategies. Accordingly, the proposed project will contribute to the GCF's Outcome-level Core Indicator 5: *Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low-emission, climate-resilient development pathways in a country-driven manner*. Output 1.1 of the proposed project will contribute to Core Indicator 5 via the establishment of commune-level technical committees (CTCs). The establishment of WUGs under Output 2.2 will strengthen institutional and regulatory frameworks for monitoring water use and availability within the target region, while training FBOs and WFS on climate-smart agriculture will support low-emission, climate-resilient development within the agricultural sector. The establishment of a Small Grants Facility and associated on-granting mechanism under Output 2.3 of the proposed project will ensure that systems are put in place to regulate the flow of funds from sustainable livelihood activities back into CCA beyond the project life cycle.

Potential for scaling up successful climate change adaptation through knowledge sharing

The proposed project will contribute to the GCF's Outcome-level Core Indicator 8: *Degree to which GCF investments contribute to effective knowledge generation and learning processes and use of good practices, methodologies and standards*. Numerous studies and assessments will be conducted within the target hubs to improve the country's climate change knowledge base, including value chain assessments, surface and groundwater quality and quantity, and community vulnerability assessments (disaggregated by gender). All the assessments will contribute to a CCA knowledge management platform, which will be used as a centralised repository accessible to stakeholders across different administrative levels. Best practices on CCA will be collated, tailored and incorporated into knowledge products, informing the design and implementation of subsequent adaptation interventions. The platform will: i) integrate findings from value chain, water resource and vulnerability assessments to create gender-responsive, location-specific adaptation guidance; ii) document successful interventions from the target hubs that can be replicated in other wilayah; and iii) feature case studies of successful CCA initiatives from the target hubs, including lessons learned and implementation challenges. This approach to knowledge management will support both vertical scaling (enhancing the impact within the target hubs) and horizontal scaling (supporting CCA in other hubs, countries or regions) of successful adaptation measures. These knowledge products will be integrated into the NAP Knowledge Management Platform, which manages knowledge related to desertification and climate change in Mauritania.

Financial and Economic modelling carried out for the economic and financial assessment indicates that the adoption of EbA practises, coupled with dune fixation activities, in each of the hubs could potentially increase crop yields by up to 30%⁴² with variations in each hub. Associated increased fodder yields will consequently improve carcass mass of herd animals by between 11% and 48%, yielding more meat for household consumption⁴³. Rainwater harvesting and auxiliary infrastructure can also enhance the availability of water to close the gap of unmet water demand in the hubs. Overall, the economic and financial model yields a benefit to cost ratio of 5.86 to 1. It is also important to note that scaling of the proposed project is complicated to the spatial arrangements of oasis-based agriculture. There is however scope for replicability of the project in regions with a similar socio-economic, climatic and ecological conditions. Replicability could be executed through extension of the proposed grant funding mechanism (Activity 2.3.2), beyond the GCF funding cycle.

D.3. Sustainable development (max. 500 words, approximately 1 page)

The project will contribute towards achieving 11 of the 17 Sustainable Development Goals (SDGs). These goals are: SDG 2 — Zero hunger; SDG 3 — Good health and well-being; SDG 5 — Gender equality; SDG 6 — Clean water and sanitation; SDG 8 — Decent work and economic growth; SDG 10 — Reducing inequality; SDG 11 — Sustainable cities and communities; SDG 12 — Responsible consumption and production; SDG 13 — Climate action; SDG 15 — Life on land; and SDG 16 — Peace, justice and strong institutions. In addition, project interventions will produce numerous environmental, social and economic co-benefits through a gender-sensitive and -responsive development approach, as described below.

The proposed project will provide numerous environmental, social and economic co-benefits through the: i) rehabilitation and conservation of natural ecosystems; ii) introduction of physical structures to reduce sand inundation; and iii) promotion of climate-resilient livelihoods. The co-benefits are presented below in Table 23.

⁴² Based on evidence from: WB. 2014. The West Africa Agricultural Productivity Program: a major boost for agriculture in Mali. WB: Washington, DC. Available at: <http://www.worldbank.org>

⁴³ Based on evidence from Akinlade, J.A., Smith, J.W., Larbi, A., Akinsoyinu, A.O., and Adekunle, I.O. (2011) 'Impact of cattle grazing and fallow systems on maize grain yield, soil chemical properties and daily weight gain in the derived savanna zone of Nigeria', The International Journal of Organic Agriculture Research and Development, 3. Available at: <https://www.ijoardjournal.org/index.php/ijoardjournal/article/view/189>

Table 4. Cross-cutting benefit streams

Benefit stream	Co-benefits		
	Environmental	Social	Economic
Sustainable economic growth and development	- Reduced exploitation of limited natural resources resulting from improved access to sustainable, alternative livelihoods - Reduced land degradation	- Reduced conflicts over water resources - Improved opportunity to resume traditional nomadic lifestyles - Reduced risk of GBV and freeing up women's time, as women will no longer have to travel far from their households to collect water - Improved school attendance for girls, who frequently leave school to fulfil the task of collecting water	Enhanced labour productivity
Enhanced biodiversity	Enhanced ecosystem health and resilience	Continuation of cultural and aesthetic value associated with the biodiversity of ecosystems	Benefits associated with the prevention of ecosystem degradation such as carbon sequestration and water regulation
Strengthened flood attenuation	Increased soil stabilisation leads to reduced soil erosion	- Reduced number of direct deaths - Reduced number of epidemics linked to waterborne diseases - Improved drinking water quality	- Reduced losses from damage to property, infrastructure and crops - Increased lifespan and capacity of water infrastructure such as weirs, gabions or wells
Improved community health	N/A	- Reduced prevalence of respiratory diseases such as lung cancer or respiratory infection resulting from exposure to airborne dust - Fewer respiratory diseases contribute to longer lifespans and improved quality of life - Improved access to clean water and sanitation	- Reduced pressure on medical services - Positive effects on household economy as a result of reduced healthcare costs
Strengthened food security	Reduced pressure on natural resources and ecosystems to meet food provision demands	- Improved nutritional health. - Reduced dependence on food aid	- Reduced reliance on external imports or food aid - Reduced pressure on medical services - Positive effects on household economy as a result of reduced healthcare costs

D.4. Needs of recipient (max. 500 words, approximately 1 page)

Mauritania is classified as a least-developed country, with a gross domestic product (GDP) of ~US\$7.9 billion and a GDP per capita of ~US\$1,700⁴⁴. As a result of high poverty rates, particularly within rural areas, the population is heavily dependent on agriculture for subsistence; therefore, most households' capacities to adopt alternative livelihoods are limited. The country's Human Capital Index (HCI) score of 38, which ranks it 150th out of 157 countries, indicates that a child born in Mauritania can only expect to attain 38% of human capital by the age of 18, even if they

⁴⁴ World Bank. 2020. Country Profile – Mauritania. Available at: https://databank.worldbank.org/views/reports/reportwidget.aspx?Report_Name=CountryProfile&Id=b450fd57&tbar=y&dd=y&inf=n&zm=n&country=MRT

receive complete health and education⁴⁵. Moreover, despite progress made over the past three decades, Mauritania's HDI value places it 157th out of 189 countries (2019), categorising it as having 'low human development'⁴⁶.

In 2019, Mauritania was ranked the 20th country most vulnerable to climate change and other global challenges, according to the Notre Dame Global Adaptation Initiative (ND-GAIN) country index. With the highest global sensitivity score (0.602), Mauritania faces substantial insecurity in both the food and water sectors⁴⁷. Out of 182 countries on the ND-GAIN readiness index, Mauritania was ranked as the 75th least ready to respond to climate change and economic shocks, with education and control of corruption being its worst-scoring readiness indicators⁴⁸.

Between 1900 and 2018, natural hazards such as droughts, disease epidemics, insect infestations and storms (strong winds and heavy rainfall) impacted more than 11.5 million people, according to the Emergency Events database (EM-Dat). Between 1990 and 2020, an average of three storms, six epidemics, eight droughts, and 19 floods have been recorded across Mauritania each year. In 2019, ~33,600 Mauritians were impacted by floods, while in 2020, approximately 609,000 people were impacted by droughts⁴⁹. Flooding alone caused an estimated US\$5.5–11.4 billion in damages and losses between 2000 and 2008⁵⁰. The annual cost of natural capital⁵¹ degradation due to climate change and unsustainable anthropogenic activities has been estimated at ~5.4% of GDP. Under future climate change scenarios, Mauritania faces substantial challenges such as: i) an overall decrease in water availability of 10–15%; ii) an increase in land degradation and erosion; iii) the southward expansion of the Sahara Desert, resulting in a decrease in arable land; iv) a decrease in rangeland availability for livestock; and v) a reduction in food availability⁵². Consequently, the impacts of climate change on food and water security, economic growth and human health are projected to intensify in the country.

The country imports 70–85% of its food requirements each year (accounting for 21% of GDP), despite agriculture being the third largest economic sector. The cumulative impact of these factors inhibits the capacity of national and local decision-makers to implement long-term strategies for responding to the adverse impacts of climate change on the country's population.

The country's high public and external debt — estimated at 79% of GDP and 68% of GDP, respectively — constrain the government's ability to allocate funding from the national budget for CCA. Moreover, limited opportunities exist for public and private sector stakeholders to access financial resources for CCA, which is expected to cost ~USD\$9.3 billion⁵³ between 2015 and 2030⁵⁴. Given that this cost is likely to increase considerably under future climate change conditions, external investment is crucial to enhance the climate resilience of Mauritania's population.

D.5. Country ownership (max. 500 words, approximately 1 page)

The proposed project is well aligned with Mauritania's national adaptation priorities; it was designed in collaboration with project stakeholders at all levels.

Mauritania's First Nationally Determined Contribution (NDC), updated in 2021, identifies 19 objectives to be achieved by 2030. The project directly addresses five of these climate ambitions, namely: i) restoring natural grasslands; ii) conducting hydrological assessments and water resources availability analyses; iii) strengthening ecosystems' resilience against climate change; iv) enhancing vulnerable populations' resilience to climate change, particularly in

⁴⁵ World Bank Group. 2020. Human Capital Index 2020 — Mauritania. Available at: https://databank.worldbank.org/data/download/hci/HCI_2pager_MRT.pdf.

⁴⁶ UNDP. 2020. Human development reports — Mauritania. Available at: <https://hdr.undp.org/en/countries/profiles/MRT#>.

⁴⁷ University of Notre Dame Global Adaptation Initiative. 2022. Country index — Mauritania. Available at: <https://gain.nd.edu/our-work/country-index/rankings/>.

⁴⁸ University of Notre Dame Global Adaptation Initiative. 2022. Country index — Mauritania. Available at: <https://gain.nd.edu/our-work/country-index/rankings/>.

⁴⁹ World Bank Group. 2022. Climate change knowledge portal — Mauritania. Available at: <https://climateknowledgeportal.worldbank.org/country/mauritania>.

⁵⁰ World Bank Group. 2018. Country partnership framework for the Islamic Republic of Mauritania for the period FY18–FY23. Available at: <https://documents1.worldbank.org/curated/en/288231531625439579/pdf/MAURITANIA-CPF-NEW-06192018.pdf>

⁵¹ Natural capital is another term for the stock of renewable and non-renewable resources (e.g. plants, animals, air, water, soils, minerals) that combine to yield a flow of benefits to people

⁵² Islamic Republic of Mauritania. 2014. Third National Communication to the UNFCCC.

⁵³ Habitat, urbanism and land management would require USD 5,000 m; fisheries, USD 1,644 m; water and sanitation, USD 1,500 m; agriculture, USD 843 m; health, USD 221 m; ecosystem conservation, USD 133 m; and livestock, USD 36.4 m.

⁵⁴ MEDD. 2021. Contribution Déterminée au Niveau National (CDN): Mauritania. Available at: https://unfccc.int/sites/default/files/NDC/2022-06/CDN-actualis%C3%A9%202021_%20Mauritania.pdf

rural areas; and v) strengthening institutional and technical capacities at local, regional and national levels to enable more effective planning, financing and implementation of adaptation measures⁵⁵.

The project's objectives are consistent with the country's National Adaptation Programme of Action (NAPA), developed in 2004, which encourages a cross-sectoral approach to achieving 27 Priority Adaptation Measures (PAMs) for adaptation to climate change. 19 of the 27 policies and measures are expected to be addressed: eight under Output 2.1 (EbA interventions and physical measures for reducing desertification) and a further 11 under Output 2.2 (sustainable land and natural resources management and improved livelihoods). The most important PAMs to which the project will directly contribute include: i) enhancing knowledge of surface water's hydrological cycle; ii) constructing flooding breakdown dikes in pluvial and oasis zones; iii) promoting water-saving techniques in oasis zones; iv) participatory reforestation for agroforestry in agricultural zones; and v) re-organising and capacity-building in communities adversely affected by climate change⁵⁶.

The project is complementary to the country's ongoing GCF-supported NAP development process, initiated in 2018. The project will build on the following NAP activities: i) enhancing the capacity of policymakers and staff from relevant government institutions (Activity 1.1.1); ii) enhancing the capacity of vulnerable groups, in collaboration with key ministries (Activity 1.1.1); and iii) developing and implementing at least one awareness-raising campaign targeting local authorities in each wilayah, civil society organisations and local communities.

The proposed project aligns with Mauritania's national spatial planning strategy, which aims to strengthen resilience using an oasis-centred administrative structure. This approach aims to prevent rural exodus and enhance resilience to the impacts of climate change-induced desertification.

Stakeholder engagement

The proposed project has been designed following extensive consultations with Mauritanian government representatives, national consultants, public authorities, civil societies, local community members and the private sector. Mauritania's NDA, the Ministry of the Environment and Sustainable Development (MEDD), has been actively involved in the project's development since inception by contributing to the consultation process and providing inputs during field missions. Additionally, as the executing entity, UNEP has facilitated dialogue between national consultants, the MEDD, other relevant institutions and community stakeholders, ensuring that the project responds to the adaptation priorities of the country. Moreover, the proposed adaptation interventions under this project have been validated by all relevant project stakeholders, including the NDA.

Community consultations were organised by UNEP within each of the proposed project's target hubs, namely Aoujeft, Rachid, Tamcheket and Nema. Specific details regarding, inter alia, the dates of consultations, types of stakeholders in attendance and number of community members consulted are provided in Annex 7: Summary of Consultations and Stakeholder Engagement Plan. Four field missions were conducted over a two-year period, during which more than 330 local community members were consulted across the target hubs.

D.6. Efficiency and effectiveness (max` . 500 words, approximately 1 page)

Cost-effectiveness of project interventions

A financial and economic analysis (FEA) was conducted to inform this decision-making⁵⁷. This analysis determined the costs and benefits associated with: i) implementing EbA interventions, including dune fixation and land rehabilitation activities, to enhance communities' resilience to climate change-induced desertification; ii) facilitating sustainable natural resource management practices and the adoption of sustainable, diversified livelihoods by developing value chains; and iii) supporting the GoM in developing local, regional and national frameworks for mainstreaming climate change and gender-responsive adaptation solutions into plans and policies.

⁵⁵ MEDD. 2021. Contribution Determinee au Niveau National (CDN): Mauritania. Available at: https://unfccc.int/sites/default/files/NDC/2022-06/CDN-actualis%C3%A9%202021_%20Mauritania.pdf

⁵⁶ Islamic Republic of Mauritania. 2004. The National Adaptation Programme of Action. Available at: <https://unfccc.int/resource/docs/napa/mau01e.pdf>

⁵⁷ The financial and economic analysis undertaken to support the proposed project is presented in Annex 3: Financial and Economic Analysis. [under development]

The economic analysis estimated the net present value⁵⁸ (NPV) of the net benefits (economic benefits less project costs) to be ~USD152 million (discounted at social discount rate of 2%⁵⁹ over 20 years), with a benefit-to-cost ratio of ~5.8 (USD5.86 generated per USD1 spent), reflecting a cost-effective investment for the GCF and GoM.

Project benefits included in the economic analysis are:

- Food crops sustained for household consumption;
- Water security sustained for household consumption;
- Fruit and beverage resources sustained for household consumption;
- Livestock grazing forage sustained for household consumption;
- Agricultural income generation sustained for trading households;
- Natural products income generation sustained for trading households; and
- Income generation opportunities created through new goods and services for trading households.

Health benefits

The NPV of health benefits sustained under the proposed GCF project is estimated at USD24.3 million. This estimate was determined by conducting a literature review to reveal the incremental healthcare cost for treating malnourished patients. The healthcare cost per capita in Mauritania was estimated using available data and the incremental cost was applied. The proposition behind this is that by implementing the GCF interventions, the livelihoods of the population in the target hubs improve, resulting in less malnourishment in hub communities. This leads to substantial cost savings, as literature suggests it is more expensive to treat patients who are/were malnourished. Further details are available in Annex 3: Financial and Economic Appraisal.

Rural livelihood benefits

A financial analysis of income potential sustained by the adoption of climate-smart agriculture, climate-resilient livestock management and alternative, natural resource-based livelihoods was carried out using best available data and research. Table 5 below provides a summary of the financial and economic results.

Table 5. Summary of the financial and economic results⁶⁰

Indicator	Unit	Total/Combined	Aoujeft	Rachid	Tamcheket	Nema
Baseline scenario (without GCF support)						
Revenue potential: Crop production - Real terms	\$/annum	17,036,661	4,327,591	3,428,625	1,875,570	7,404,875
Revenue potential: Livestock production - Real terms	\$/annum	53,166,317	2,037,407	20,324,221	11,028,480	19,776,209
Revenue potential: Water supply - Real terms	\$/annum	76,992	4,614	17,401	14,609	40,368
Financial IRR	%	12%				
Project scenario (with GCF support)						
Revenue potential: Crop production - Real terms	\$/annum	22,147,660	5,625,869	4,457,212	2,438,241	9,626,338
Revenue potential: Livestock production - Real terms	\$/annum	59,014,612	2,261,522	22,559,885	12,241,613	21,951,592
Revenue potential: Water supply - Real terms	\$/annum	151,959	9,107	34,344	28,833	79,675
NPV of Additional GDP Created: Crop production	\$	59,481,666	15,109,319	11,970,674	6,548,349	25,853,324
NPV of Additional GDP Created: Livestock production	\$	68,062,303	2,608,242	26,018,603	14,118,407	25,317,050
NPV of Additional GDP Created: Water supply	\$	872,455	52,286	197,184	165,542	457,444
NPV of Multiplier effect	\$	29,933,811	4,142,143	8,901,247	4,855,999	12,034,421

⁵⁸ Net present value (NPV) is the difference between the present value of cash inflows and the present value of cash outflows over a specified period of time. NPV is used in capital budgeting and investment planning to analyse the profitability of a projected investment or project.

⁵⁹ The use of the 2% social discount rate is based on economists preference revealed in research available at: <https://www.lse.ac.uk/granthaminstitute/explainers/what-are-social-discount-rates/>

⁶⁰ Annex 3: Financial and Economic Appraisal

NPV of Health Benefit	\$	24,337,247	1,458,515	5,500,464	4,617,809	12,760,459
NPV of Total Economic & Ecosystem Service Benefit	\$	182,687,481	23,370,505	52,588,172	30,306,107	76,422,699
NPV of Project Costs	\$	31,161,479				
Financial IRR	%	22%				
ERR	%	29%				
Benefit-to-cost Ratio	%	5.86				

This requested GCF grant investment will be supported by in-kind co-financing of US\$2.657 million from the MEDD and regional authorities in the target hubs. Co-finance from the MEDD and regional authorities will be used to cover baseline costs, including operations and maintenance (which may be revised during project implementation) necessary to ensure sustainable land and water resources management. With the support of GCF grant funding, the proposed project will deliver several adaptation benefits using an integrated approach to EbA and sustainable natural resources management, contributing to a paradigm shift in the country's approach to addressing the impacts of climate change-induced desertification⁶¹.

Beyond the direct co-financing, the proposed project is strategically positioned to benefit from parallel financing totalling ~US\$43.8 million, drawn from complementary initiatives within the target regions. These include the Integrated Management of Protected Areas in the Arid Regions of Mauritania (IMPADRA) with around US\$20 million, the Sustainable Management of Natural Resources project (PROGRES) with an estimated US\$6.6 million, the Africa Integrated Climate Risk Management Programme with approximately US\$14.7 million, and a climate change adaptation project with around US\$6 million in funding⁶². The combined investment of ~US\$63.8 million will be used efficiently and effectively to achieve two main project objectives. First, local, regional and national institutions will gain improved access to climate-risk information for strategic land-use planning and natural resource management that will, in turn, facilitate continued investment in ecosystem restoration, EbA, diversified livelihoods and sustainable natural resources management. Second, communities in rural-urban hubs along the Sahara-Sahel boundary will have sustainable access to food and water resources, reducing pressure from the impacts of climate change-induced desertification.

⁶¹ A detailed description of project activities is presented in Sections B.3 and E.6 of the Funding Proposal.

⁶² Please refer to Table 70 in Annex 2 for a detailed list of parallel finance opportunities.

LOGICAL FRAMEWORK

E.1. Project/Programme Focus

- ☐ Reduced emissions (mitigation)
- ☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	There has been some progress in addressing climate change adaptation (CCA) and natural resource management in Mauritania through national and local climate change committees, plans and strategies. National cross-sectoral coordination initiatives implemented towards improving climate change adaptation include the establishment of Regional Delegations for Environment and Sustainable Development (DREDDs) in each wilayah, the National Council for the Environment and Sustainable Development (CNEDD) and corresponding Regional Councils for Environment and Sustainable Development (CREDDs) in each wilayah, and focal points for climate change designated within most ministries within the GoM. Locally, coordination has been improved through several donor-funded initiatives that have led to the strengthened capacity of Associations for Natural Resources (AGLCs) to implement EbA as well as the establishment of Climate Action Plans. While coordination has improved through initiatives like DREDDs and AGLCs, clearer mandates are needed for both state and non-state actors to integrate CCA into sub-national and local legislation and policy. The use of knowledge on climate change impacts and adaptation	<u>Medium</u>	The project aims to achieve the following outcomes across the four target wilayah, which include 43 communes outside rural-urban hubs and ~802,600 residents: - Enhanced capacity and integration — clearer mandates for both state and non-state actors regarding integrating CCA into sub-national and local legislation, policy and improved adaptation funding. This target addresses the baseline's coordination gaps by providing specific responsibilities and frameworks for integrating CCA. - Improved incorporation — incorporation and mainstreaming of climate change adaptation and natural resource	Community and decision-makers participation will be secured through the CTCs, WUGs, commune-level livestock cooperatives and other community level groups. Once formally recognised by the Ministry of the Interior, CTCs can be established in all remaining communes, to strengthen coordination and knowledge-sharing — both within and across administrative levels — across Mauritania. Newly established CTCs will leverage the NAP Knowledge Management Platform to access and apply lessons learned and best practices from the proposed project. Through this platform, CTCs will be able to draw upon documentation of successful interventions, practical implementation guides and monitoring frameworks that have been tested in the initial target hubs. The platform will be particularly valuable for sharing region-specific approaches to climate change adaptation, allowing CTCs to streamline proven methodologies to their local interventions while avoiding implementation challenges identified by the proposed project, resulting in more efficient resource allocation. This will enable interventions to be upscaled in the target hubs, as well as implemented in the 43 non-target communes. Community members will be trained to install and maintain dune fixation interventions, thereby facilitating the scaling up of these measures across other communes in the hubs.

	responses is limited, creating a gap between research and policymaking. Additionally, the technical and financial capacity of relevant government departments to mainstream climate change considerations into sectoral policies and programmes is still limited. This insufficient capacity is partly a result of limited available information on climate change as well as limited financial resources. As a result, the potential for adaptation options to be upscaled beyond isolated, primarily donor-funded projects is currently limited.		management into regional and local budgets, strategies and plans. • Strengthened knowledge integration — enhanced use of decision-makers' knowledge on climate change impacts and adaptation responses, bridging the current gap between research and policymaking. • Improved coordination — regional and local coordination mechanisms for implementing climate change adaptation. • Increased financial access — greater access to financing to upscale EbA and other climate change adaptation activities. The project will facilitate access to funding mechanisms, enabling the scaling up of adaptation initiatives.	The upscaling of climate-resilient agricultural practices will be accomplished within hubs through traditional information-sharing systems. Spatial upscaling of ecosystem rehabilitation within the target hubs will be supported by establishing nurseries and generating financial incentives from sustainable, agricultural-based livelihoods and payment for ecosystem services. Establish an on-granting mechanism to provide financial stability and facilitate continued investment in upscaling successful EbA activities and sustainable livelihoods. Transfer of the model to the MEDD. Build its capacity in fund management. Skillsets of decision-makers will be developed through: i) retaining training materials used in project activities in government institutions after the project termination; ii) interviewing and retaining knowledge from trainers used in project training activities; and iii) incorporating CCA and sustainable natural resource management practices into institutional standard operating procedures. These steps will capacitate decision-makers to streamline project EbA interventions into policy. ~1,800 national and sub-national decision-makers will be trained on how to design, implement and manage CCA interventions within the target regions, including members of CTCs and WUGs; ~34 individuals will benefit from strengthened capacities to use the NAP Knowledge Management Platform; and ~720 individuals will be capacitated to adopt sustainable natural resource-based livelihoods. Establishing livestock cooperatives will support the expansion of sustainable livestock management.
Replicability	Mauritania is facing numerous challenges in adapting to climate change: • Financial constraints — Adaptation investments totalling USD9.4 billion are required between 2015 and 2030, but high public debt (99% of GDP in 2016) limits public financing for CCA. • Policy and institutional gaps — Despite initiating steps towards climate resilience, gaps remain in policy and institutional frameworks at all levels, with	<u>Low</u>	The project's benefits are potentially extensible to the remaining eight wilayah in Mauritania, which house ~2.7 million people. Replicability depends on the below. • Scalable governance framework: an adaptable	Mainstreaming adaptation into regional plans, policies and budgets. Once formally recognised by the Ministry of the Interior, CTCs can be established in all remaining communes, to further strengthen coordination and knowledge-sharing, both within and across administrative levels. Newly established CTCs will leverage the NAP Knowledge Management Platform to access

	limited plans for water, pasture, and forest management and insufficient mainstreaming of CCA in urban plans for the four target hubs. • Knowledge and information deficits — Limited assessments on climate change impacts and vulnerabilities are available, particularly within the target hubs, leading to a notable gap between research and policymaking on CCA.		governance framework that defines roles and responsibilities for integrating CCA into sub-national legislation, policy, and funding mechanisms. This framework will be designed to be adjusted to suit the specific contexts of different regions, ensuring consistent CCA implementation. • Knowledge mobilisation system: a system for gathering, analysing and disseminating climate information to support evidence-based decision-making. This system will enable efficient knowledge transfer and adaptive management across diverse regions, facilitating informed adaptation planning and implementation. • Integrated planning and budgeting approach: a standardised approach for incorporating CCA into regional and local planning and budgeting processes. This approach will ensure that climate considerations are systematically integrated into development plans and resource allocation across all wilayah.	and apply lessons learned and best practices from the proposed project. The platform will be valuable for sharing region-specific approaches to climate change adaptation, allowing new projects in the Sahel to adapt proven methodologies to their local contexts while avoiding implementation challenges identified by the proposed project.
--	---	--	--	---

			- Sustainable financing mechanisms: partnerships with financial institutions and the development of innovative financing mechanisms to support the upscaling of EbA and other CCA activities. This will ensure long-term financial sustainability and enable broader adoption of effective adaptation measures. - The project's success will be measured through a monitoring and evaluation framework that captures lessons learned and informs adaptive management strategies, ensuring its long-term sustainability and impact during replication.	
Sustainability	Nationally, public investment in CCA includes the Presidential Priority Program (PPP). This investment, however, is currently limited to less than 5% of the country's annual GDP. At the wilayah-level regional funds for sustainable development also exist, but these vary considerably from year to year, making long-term planning for CCA investment difficult. Moreover, there are limited guidelines and policies dictating what percentage of this investment should be channelled towards CCA. Private sector investment has somewhat alleviated gaps in public CCA investment. This investment includes PES schemes introduced to some areas by the World Bank and IFAD. These PES schemes, however, are limited in their physical extents.	<u>Low</u>	The project will create a strengthened enabling environment that will support the long-term sustainability and scaling of CCA measures in Mauritania. This enabling environment will include: improved vertical (across national, regional and local levels) and horizontal (between government departments at the hub level) coordination mechanisms between government institutions at the local, regional and national levels through	Mainstreaming adaptation into regional plans, policies and budgets and the formal recognition of CTCs established under the project will make it more likely that community participatory processes will be supported and continued. The continued operation and maintenance of the proposed project's EbA interventions will be promoted by the implementation of the on-granting mechanism. Skills development and retention by: i) institutionalising training programmes; ii) embedding CCA and sustainable natural resource management practices into standard operating procedures; and iii) fostering a culture of continuous learning and adaptation within participating organisations. ~1,800 national and sub-national decision-makers will be trained on how to design, implement and manage CCA interventions within the target regions, including members of CTCs and WUGs; ~34 individuals will benefit from

	Insufficient capacity among local authorities and community members to implement, operate and maintain introduced practices.		commune-level technical committees (CTCs); • updated policies, plans and strategies that mainstream CCA and natural resource management in a gender responsive manner; and • strengthened knowledge management systems aimed at collecting, tailoring and disseminating CCA best practices.	strengthened capacities to use the NAP Knowledge Management Platform; and ~720 individuals will be capacitated to adopt sustainable natural resource-based livelihoods. Leveraging lessons from past investments in Mauritania's oasis ecosystems, particularly the IFAD-PDDO project (2005-2014), the project will use the PDDO's integrated approach to land rehabilitation — linking water availability with sustainable land management. Additional best practices, including <i>inter alia</i> infiltration weirs, localised irrigation, crop diversification and promotion of collective action, will be incorporated into the project design. To ensure long-term adoption, extension services and capacity-building programmes will be provided. The project will also address the challenges of incentivising new technologies by considering local needs and promoting beneficial, locally-aligned practices. Community involvement will be prioritised through participatory design, implementation and management. Continuous monitoring and evaluation will inform adaptive management and ensure project outcomes. By incorporating these lessons and best practices, this project aims to create a resilient and sustainable oasis ecosystem that provides long-term benefits for local communities.
--	--	--	---	---

E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)

GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note ⁶³
				Mid-term	Final ⁶⁴	
<u>ARA1 MOST VULNERABLE</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>		0 people with increased resilience to climate change impacts relating to droughts and sand inundation	42,615 people (22,176 women, 20,439 men) benefitting from increased resilience to climate change impacts relating to droughts and sand inundation. This includes:	85,229 people (44,352 women, 40,877 men) benefitting from increased resilience to climate change impacts relating to droughts and sand inundation. This includes:	<u>Assumptions</u> Capacity building, technologies and practices introduced are continued beyond the project lifespan and are

⁶³ Beneficiary numbers are not additive across indicators, as the same individuals may be counted under multiple indicators.

⁶⁴ The final target means the target at the end of project/programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective.

<u>PEOPLE AND COMMUNITIES</u>		Baseline and annual reports informed by: i) household surveys and questionnaires; and ii) field surveys to confirm beneficiary numbers. Non-project generated sources include government surveys.		• 7,126 (3,532 women, 3,594 men) in Aoujeft • 13,941 (7,368 women, 6,573 men) in Rachid • 7,011 (3,793 women, 3,219 men) in Tamcheket and • 14,537 (7,484 women, 7,054 men) in Nema.	• 14,252 (7,064 women, 7,188 men) in Aoujeft • 27,881 (14,736 women, 13,145 men) in Rachid • 14,022 (7,585 women, 6,437 men) in Tamcheket and • 29,074 (14,967 women, 29,074 men) in Nema.	effective at delivering tangible adaptation benefits.
	<u>Supplementary 2.1: Beneficiaries (female/male) adopting improved and/or new climate-resilient livelihood options</u>		0 people with increased resilience to climate change impacts relating to droughts and sand inundation	42,615 people (22,176 women, 20,439 men) benefitting from increased resilience to climate change impacts relating to droughts and sand inundation. This includes: • 7,126 (3,532 women, 3,594 men) in Aoujeft • 13,941 (7,368 women, 6,573 men) in Rachid • 7,011 (3,793 women, 3,219 men) in Tamcheket and • 14,537 (7,484 women, 7,054 men) in Nema.	85,229 people (44,352 women, 40,877 men) benefitting from increased resilience to climate change impacts relating to droughts and sand inundation. This includes: • 14,252 (7,064 women, 7,188 men) in Aoujeft • 27,881 (14,736 women, 13,145 men) in Rachid • 14,022 (7,585 women, 6,437 men) in Tamcheket and • 29,074 (14,967 women, 29,074 men) in Nema.	Livelihood options are successful and generate income. Suitable markets exist that are accessible to target communities.
	<u>Supplementary 2.5: Beneficiaries (female/male) adopting innovations that strengthen climate change resilience</u>	Baseline and annual reports informed by: i) household surveys and questionnaires; and ii) field surveys to confirm number of beneficiaries with observable increased resilience to sand inundation.	0 people with increased resilience to climate change impacts relating to droughts and sand inundation	42,615 people (22,176 women, 20,439 men) benefitting from increased resilience to climate change impacts relating to droughts and sand inundation. This includes: • 7,126 (3,532 women, 3,594 men) in Aoujeft • 13,941 (7,368 women, 6,573 men) in Rachid • 7,011 (3,793 women, 3,219 men) in Tamcheket and • 14,537 (7,484 women, 7,054 men) in Nema.	85,229 people (44,352 women, 40,877 men) benefitting from increased resilience to climate change impacts relating to droughts and sand inundation. This includes: • 14,252 (7,064 women, 7,188 men) in Aoujeft • 27,881 (14,736 women, 13,145 men) in Rachid • 14,022 (7,585 women, 6,437 men) in Tamcheket and • 29,074 (14,967 women, 29,074 men) in Nema.	Dune fixation measures are successful against sand encroachment. And are sufficiently maintained beyond the project's lifespan.
<u>ARA2 Health, well-being, food</u>	<u>Supplementary 2.3: Beneficiaries (female/male)</u>	Baseline and annual reports informed by: i) household surveys and questionnaires; and ii) field surveys to confirm	0 people with increased resilience to climate change impacts relating to	42,615 people (22,176 women, 20,439 men) benefitting from increased resilience to climate change impacts relating to	85,229 people (44,352 women, 40,877 men) benefitting from increased resilience to climate change impacts relating to	Water resource attenuation interventions are successful.

<u>and water security</u>	<u>with more climate-resilient water security</u>	improved access by beneficiaries to reliable and safe water supply. Non-project generated sources include government health and population surveys.	droughts and sand inundation	droughts and sand inundation. This includes: • 7,126 (3,532 women, 3,594 men) in Aoujeft • 13,941 (7,368 women, 6,573 men) in Rachid • 7,011 (3,793 women, 3,219 men) in Tamcheket and • 14,537 (7,484 women, 7,054 men) in Nema.	droughts and sand inundation. This includes: • 14,252 (7,064 women, 7,188 men) in Aoujeft • 27,881 (14,736 women, 13,145 men) in Rachid • 14,022 (7,585 women, 6,437 men) in Tamcheket and • 29,074 (14,967 women, 29,074 men) in Nema.	
<u>ARA4 Ecosystems and ecosystem services</u>	<u>Core 4: Hectares of natural resources brought under improved low-emission and/or climate-resilient management practice</u>	Field surveys and GIS mapping of changes in land cover. Project records on restoration, conservation and sustainable landscape management agreements.	0 hectares restored, conserved or managed.	1,087 ha of natural ecosystem within the four target hubs protected and strengthened in response to climate change and variability (~724 ha per year from year 2 of implementation).	2,123 ha of natural ecosystem within the four target hubs protected and strengthened in response to climate change and variability (~724 ha per year from year 2 of implementation).	Restoration and sustainable management interventions are successful and supported by local stakeholders through the reduction of non-climatic drivers of ecosystem degradation.
	<u>Supplementary 4.1: Hectares of terrestrial forest, terrestrial non-forest, freshwater and coastal marine areas brought under restoration and/or improved ecosystems</u>	Field surveys and GIS mapping of changes in land cover. Project records on restoration and conservation agreements.	0 hectares rehabilitated, restored or conserved.	758 ha of natural ecosystem within the four target hubs restored or under community-based conservation (~505 ha per year from year 2 of implementation). This includes: • 216 ha in Aoujeft • 197 ha in Rachid • 222 ha in Tamcheket • 123 ha in Nema	2,275 ha of natural ecosystem within the four target hubs restored or under community-based conservation (~505 ha per year from year 2 of implementation). This includes: • 649 ha in Aoujeft • 592 ha in Rachid • 666 ha in Tamcheket • 370 ha in Nema	Restoration interventions are successful and supported by local stakeholders through the reduction of non-climatic drivers of ecosystem degradation.
	<u>Supplementary 4.2: Number of livestock brought under</u>	Baseline and annual reports informed by: i) community surveys and questionnaires; and ii) field surveys to quantify	0 livestock under climate-resilient management practices.	~1,440 livestock under resilient management practices within the four target hubs.	~2,880 livestock under resilient management practices within the four target hubs.	Implemented livestock management practices are sustainable and assist in improving

	<u>sustainable management practices</u>	herd numbers. Non-project generated sources include government agricultural surveys.				rangeland resilience to climate change.
--	---	--	--	--	--	---

E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)

Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
<u>Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner</u>	Mauritania has made some progress towards addressing its resilience to climate change impacts through alignment with several multilateral environmental agreements — including the UNFCCC, CBD and CCD⁶⁵ — as well as several national planning frameworks and strategies. Despite these efforts, there are still considerable gaps in the policy and institutional environments at the national, regional and local levels regarding climate change and natural resource management. Climate change knowledge in general has not been fully institutionalised across Mauritania. Government officials at both the provincial and local levels often do not fully understand the impacts of climate change and how to effectively develop strategies to improve resilience. Additionally, staff from key sectoral organisations at the local level often have limited technical capacity to support CCA initiatives.	<u>low</u>	• Create clearer mandates for both state and non-state actors regarding integrating CCA into sub-national and local legislation, policy and improved adaptation funding. • Enhance decision-makers' use of knowledge on climate change impacts and adaptation responses, bridging the current gap between research and policy-making. • Improved incorporation and mainstreaming of climate change adaptation and natural resource management into regional and local budgets, strategies and plans. • Robust regional and local coordination mechanisms for implementing climate change adaptation. • Improved access to financing to upscale EbA and other	Climate change and sustainable management of natural resources will be mainstreamed into sub-national and local plans, policies and frameworks the establishment of commune-level technical committees (CTCs) within each target commune with a focus on coordination of CCA implementation. Establish an on-granting mechanism to provide financial stability and facilitate continued investment in upscaling successful EbA activities and sustainable livelihoods. Transfer of the model to the MEDD. Build its capacity in fund management.	<u>Multiple sub-national areas within a country</u>

⁶⁵ Ministry of Rural Development and of Environment. 2004. National Adaptation Programme of Action to Climate Change. Available at: <https://unfccc.int/resource/docs/napa/mau01e.pdf>

			climate change adaptation activities.		
<u>Core indicator 8:</u> <u>Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards</u>	There are limited assessments on the impact of climate change and vulnerabilities of communities and ecosystems throughout most of Mauritania and specifically in the four target hubs. Information on losses and damages from climatic hazards, for example, are not quantified and readily available. Although some lessons learned have been generated from previous adaptation projects, there is generally limited knowledge on the implementation and benefits of CCA measures and natural resource management. Economic appraisals of potential cost-effective adaptation measures such as integrated land rehabilitation, water resource management, resilient crop species and livestock breeds have not been undertaken in Mauritania that would inform and promote the upscaling of these interventions.	<u>low</u>	Enhanced access of decision-makers to relevant tools, knowledge and data on climate change impacts and appropriate cost-effective adaptation responses. Removal of the current gap between research and policymaking on CCA to enable effective and informed responses to the impacts of climate change across the country.	Through the proposed project, several studies and assessments will be undertaken to improve the knowledge base on climate change impacts and the viability of sustainable adaptation interventions. Data from assessments, along with lessons learned and best practices from the proposed project, will be stored, analysed and shared through the NAP Knowledge Management Platform (Output 1.1). Using the platform developed under the National Adaptation Plan, trained CTC members and government staff will systematically document policy findings, coordination mechanisms and methods for integrating climate considerations into local plans and budgets. This documentation will support the replication of successful approaches beyond the initial 12 communes. Communities will be capacitated to install, implement and maintain adopted interventions or practices on dune fixation, restoration, climate-smart agriculture and water use management through training programmes, workshops and awareness-raising campaigns.	<u>Multiple sub-national areas within a country</u>

E.5. Project/programme specific indicators (project outcomes and outputs)

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Outcome 1. Increased implementation and upscaling of Ecosystem-based Adaptation (EbA) measures across the Sahara-Sahel boundary in Mauritania.	Percentage increase in institutional commitments and allocated budget for EbA interventions in government strategies and plans.	Analysis of annual budget allocations, policy documents and strategic planning documents that reflect increased commitment and financial support for EbA interventions.	TBD ⁶⁶	15% increase in institutional commitments and allocated budget for EbA interventions.	30% increase in institutional commitments and allocated budget for EbA interventions.	All relevant government staff actively engage with the CTCs, best practices and lessons learned, and policies, plans and frameworks. Knowledge and coordination gained through the project are retained within the institution and not lost through staff turnover.
Output 1.1. Governance structures are strengthened to support the implementation of EbA measures and the integration of climate change considerations and EbA into government plans, policies and budgets.	Change in the technical capacity of the MEDD, Ministry of Sustainable Development; Ministry of Agriculture; Ministry of Livestock Farming; Ministry of Water and Sanitation; and Ministry of Housing, Urbanism and Regional Planning to plan and budget for EbA in desert ecosystems and oases.	Certifications at workshops/ surveys showing increase in awareness over time; and Technical and institutional capacity surveys ⁶⁷	Score card: Level 0	Score card: Level 2	Score card: Level 3	Training and capacity building is applied by national government officials, regional government representatives and members of the CTC to mainstream CCA and natural resource management into policies, plans and budgets.

⁶⁶ Baseline Survey to be conducted in Year 1 of project implementation and reported in the first annual progress report.

⁶⁷ Score card with four levels of technical and institutional capacity of public-sector entities to understand and implement sustainable natural resource management practices in desert ecosystems. Level 0: Little to no technical capacity to understand and implement sustainable natural resource management practices in desert ecosystems, Level 1: Medium technical and institutional capacity of public sector entities to understand and implement sustainable natural resource management practices in desert ecosystems, Level 2: High technical and institutional capacity of public sector authorities resulting in high confidence for implementing sustainable natural resource management practices in desert ecosystems, and Level 3: High technical capacity resulting in public sector entities having implemented sustainable natural resource management practices in desert ecosystems. Score card survey to be developed by M&E officer.

Output 1.2. Knowledge products developed and disseminated to support decision making and upscaling.	Number of knowledge management products developed.	Review of knowledge management products.	0	36 knowledge management products developed.	72 knowledge management products developed.	Discussions with CTCs identify lessons learned and best practices, and these are captured in implementation guides and best practice reports.
Outcome 2. Communities in four rural-urban hubs along the Sahara-Sahel boundary in Mauritania have increased livelihood and water security as a result of investment in EbA measures, water access, and sustainable land- and natural resources management practices.	Number of individuals with improved access to sustainable sources of food and water, and reduced exposure to climate change-induced desertification.	Household surveys Focus group discussions	0 people	42,615 people (22,176 women, 20,439 men) This includes: • 7,126 (3,532 women, 3,594 men) in Aoujeft • 13,941 (7,368 women, 6,573 men) in Rachid • 7,011 (3,793 women, 3,219 men) in Tamcheket and • 14,537 (7,484 women, 7,054 men) in Nema.	85,229 people (44,352 women, 40,877 men). This includes: • 14,252 (7,064 women, 7,188 men) in Aoujeft • 27,881 (14,736 women, 13,145 men) in Rachid • 14,022 (7,585 women, 6,437 men) in Tamcheket and • 29,074 (14,967 women, 29,074 men) in Nema.	CCA interventions are implemented and maintained appropriately to mitigate the impacts of drought and sand inundation.
Output 2.1. Green-grey dune fixation infrastructure is established to control sand encroachment, enhance the provision of ecosystem services and slow the rate of desertification within the four target hubs	Change in the rate of sand encroachment within target areas.	Field surveys Key informant interviews	~10–20 m per year ⁶⁸	At least 5 m reduction in rate of sand encroachment from the baseline	At least 10 m reduction in rate of sand encroachment from the baseline	Nature-based and physical solutions are implemented and maintained appropriately to mitigate the impacts of drought and sand inundation.
Output 2.2. Improved access to water for agricultural and land rehabilitation activities.	Increased water availability on historical transhumance routes through boreholes and solar pumps	Field surveys of number of boreholes (solar pumps)	0 boreholes (or well-points) with solar pumps	24 boreholes and solar pumps	48 boreholes and solar pumps	Groundwater is available for extraction.

⁶⁸Jensen, A.M. 1993. Protection des routes contre l'ensablement en Mauritanie. New York, USA. United Nations Office to Combat Desertification and Drought (UNSO).

	Additional volume of water (litres) stored through rooftop rainwater harvesting.	Household surveys of stored water	0 L storage capacity through rainwater harvesting structures	720,000 L storage capacity through rainwater harvesting structures	720,000 L storage capacity through rainwater harvesting structures	Water resource attenuation interventions are successful.
	Number of people with improved access to water for use in market gardens, livestock farming and sustainable nature-based livelihoods.	Household surveys Focus group discussions	0 people	42,615 people (22,176 women, 20,439 men) This includes: • 7,126 (3,532 women, 3,594 men) in Aoujeft • 13,941 (7,368 women, 6,573 men) in Rachid • 7,011 (3,793 women, 3,219 men) in Tamcheket and • 14,537 (7,484 women, 7,054 men) in Nema.	85,229 people (44,352 women, 40,877 men). This includes: • 14,252 (7,064 women, 7,188 men) in Aoujeft • 27,881 (14,736 women, 13,145 men) in Rachid • 14,022 (7,585 women, 6,437 men) in Tamcheket and • 29,074 (14,967 women, 29,074 men) in Nema.	All individuals within target communities have equal access to communal water infrastructure established under the proposed project. Communal water infrastructure will be located within walking distance from target populations — along historical transhumance routes and in <i>graras</i> , which all beneficiaries will have access to.
Output 2.3. Climate-resilient agricultural livelihoods based on sustainable land- and natural resource-use are developed and/or strengthened to reduce land degradation and support climate-resilient income-generation by community members within the target regions.	Change in livestock, crop and horticultural yields	Household surveys Key informant interviews	TBD ⁶⁹	5% increase from baseline	15% increase from baseline	Livelihoods are economically viable within the hubs, and are accepted, adopted and maintained by communities.

⁶⁹ Baseline survey to be conducted in Year 1 of project implementation and reported in the first annual progress report.

	Amount of funds disbursed and number of EbA sub-projects funded through the on-granting mechanism.	Financial reports from the on-granting mechanism Project funding records and documentation of funded EbA projects.	0	At least 50% of allocated on-granting mechanism funds disbursed to support a minimum of 5 EbA projects	100% of allocated on-granting mechanism funds disbursed to support at least 10 EbA projects	There is adequate demand for funding from potential EbA project implementers and that projects align with the strategic goals of the government's climate change adaptation and sustainable natural resource management efforts.
Project/programme co-benefit indicators						
Co-benefit 1: Improved health of target communities	Change in the number of recorded respiratory, nutritional or waterborne diseases in the target communities.	Household surveys Medical statistics	TBD ⁷⁰	5% change in the number of recorded respiratory, nutritional or waterborne diseases	15% change in the number of recorded respiratory, nutritional or waterborne diseases	A recorded change in health across the communities is directly attributable to the proposed projects interventions and not from other factors.
Co-benefit 2: Increased biodiversity in the target communes	Change in species richness in restored and conserved ecosystems.	Field surveys	TBD ⁷¹	10% increase in species richness	20% increase in species richness	Change in biodiversity is representative of local natural ecosystems.
Co-benefit 3: Flood attenuation	Reduction in damages and losses caused by floods	Local government records Community feedback	~US\$78,000 per commune ⁷²	10% reduction in annual damages and losses caused by floods within the target communes	25% reduction in annual damages and losses caused by floods within the target communes	A reduction in flood-related damages and losses is directly linked to the implementation of flood attenuation measures under the project, such as the installation of stone bunds, gabions, dikes and weirs

⁷⁰ Baseline survey to be conducted in Year 1 of project implementation and reported in the first annual progress report.

⁷¹ Baseline survey to be conducted in Year 1 of project implementation and reported in the first annual progress report.

⁷² Average annual damages and losses caused by flooding are estimated at ~US\$17 million. On average, this amounts to ~US\$78,000 per commune (216 communes across Mauritania). The exact costs per commune will be determined via a baseline survey in Year 1 and monitored over the course of the project.

Co-benefit 4: Improved food security	Increase in agricultural yield of communal crops	Agricultural records Household surveys	TBD ⁷³	10% increase in agricultural yield	20% increase in agricultural yield	The increase in agricultural yield is attributed to project interventions, including the adoption of climate-smart agricultural practices, improved irrigation and access to quality seeds.
	Increase in the diversity of crops grown	Agricultural records Household surveys	TBD ⁷⁴	10% increase in crop diversity	20% increase in crop diversity	The increase in crop diversity is attributed to improved access to climate-resilient varieties, achieved via the establishment of seed banks.

E.6. Project/programme activities and deliverables

Activities	Description	Sub-activities	Deliverables
Outcome 1. Increased implementation and upscaling of Ecosystem-based Adaptation (EbA) measures across the Sahara-Sahel boundary in Mauritania.			
<u>Output 1.1. Governance structures are strengthened to support the implementation of EbA measures and the integration of climate change considerations and EbA into government plans, policies and budgets.</u>			
<i>Activity 1.1.1. Establish coordination platforms to facilitate knowledge-sharing, natural resource management, sustainable land-use planning and the implementation of EbA activities at the regional and local levels.</i>	Activity 1.1.1 will enhance coordination and technical capacity for CCA across the four target hubs by forming commune-level technical committees (CTCs) in 12 priority communes. Each CTC will be chaired by the respective commune's mayor, with members comprising of: i) DREDD representatives; ii) AGPO representatives; iii) technical experts in hydrology, agriculture, livestock husbandry and other pertinent fields; iv) civil society representatives; and v) local community members. These CTCs will be capacitated through training workshops to enable them to design and implement site-specific CCA interventions. To further support CTCs'	1.1.1.1: Establish commune-level technical committees (CTCs) within each priority commune in the four target hubs. 1.1.1.2: Deliver training workshops to CTCs, enhancing members' capacities to implement and manage project activities, support the integration of climate change in regional- and commune-level policies, plans and budgets, and facilitate knowledge sharing between regional- and local-level stakeholders. 1.1.1.3: Train CTC members and government staff to use the NAP Knowledge Management Platform and facilitate the collection and dissemination of climate information and adaptation best practices by CTC members.	1.1.1.1.a. 12 CTCs established across the priority communes (at least 50% of members are women). 1.1.1.2.a. 120 CTC members trained (10 per priority commune; at least 30% women) 1.1.1.3.a. 10 government staff trained (2 national; 8

⁷³ Baseline survey to be conducted in Year 1 of project implementation and reported in the first annual progress report.

⁷⁴ Baseline survey to be conducted in Year 1 of project implementation and reported in the first annual progress report.

	capacities to implement and manage CCA interventions at the commune-level, existing wilayah-, moughataa-, and commune-level development plans, policies and budgets will be reviewed and policy briefs for integrating climate considerations and gender-responsiveness into these documents prepared. A workshop will then be convened to present and validate the prepared policy briefs, inviting representatives from relevant government ministries and departments.	1.1.1.4: Conduct a review of existing wilayah-, moughataa- and commune-level development plans, policies and budgets and prepare policy briefs for the integration of climate considerations and gender-responsiveness into these documents. 1.1.1.5: Convene training workshops for Regional and Communal Councils and relevant sectors to support the integration of climate change in regional- and commune-level policies, plans and budgets, including through the presentation of the policy briefs prepared under Sub-activity 1.1.1.4.	sub-national; at least 30% women) 1.1.1.3.b. 24 CTC members trained (2 per commune; at least 30% women) 1.1.1.4.a. One review report of existing wilayah-, moughataa- and commune-level development plans, policies and budgets completed. 1.1.1.5.a. One workshop undertaken (30% women participants) to present the policy briefs prepared under Sub-activity 1.1.1.4.
<u>Output 1.2. Knowledge products developed and disseminated to support decision making and upscaling.</u>			
<i>Activity 1.2.1. Develop knowledge products on project lessons learned and best practices through a participatory process engaging the communities.</i>	Activity 1.2.1 will focus on systematically capturing community perspectives and best practices that emerge throughout project implementation. By integrating participatory approaches, the project will facilitate knowledge exchange between local stakeholders and the Project Management Unit (PMU), enabling adaptive management and the dissemination of successful adaptation strategies that can be referenced when upscaling or replicating climate change adaptation strategies across Mauritania, or when developing new climate change adaptation projects. Bi-annual gender-inclusive discussions will be held between the PMU and communities in the target hubs. These discussions will provide a structured forum for	1.2.1.1. Hold bi-annual gender-inclusive discussions between representatives from the PMU, CTCs and communities in the target hubs on project intervention successes and challenges, and develop these discussions into community engagement reports. 1.2.1.2. Identify lessons learned and best practices used in project interventions, and develop these into implementation guides and best practice reports.	1.2.1.1.a. 48 community engagement reports developed. 1.2.1.2.a. 4 implementation guidelines and 4 best practice reports compiled.

	community members to share their experiences, voice concerns and highlight successes and challenges related to project interventions. The insights gained from these discussions will be synthesised into community engagement reports, which will serve as a record of local experiences and provide valuable feedback for refining project implementation strategies. In parallel, the project will undertake a systematic identification of best practices and lessons learned emerging from project interventions. These knowledge products will be documented in best practice reports, as well as regular M&E reports, capturing successful approaches, lessons learned and strategies that have proven effective in enhancing climate resilience at the community level.		
<i>Activity 1.2.2. Enhance the dissemination of adaptation knowledge to sub-national decision-makers and communities to support upscaling.</i>	Activity 1.2.2 will facilitate the dissemination of knowledge products through both national and international platforms. The primary mechanism for this will be the NAP knowledge management platform, which is currently under development in Mauritania, where monitoring reports, GCF evaluation reports, community engagement reports, records of the validation workshop, final policy briefs and best practices generated under Activity 1.2.1 will be uploaded and stored. To extend the reach of these knowledge products beyond digital platforms, the project will strengthen the dissemination of stored materials through community-based communication channels. A communications expert will be consulted to design knowledge-management campaigns that are adapted to the local context. These campaigns will include targeted knowledge-sharing efforts via local radio stations, TV programmes and printed materials such as flyers distributed by DREDD officers.	1.2.2.1. Upload knowledge products (e.g. implementation guides, monitoring and evaluation reports, community engagement reports, policy briefs, lessons learned and best practice reports) onto the NAP knowledge management platform. 1.2.2.2. Package knowledge in the NAP knowledge management platform into formats that are accessible at local level (e.g. brochures, TV and radio programmes, awareness-raising materials). 1.2.2.3. Disseminate locally-accessible knowledge products in target and non-target communes across the four project wilayahs to catalyze upscaling, with support from the DREDDs and CTCs.	1.2.2.1.a. 20 reports are shared on the NAP knowledge management platform. 1.2.2.2.a. 20 locally-accessible knowledge products are developed. 1.2.2.3.a. Knowledge products are disseminated to all 12 target communes.
Outcome 2. Communities in four rural-urban hubs along the Sahara-Sahel boundary in Mauritania have increased livelihood and water security as a result of investment in EbA measures, water access, and sustainable land- and natural resources management practices.			
<u>Output 2.1. Green-grey dune fixation infrastructure is established to control sand encroachment, enhance the provision of ecosystem services and slow the rate of desertification within the four target hubs.</u>			

<i>Activity 2.1.1. Establish 2,123 hectares of EbA dune-fixation infrastructure and 120 kilometres of protective fencing across the four target hubs, to facilitate the rehabilitation and maintenance of degraded landscapes and enhance ecosystem services related to dune stabilisation and the supply of natural resources.</i>	Under this activity, EbA dune-fixation measures and physical dune-fixation infrastructure will be established to stabilise sand dunes and restore important ecosystem functions across the proposed project's four target hubs. EbA measures will include ecosystem restoration and tree-planting interventions, which will be designed to slow sand encroachment and subsequent desertification under future climate change conditions. Physical dune-fixation infrastructure will include the installation of artificial check-dunes, windbreak hedges and fence lines, to serve as physical barriers to sand inundation in the short-term. In addition to preventing sand inundation, landscape rehabilitation will improve the supply of other ecosystem services, such as flood attenuation and fodder supply within the proposed project's target communes. The implementation of dune fixation measures will be aided and accompanied by local-level restoration plans and the establishment of community-managed conservation areas in identified sites, which will be overseen by CTCs and established within each target commune. CTCs will be mandated to manage the implementation, operation and maintenance of project interventions that increase communities' resilience to the impacts of climate change-induced desertification under Activity 2.1.1. Once the locations for community-managed conservation areas have been signed off on by all relevant stakeholders, local labour will be recruited from nearby communes to install fences around these areas. Additionally, guards will be recruited from nearby villages to facilitate the stewardship of natural resources contained within fenced rehabilitation conservation areas and rehabilitation sites.	2.1.1.1: Support CTCs to co-develop commune-level land rehabilitation plans in collaboration with village-level stakeholders, project management teams and DREDD representatives. 2.1.1.2: Install ~2,123 ha of dune-stabilisation infrastructure (biological and mechanical dune-fixation) at strategic sites across the four target hubs, to protect critical areas against the impacts of sand inundation. 2.1.1.3: Install ~120 km of fence line around dune-fixation sites established under sub-activity 2.1.1.3, to protect biological dune-fixation infrastructure against damage from livestock and unsustainable use of natural resources. 2.1.1.4: Train livestock herders within the target communes to implement climate-resilient livestock management practices, such as rotational grazing, transhumance, supplementary feeding and 'livestock collar re-seeding'.	2.1.1.1.a. 12 site-specific commune-level rehabilitation plans developed (one for each of the target communes). 2.1.1.2.a. ~1,138 ha of buffer zones (green belts) installed at strategic sites across the four target hubs. 2.1.1.2.b. ~985 ha of biological- and mechanical dune-fixation infrastructure installed at vulnerable sites across 12 priority communes. 2.1.1.3.a. ~120 km of fence line installed around dune fixation sites established under Sub-activity 2.1.1.3. 2.1.1.4.a. 240 livestock farmers trained (20 per priority commune; 50% women).
--	--	--	---

	Activity 2.1.1 will additionally promote the implementation of climate-resilient sustainable livestock management practices to assist with the protection of degraded ecosystems against over-grazing. This will include the replacement of local livestock breeds with climate-resilient breeds, the growth of climate-resilient fodder varieties for supplementary feeding during the lean season and the establishment of 'set-aside' rotational grazing systems to rehabilitate degraded ecosystems and rangelands.		
<u>Output 2.2. Improved access to water for agricultural and land rehabilitation activities.</u>			
<i>Activity 2.2.1. Establish community-managed Water User Groups (WUGs) and commune-level water monitoring and regulation plans.</i>	Under Activity 2.2.1, 12 WUGs will be established. WUG members will comprise local community members and representatives from livestock cooperatives, farmer-based organisations and women's groups in each target commune. WUGs will be trained to monitor water availability and usage within their respective communes. Additionally, members will be trained to oversee and manage the construction, operation and maintenance of infrastructure for capturing surface water, managing runoff flow, increasing groundwater recharge and reducing flood risks. A team of national experts will conduct hydrogeological studies to quantify and map groundwater and surface water resources within each of the target hubs. This will inform the selection of strategic sites to install: i) groundwater dams, for improving aquifer recharge; ii) gabions, dikes and stone bunds for reducing flood risks; and iii) reservoirs, boreholes, well points and rainwater-harvesting systems to increase rainwater capture and storage. The selection of target sites for water management infrastructure will be summarised in the form of commune-level water management plans.	2.2.1.1. Establish community-managed Water User Groups (WUGs) within each target commune and train members to implement and maintain water-related activities introduced under Activity 2.2.2 of the project. 2.2.1.2. Support CTCs to raise awareness about sustainable water usage and co-develop commune-level water monitoring and regulation plans in collaboration with WUGs. 2.2.1.3. Conduct hydrogeological studies (or consult existing hydrogeological maps, where applicable) and engage with WUG members to identify priority sites to install water-management infrastructure (Sub-activities 2.2.2.1, 2.2.2.2 and 2.2.3.1) in each target commune. Summarise the finding of site selection into a commune-level water management plan for each priority commune.	2.2.1.1.a. 12 WUGs established across 12 priority communes in the four target hubs (one per commune; 50% of members are women). 2.2.1.2.a. 12 commune-level water monitoring and regulation plans developed (one per commune). 2.2.1.2.b. 1,800 people trained on water monitoring and regulation (150 per commune; 50% women) 2.2.1.2.c. 12 annual awareness raising events hosted across

	forming commune-level water user groups (WUGs) in the 12 priority communes. Each WUG		the four target communes. 2.2.1.3.a. One hydrogeological report identifying priority sites for the installation of water management infrastructure in each priority commune is produced
<i>Activity 2.2.2. Install physical water management infrastructure — including weirs, gabions, dikes, stone- and earthen bunds, groundwater dams and water access points for pastoralists.</i>	Under Activity 2.2.2, improved water resource and demand management practices will be introduced in the target hubs. These demand-management practices will be supplemented by sustainable water harvesting and storage solutions, to address the declining water security experienced by rural communities as a result of climate change in the four hubs. The proposed GCF project will introduce water-harvesting, flow-regulation and storage solutions — including solar-powered groundwater pumps, rainwater harvesting systems (with cisterns), gabions, dikes and bunds — that will increase the recharge of critical groundwater reserves and improve access to water to support dune fixation and livelihood interventions introduced under the proposed project. Additionally, the proposed GCF project will rehabilitate and build new pastoral wells along historical transhumance routes within the project's four target hubs.	2.2.2.1: Install physical water management infrastructure — including weirs, gabions, dikes, stone- and earthen bunds, groundwater dams and solar-powered pumps — at strategic sites within each target commune, to improve water access and availability, increase groundwater recharge rates and reduce flood risks in the target hubs. 2.2.2.2: Establish water access points along historical transhumance routes and in <i>graras</i> , to improve nomadic pastoralists' access to water and reduce sedentarisation among livestock herders.	2.2.2.1.a. A maximum of 16 groundwater dams established (4 per hub). 2.2.2.1.b. A maximum of 8 km of gabions and 8 km of clay dikes established (2 km per hub, respectively). 2.2.2.1.c. A maximum of 320 ha of stone bunds established (80 ha per hub). 2.2.2.2.a. A maximum of 48 boreholes and solar-powered pumps have been established across the four hubs (4 per priority commune).
<i>Activity 2.2.3 Install 12 rainwater-harvesting systems and communal cisterns (5,000 L per system) within each target commune</i>	Under Activity 2.2.3 of the proposed project, the installation of RWH systems and communal cisterns will increase the capture of surface water runoff during extreme rainfall events and increase target communities' capacities to store water for use during prolonged periods of	2.2.3.1. Install 12 rainwater-harvesting systems and communal cisterns (5000L per system) within each target commune across the four hubs, to improve access to water for agricultural livelihood activities.	2.2.3.1 a. 144 rainwater-harvesting systems and communal cisterns established (12 per priority commune).

	drought. Within each commune, the exact installation sites RWH systems and cisterns will be determined by WUG members, in consultation with CTC representatives and DREDD representatives; however, to promote fair access, sites located near communal buildings (town halls, schools and health posts) and agricultural areas (including market gardens and nurseries) will be given preference, to ensure water is used for agricultural activities and sustainable livelihoods introduced under the proposed project.		
<u>Output 2.3. Climate-resilient agricultural livelihoods based on sustainable land- and natural resource-use are developed and/or strengthened to reduce land degradation and support climate-resilient income-generation by community members within the target regions.</u>			
<i>Activity 2.3.1. Facilitate the adoption of climate-smart agricultural practices and sustainable diversified livelihoods by farmers and community members within the target communes</i>	Activity 2.3.1 aims to facilitate the adoption of sustainable climate-smart agricultural and livestock practices, as well as sustainable alternative livelihoods within the twelve target communes. Agricultural interventions will be focused on, <i>inter alia</i>, crop diversification, upscaling water-efficient irrigation systems and the adoption of small-scale fruit and vegetable production. Climate-resilient livestock management practices to be supported by the GCF project will include: i) the development of rotational grazing plans; ii) the use of climate-resilient fodder species and livestock breeds; and iii) the establishment of water access points along historical transhumance routes. The GCF project will further decrease communities' dependence on rainfall-sensitive economic sectors by facilitating the adoption of alternative, sustainable livelihoods such as apiculture (beekeeping) and market gardening. Climate-smart agricultural and horticultural measures will be supported by the establishment of nurseries and seed banks to provide climate-resilient crop varieties. Stock material consisting of plant cuttings and seeds will be supplied to the	2.3.1.1: Establish nurseries and seed banks in each target commune to supply activities related to land rehabilitation and dune fixation (Sub-activity 2.1.1.2), climate-smart agricultural practices (Sub-activity 2.2.1.2) and horticultural activities such as market-gardening. 2.3.1.2. Collect cuttings and seeds from agricultural crop species, as well as indigenous grass and tree species, to serve as stock material for nurseries and seed banks established under Sub-activity 2.3.1.1. 2.3.1.3: Train farmers within the target communes to practice climate-resilient crop agriculture and use improved agricultural technologies, including drip irrigation kits, solar powered pumps, integrated pest management strategies, zai pits and half moons. 2.3.1.4. Conduct site visits and provide technical support to facilitate the uptake of sustainable livelihood activities — including horticulture (market-gardening), apiculture, poultry farming, livestock feed production and the collection and sale of non-timber forest products — by community members within the target communes. 2.3.1.5: Supply farmers and horticulturalists with water-efficient irrigation equipment and climate-resilient crop varieties to support	2.3.1.1.a. 12 nurseries established (one per priority commune) 2.3.1.2.a. 12 seed banks established (one per priority commune) 2.3.1.3.a. 240 farmers trained (20 per priority commune; at least 50% women) 2.3.1.5.a. 240 households supplied with drip irrigation kits and climate-resilient crop varieties (20 per commune; at least 50% female headed). 2.3.1.6.a. 12 weekly markets established across the 12 priority communes (one per commune)

	nurseries and seed banks by extension officers from the MEDD and MDR. Additionally, extension officers will train nursery operators and other relevant stakeholders (CTC, FBO and WFG representatives) to maintain sustainable stocks (or seed banks) of these varieties in the long-term. Under Activity 2.2.1, technical support will be provided to community members through training on climate-smart agricultural practices, sustainable livestock practices and alternative livelihoods. This will consist of providing training and supplying equipment such as water-efficient irrigation kits to recipient community members. The income generation potential of introduced or enhanced livelihoods will be strengthened by improving access to urban markets and developing market chains within each target commune.	the uptake of agricultural activities adopted under Sub-activities 2.2.1.2. 2.3.1.6: Improve access to urban markets and develop value chains for offloading agricultural produce within each target commune, to enhance income generated from sustainable agricultural livelihoods.	2.3.1.6.b. Community carts' system established within each target commune for transporting produce to markets.
<i>Activity 2.3.2 Establish a small grants facility to facilitate continued investment in upscaling successful EbA activities and sustainable livelihoods.</i>	Under Activity 2.3.2, a project Small Grants Facility (SGF) will be established to enhance local communities' access to finance to upscale successful EbA interventions and sustainable, agriculture-based livelihoods introduced under the project. The on-granting mechanism will promote and support community-led initiatives, fostering the autonomous uptake of EbA practices by community members. Regional government budgets in the four target wilayah will be inspected and based on this, budget briefs with recommendations for channelling a percentage of sustainable development funding towards the project SGF will be prepared, to promote government investment in CCA.	2.3.2.1: Establish and operationalise a Small Grants Facility (SGF) to facilitate continued investment in upscaling successful EbA activities and sustainable livelihoods introduced under the project. 2.3.2.2: Prepare budget briefs for directing regional government funds into the SGF established under Sub-activity 2.3.3.1, to promote government investment in climate change adaptation. 2.3.2.3: Develop monitoring and reporting mechanisms to ensure the traceability and risk management of funds between the SGF and local-level stakeholders. 2.3.2.4: Produce annual monitoring and evaluation reports for sub-projects funded via the on-granting mechanism established under Sub-activity 2.3.2.1..	2.3.2.1.a. One Small Grants Facility established. 2.3.2.1.b. Finalised Funding Proposal template developed and selection criteria for the evaluation of proposals submitted. 2.3.2.2.a. Four wilayah-level budget briefs prepared. 2.3.2.2.b. One long-term financing and scaling strategy developed.

			2.3.2.3.a. Sub-project monitoring and reporting mechanisms developed. 2.3.2.4.a. Annual sub-project monitoring and evaluation reports produced.
--	--	--	---

E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

A full time Monitoring Officer will be employed to conduct and coordinate monitoring and evaluation (M&E) activities for the project. The Monitoring Officer will design and operationalise a performance monitoring framework to track the project's progress towards achieving its targets. This will include: i) measuring performance against the project indicators (Sections E.3–5) to evaluate the progress of the project; ii) reporting the project's performance to the PSC and PMU; and iii) providing technical support to the PC. Additionally, the Monitoring Officer will be responsible for overseeing and monitoring the application of gender-disaggregated indicators and ensuring that all gender targets are met. At particular milestones throughout the project — including annual performance reports, interim evaluations and at project conclusion — the project team will undertake evidence-gathering exercises to verify the progress. These internal reports will be further validated by reviews at the interim and end of the project implementation, as described below. UNEP will be responsible for managing the interim and the final evaluation of the project.

Interim Evaluation

The UNEP Task Manager will oversee the process of hiring an external independent consultant to carry out the interim evaluation, who will provide an assessment of project performance against its targets at the project's mid-point. This will be a formative exercise and will include analysing whether the project is on track, what problems and challenges the project is encountering, and what corrective actions are required so that the project will achieve its intended outcomes by project completion in the most efficient and sustainable way. The PSC will participate in the interim evaluation and develop a management response to the evaluation recommendations along with an implementation plan. It is the responsibility of the UNEP Task Manager to monitor whether the agreed recommendations are being implemented during the remainder of the project's operational life.

Final Evaluation

An independent consultant will be contracted by the UNEP Evaluation Office to conduct the final evaluation at the end of project implementation. UNEP's Evaluation Office (EO) will be responsible for overseeing the final evaluation at the end of project implementation — which is a summative evaluation — and will liaise with the UNEP Task Manager throughout the process. An independent assessment of project performance against standard evaluation criteria (e.g., strategic relevance, effectiveness, efficiency, likelihood of impact and sustainability) will be made based on documentary evidence, stakeholder interviews and, in most cases, a field mission. The evaluation criteria will be aligned with the criteria contained in the GCF Evaluation Policy. Each evaluation criterion will be rated using a six-point rating scheme and a weighted average will be determined to provide an overall performance rating for the project as a whole. Where there are any differences in ratings between the evaluation team and the Evaluation Office, a final determination will be made by the Evaluation Office when the evaluation report is finalised.

The draft final evaluation report will be sent to project stakeholders during a commenting process managed by the Evaluation Office. Formal comments on the report will be shared by the Evaluation Office in an open and transparent manner. This evaluation report will be publicly disclosed and will be followed by a recommendation on compliance processes. The costs for results monitoring and performance evaluation are included in the project budget (Annex 3).

The EE (Ministry of Environment and Sustainable Development) will submit semi-annual progress reports and quarterly financial statements to the PMU and the PMU will consolidate the reports and submit them to UNEP as the AE. In turn, UNEP will submit annual performance reports and semi-annual financial reports to GCF. The detailed reporting timelines are as follows:

Under the PCAs, the EE will be required to report to UNEP as follows:

- a. Progress reports: by 30 July for January to June;
- b. Annual Performance Reports on or before the start of second quarter (Q2) of each year, starting from year 2;
- c. Quarterly financial reports by 15 January, 15 April, 15 July, and 15 October;
- d. Annual audited statements by 30 April;
- e. Final report: within 3 months of Project completion.

UNEP (AE) reports to the GCF:

- a. Annual Performance Reports by 1 March;
- b. Semi-annual Financial Information by 1 March and 30 September;
- c. Interim Evaluation report: halfway through Programme;
- d. Final APR: within 6 months of Project completion;
- e. Final Evaluation report: within 12 months of Project completion.

Monitoring will also be undertaken by the AE through supervision visits and field missions to track implementation progress and challenges and strategically plan the way forward.

RISK ASSESSMENT AND MANAGEMENT		
F.1. Risk factors and mitigations measures (max. 3 pages)		
Selected Risk Factor 1		
Category	Probability	Impact
<u>Technical and operational</u>	<u>Medium</u>	<u>Medium</u>
Description		
High staff turnover within government institutions could impede capacity building, the retention of skills and knowledge, and knowledge-sharing across relevant institutions.		
Mitigation Measure(s)		
Communities and local authorities at different levels will be involved in the implementation of the proposed project. This will include the establishment of hub-level coordinating platforms, community-level decision-making groups, livestock cooperatives and community-level water-user groups through which information on — and ownership of — project interventions will be shared and maintained. As a result, knowledge regarding project planning and implementation processes will be spread across a range of individuals, thereby reducing the impact on the project if individuals leave their respective organisations.		
Under the proposed project, institutional stability will be provided by the Project Steering Committee (PSC) established as part of the project management arrangements (see Section B.4 for more detail).		
Selected Risk Factor 2		
Category	Probability	Impact
<u>Technical and operational</u>	<u>Medium</u>	<u>Medium</u>
Description		
Climate-related hazards such as droughts, floods or sandstorms or other environmental hazards may disrupt or damage nature-based or physical structures aimed at increasing community resilience to climate change during their implementation. The survival of planted seedlings used for agricultural practices or ecosystem restoration is also likely to be threatened by climate threats such as drought as well as pests such as herbivorous mammals or insects.		
Mitigation Measure(s)		
Project interventions will be closely and continuously monitored to identify potential climatic and environmental hazards so that appropriate measures can be enacted should such an event occur. Agricultural planting and ecosystem restoration will be undertaken using drought-resilient species and will be undertaken during the cooler months. Water-efficient irrigation techniques such as bottomless cylinders and the use of mulch to reduce evaporation will also be considered to maintain seedling health. Where needed, supplementary planting of seedlings will be undertaken to replace lost plants. Enclosures and guards will be utilised to protect restored areas from herbivorous pests, with guards specifically aimed at prevent animal invasion and identifying any damages to the enclosures.		
Selected Risk Factor 3		
Category	Probability	Impact
<u>Governance</u>	<u>Medium</u>	<u>Medium</u>
Description		
Limited coordination between implementing partners reducing the efficiency of interventions.		
Mitigation Measure(s)		
A project management unit and steering committee will be established to coordinate project implementation (Further details on the implementation arrangements are presented in Section C.7). At the hub level, coordination and management committees will ensure coordination among government institutions. At the community-level, user groups, decision-making groups and cooperatives will ensure coordination between communities and local authorities.		

The relevant implementing partners that were involved in the design of the project activities and have agreed on all roles and responsibilities. A monitoring and evaluation plan has been designed for the project, which will provide a platform for monitoring the performance of implementing partners and will ensure that these institutions and agencies are fulfilling their responsibilities during the proposed project's implementation period. This will be supported by regular meetings/engagements between implementing partners to facilitate information sharing.

Selected Risk Factor 4

Category	Probability	Impact
<u>Governance</u>	<u>Medium</u>	<u>Medium</u>

Description

Limited participation by women and other marginalised groups in project interventions and decision-making could lead to the needs of these groups not being considered or represented in project design or implementation. In particular, similar previous projects have shown that limited female participation in formal training activities is a risk as a result of long travel distances and limited transport options for women.

Mitigation Measure(s)

During project development, women and marginalized groups were consulted to ensure that their specific requirements are addressed. A GAP analysis was undertaken to ensure that interventions are gender-sensitive and that women and marginalized groups are represented in awareness-raising, training and other measures that benefit communities. Additionally, all members of the project management structures will receive gender sensitivity training to ensure that decisions that affect project implementation are socially and gender sensitive. Particularly for training activities, formal training workshops will be organised on-site to reduce the need for women to travel.

Selected Risk Factor 5

Category	Probability	Impact
<u>Governance</u>		<u>Medium</u>

Description

Changes in political leadership at local level, as well as within ministries and other participating governmental institutions could result in delays in implementation.

Mitigation Measure(s)

Even with more frequent changes in governing parties through elections and via personnel shifts in ministries, which are to be anticipated, the project will be fully anchored through the involvement of multiple ministries and institutions, ensuring project continuity without major interruptions. Specific mitigation measures which have been built into the projects design are mentioned below.

- Regular engagement with and capacity building of non-political actors — such as government agency officials, staff and local community members — will be prioritised during project implementation.
- Formulation of local ordinances (including community-level user groups, decision-making groups and cooperatives).
- Mainstreaming of project interventions into local planning, policy-making and budgeting systems will ensure that political challenges will have negligible impact on the sustainability of project interventions.

Selected Risk Factor 6

Category	Probability	Impact
<u>Other</u>	<u>Low</u>	<u>Medium</u>

Description

Global shocks such as the Covid-19 pandemic or regional conflicts could influence government priorities, resulting in delays in project implementation and reducing the likeliness of project sustainability.

Mitigation Measure(s)

While the project will not prevent possible global shocks, implementation arrangements developed for the project that consists of multiple responsible parties at different responsibility levels, co-finance commitments and a detailed workplan will assist in ensuring project activities are met. The experience of the AE and international best practice on mitigating the effects of global shocks — including lessons learned from the Covid-19 pandemic — will be drawn on to inform the response of the PSC and implementing partners to similar shocks if they were to occur.

Selected Risk Factor 7

Category	Probability	Impact
<u>Other</u>	<u>Medium</u>	<u>Medium</u>

Description

Resistance by water users, land holders and local communities to adaptation measures, alternative livelihoods and climate-resilient agricultural practices may result in on-the-ground interventions not being effective or sustainable.

Mitigation Measure(s)

Consultations with community stakeholders — including water users, land holders, farmers and pastoralists — during project development included raising awareness on the proposed interventions to address drought and sand inundation. These consultations additionally aimed to gain support and buy-in from the communities by participating with them to identify their specific vulnerabilities and adaptation preferences. Details on these consultations are presented in Annex 7: Stakeholder consultations and engagement plan.

During project implementation, awareness-raising campaigns on climate-resilient agricultural practices and sustainable natural resource management will encourage compliance for these interventions through the sharing of information on their benefits and by fostering ownership of the project by local stakeholders. Adequate training on the introduced methods will also be provided to ensure that beneficiaries are able to implement them effectively. Additionally, community-level user groups, decision-making groups and cooperatives within each target hub will ensure that local stakeholders (including local communities) are represented at both decision-making and implementation levels. This will ensure that their concerns are considered and that they are not excluded from accessing benefits, promoting compliance with the adaptation measures implemented by the project.

Selected Risk Factor 8

Category	Probability	Impact
<u>Other</u>	<u>Medium</u>	<u>Medium</u>

Description

Selected sites for on-the-ground interventions are not chosen effectively and do not address the requirements of vulnerable communities, or the distance between sites makes project implementation difficult.

Mitigation Measure(s)

Site selection within the target areas has been undertaken during project development using international and national expertise and through participatory engagements with local stakeholders. Risk assessments and a SWAT analysis undertaken during the project design phase have additionally contributed to an informed selection of the sites, which have undergone a detailed multi-criteria analysis agreed upon by major stakeholders involved in project development. If sites are subsequently found to be unviable or undesirable during project implementation, coordination between the PMU and Implementing Partners will assist in guiding a flexible and iterative response to identifying alternative sites.

The cost relating to the large travel distances between sites has been accounted for in the project's budget. Where possible, local implementing representatives will be used to coordinate the day-to-day implementation of the interventions to reduce the need for unnecessary travel between sites.

GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

Project categorisation

The environmental and social risks associated with this project have been evaluated in accordance with the GCF's Environmental and Social Safeguards. The proposed project is classified as Category B (moderate risk), based on the screening process and regulations used by the GCF. Three of the 8 safeguard categories are deemed moderate risk:

SS 1: Biodiversity, Ecosystems and Sustainable Natural Resource Management: The inclusion of *Prosopis Juliflora* as a potential species for biological dune-fixation is considered to be necessary risk within very specific contexts (such as highly mobile dune environments where indigenous species are not able to establish). However, there is a high risk of proliferation, and the potential difficulty of removing these plants is well established.

SS 4: Community Health, Safety and Security: There are some health and safety risks inherent to undertaking construction activities, including for dune fixation and water infrastructure installation. These risks apply to project personnel, contracted labour as well community members residing in, contributing to, or traveling through, the vicinity of the construction activities. There are also some minor risks that rainwater harvesting and other water provisioning infrastructure could pose a health and safety risk to community members, related to the transmission of vector-or-water-borne diseases that are associated with standing water or poor hygiene practices.

SS 8: Labor and working conditions: gender inequality is prevalent in Mauritania. While the project does include a Gender Assessment and Gender Action Plan (GAP), there may be certain instances where women will be inhibited from participating in specific employment opportunities due to cultural or social prohibitions.

Justification for this classification is provided in Annex 6: Environmental and Social Management Plan (ESMP), which contains an overview of the general project implementation risks that have been identified to date. A detailed risk rating for each activity is provided in Chapter 5 of Annex 6.

Stakeholder consultation

To date, extensive stakeholder consultations have informed the project preparation process. The full Stakeholder Engagement Plan is presented in Annex 7: Stakeholder Engagement Plan, which includes the full list of stakeholders consulted and a summary of their feedback. The results of stakeholder consultations have closely informed the design of the proposed project and development of the ESMP. These consultations are expected to continue during the implementation phase, to ensure regular communication between project personnel and affected communities. Accordingly, a participatory approach will be used when conducting E&S risk screening exercises, such that local community members' input is incorporated into activity-level risk registries and associated risk management strategies. As a minimum, it is recommended that specific Stakeholder Engagement Plans be developed for each project Outcome, since each activity set will have different consultation requirements, stakeholders and E&S risks.

Project grievance redress mechanism

A Grievance Redress Mechanism (GRM) has been developed for this project and are described in detail in Chapter 10 of the ESMP (Annex 6). The proposed mechanism is compatible with Mauritanian legislation and compliant with UNEP and GCF standards, enabling any project stakeholders with complaints or grievances related to the project to communicate their concerns through an appropriate process. The GRM will be available to stakeholders at all stages of the project and will provide an accessible, rapid, fair and effective response to concerned community members — including any vulnerable groups that may lack access to formal legal systems or support. The GCF Independent Redress Mechanism will also be available to stakeholders and affected persons who have grievances or issues relevant to the project's activities.

All project grievances (both internal and external) will be resolved using the steps below, with the hub-level CTCs assuming most of the duties and responsibilities in this regard:

Step 1. Receipt of complaint/grievance;

Step 2. Acknowledgement of complaint/grievance;

Step 3. Assessment, investigation and resolution of complaint/grievance;

Step 4. Closure of investigation and redress process; and

Step 5. Outcome of the corrective action is verified with the complainant.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

A Gender Assessment and Action Plan (GAAP) for this project provides an overview of gender-disaggregated challenges, specifically those related to: i) women's rights; ii) the gendered division of labour; iii) political participation; v) access to land; vi) access to justice; vii) gender-based violence (GBV); and viii) climate change impacts.

Among the proposed project's target population, climate change disproportionately affects women and other socio-economically marginalised groups in Mauritania as a result of unequal access to productive resources^{75,76}. Although Mauritania has ratified the International Convention on the Elimination of all forms of Discrimination Against Women (CEDAW)⁷⁷, the GoM has not yet to lift reservations applying to articles 13(a) and 16 of CEDAW, which relate to: i) eliminating discrimination in family benefits; and ii) requiring equality in marriage and family matters.

Gender inequality is apparent at the household level, with the country's Personal Status Code (PSC) established in 2001, serving as the primary codified law governing marriage and family relations in Mauritania⁷⁸. The PSC provides a marital framework based on male guardianship over women and children, where men are considered the lawful heads of their respective households. However, ~50% of households in the focal hubs of Aoujeft, Rachid and Tamcheket are headed by women, mainly resulting from divorce, widowhood, and male migration to urban areas for work⁷⁹. Statistics detailing the gender of household heads in Nema are not currently available. Additionally, the PSC states that men hold decision-making authority, while women are expected to assist in managing the family. Mauritania's national Labour Code reinforces these conventional gender roles by prohibiting women from working in occupations considered dangerous, arduous or immoral⁸⁰. Although the GoM has ratified International Labour Organisation conventions on discrimination and equal remuneration in the workplace, women's labour force participation remains low (31%⁸¹), and they face a considerable gender pay gap⁸².

Educational disparities exist, with lower primary school completion rates for girls (~56%) compared to boys (~64%)⁸³. Early marriage and childbearing jeopardise girls' chances of completing school, resulting in lower literacy rates for women (~57%) compared to men (~70%)⁸⁴. Gender discrepancies in access to finance limit women's capacities to adapt to climate change. While the Constitution grants equal rights to men and women open bank accounts, access credit and register businesses, cultural norms and gender roles limit women's participation. Women also face 'time poverty' as they spend considerable time on non-remunerated household activities limiting their time available to diversify their livelihoods or participate in projects designed to improve their resilience.

In accordance with the GCF Updated Gender Policy and Gender Action Plan 2020–2023 (GCF/B.24/15), the proposed project will address gender dimensions within the four target hubs by incorporating the following measures into activities designed under the proposed project:

- the establishment of a gender-balanced project management unit (PMU);
- the recruitment of a Gender Officer within the PMU, to support and monitor implementation of the GAAP;
- compulsory gender training for the PMU;
- the inclusion of gender-mainstreaming tasks into all relevant consultancy and partnership ToRs;
- the identification and strengthening of at least six women-led civil society organisations (CSOs), to assist with knowledge dissemination and recruiting women for activities and training workshops;

⁷⁵ Productive resources are the natural-, human- and capital resources used to produce goods and services.

⁷⁶ Islamic Republic of Mauritania. 2004. The National Adaptation Programme of Action (NAPA) to Climate Change. Available at: <https://unfccc.int/resource/docs/napa/mau01e.pdf>

⁷⁷ United Nations Treaty Collection. 2022. Convention on the Elimination of All Forms of Discrimination against Women (CEDAW) — Mauritania. Available at:

<https://treaties.un.org/Pages/Declarations.aspx?index=Mauritania&lang=en&chapter=4&treaty=334#EndNotes%20Section>

⁷⁸ Musawah. 2017. Mauritania: Overview of Muslim family laws & practices. Available at: <https://www.musawah.org/wp-content/uploads/2019/03/Mauritania-Overview-Table.pdf>

⁷⁹ Annex 2 — Appendix 1: ARIA/ACTERRA/AGRER. 2021. Climate Risk Analysis Report.

⁸⁰ OECD. 2019. Social Institutions and Gender Index: Mauritania. Available at: <https://www.genderindex.org/wp-content/uploads/files/datasheets/2019/MR.pdf>

⁸¹ WB. 2020. Labor force, female (% of total labor force) — Mauritania. Available at: <https://data.worldbank.org/indicator/SL.TLF.TOTL.FE.ZS?locations=MR>

⁸² OECD. 2019. Social Institutions and Gender Index: Mauritania. Available at: <https://www.genderindex.org/wp-content/uploads/files/datasheets/2019/MR.pdf>

⁸³ UNICEF. 2019. State of the World's Children — 2019. Available at: <https://www.unicef.org/media/63016/file/SOWC-2019.pdf>

⁸⁴ WB. 2021. Quatrième rapport sur la Situation Économique en Mauritanie: Un meilleur avenir: accélérer la relance économique en misant sur le potentiel des femmes © World Bank. Available at: <https://documents1.worldbank.org/curated/en/961341622141230195/pdf/Rapport-sur-la-Situation-Economique-en-Mauritanie-Un-Meilleur-Avenir-Accelerer-la-Relance-Economique-en-Misant-sur-le-Potentiel-Femmes.pdf>

- specific strategies for facilitating women's participation in monitoring, evaluation and learning initiatives; and
- strategies for involving women in decision-making processes — particularly those relating to their land, resources, activities, families and livelihoods — to overcome constraints on access to natural resources.

The project's approach to addressing gender-related challenges and promoting gender equality are presented in the Gender Action Plan (Annex 8), with specific gender-sensitive benefits summarised below.

- Training modules targeting men and women, empowering policymakers, decision-makers, and community leaders to mainstream gender considerations into adaptation planning and implementation (Output 1.1).
- Enhancing women's leadership capacity through participation in training and skills-building activities, promoting representation in decision-making on climate adaptation and resource management interventions (Output 1.1).
- Developing gender-responsive policies, including one focused on gender in natural resource management and land-use planning (Output 1.1), to ensure gender-responsiveness is incorporated into national priorities, planning and budgets.
- Integrating women into the design and dissemination of awareness campaigns, tailoring core messages so they are gender-responsive (Output 1.2).
- Including women in the selection of priority sites for rehabilitation and design of awareness-raising campaigns, addressing unique threats and challenges they face (Output 2.1).
- Empowering women in target communities through training and active participation in developing livelihood activities, enabling them to independently support themselves and their families (Output 2.3).
- Enabling women to actively participate in decision-making processes relating to their lands, resources, activities, families and livelihoods (Output 2.3). This will allow them to contribute to directly address impacts that increase their vulnerability to climate change.

Stakeholder consultations and engagement with women's CSOs during the project design process aided in identifying gender-disaggregated climate change impacts within the Mauritanian context. Continued engagement with these groups during the project implementation stage will further promote gender equality within the proposed project's target regions.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

The financial management and procurement within the project will be guided by UN financial regulations, rules and procedures, as well as UNEP's Programme Manual. The financial rules of UNEP, which follow International Public Sector Accounting Standards (IPSAS), are promulgated pursuant to the Financial Regulations and Rules of the UN. Within this context, funding-allocation mechanisms are managed as per the UN framework, including eligibility criteria, proposal evaluation processes, quality assurance and control, project monitoring and supervision. UNEP is audited annually by the UN Board of Auditors and has established dedicated trust funds for GCF resources.

The funding of a project will be established through a distinct grant within the Trust Fund, with the project itself being set up in the UNEP Enterprise Resource Planning (ERP) as a 'Work Breakdown Structure' (WBS), which is itself further broken down into 'Work Breakdown Structure Elements' (WBSEs), organised by output and outcome. The grant of the project will be linked to the WBSEs to fund the activities contributing to the delivery of specific outputs (as per the disbursement schedule detailed in the term sheet). The fee income will be managed through a dedicated grant independently and separately of the GCF project grant funds. In line with UNEP procedures, the project will appoint a Financial and Administrative Officer within the PMU who will be responsible for monitoring, reporting on and approving requests for funds on a quarterly basis for the activities executed by the MEDD. Reports to summarise the disbursement and projected demands for project funding will be prepared and submitted to a UNEP Task Manager who will conduct project supervision, in line with reporting standards and methodologies applied in past projects, such as those implemented using GCF modalities.

The UN financial regulations and rules require the segregation of duties, and safeguards to ensure compliance with UN financial rules and regulations. In addition, a Fund Management Officer will be appointed to assist UNEP's Project Officer with all financial monitoring and supervision functions. All procurement will be undertaken in line with UN procurement regulations, rules and policies. In accordance with UNEP's modalities for project implementation, funds will be transferred in tranches to the EE (MEDD), once the EE has satisfied all conditions defined under the legal/cooperation instrument — including the Project Cooperation Agreement (PCA), which will be signed between UNEP's Climate Change Division and the MEDD. The PCA will include specific obligations for the EE on financial management and reporting and will require periodic financial reporting to follow international financial and auditing standards. The PCA specifically requires the audit be undertaken by a recognised firm of certified public accountants

or, for governments, by a government auditor. This auditor should state whether the GCF proceeds were covered by the scope of the audit.

As a GoM authority, the MEDD follows the government's financial and procurement rules and standards. During the proposed project's inception phase, UNEP will undertake a thorough assessment of the MEDD's capacity to undertake procurement in line with UN regulations, rules and processes. This assessment will guide the procurement monitoring plan which will be agreed upon between UNEP and the MEDD. The assessment will be undertaken following project approval but prior to project implementation. It will include assessments of elements of governance and public accountability such as, inter alia, review of the existence and quality of policies, legal and institutional framework, and systems supporting transparency, accountability and controls, especially in the use of donor funds. The outcome of the assessment will determine the threshold for procurement that is undertaken by the MEDD, which will be reflected in the procurement monitoring plan. This plan is an annex to the PCA and will be shared with GCF upon signature, as required by the Funded Activity Agreement (FAA). The procurement monitoring plan will also be reviewed periodically.

The project's investments in equipment will be undertaken in accordance with UN procurement procedures. Finally, in line with the UNEP Programme Manual, the MEDD will be mandated to provide an annual compliance audit covering all aspects of the project execution including review of all expenditures incurred during the financial year.

G.4. Disclosure of funding proposal

☒ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☐ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

8. ANNEXES

H.1. Mandatory annexes

- ☐ Annex 1 NDA no-objection letter(s) [\(template provided\)](#)
- ☒ Annex 2 Feasibility study - and a market study, if applicable
- ☒ Annex 3 Economic and/or financial analyses in spreadsheet format
- ☒ Annex 4 Detailed budget plan [\(template provided\)](#)
- ☒ Annex 5 Implementation timetable including key project/programme milestones [\(template provided\)](#)
- ☒ Annex 6 E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3):
[\(ESS disclosure form provided\)](#)
 - ☐ Environmental and Social Impact Assessment (ESIA) or
 - ☒ Environmental and Social Management Plan (ESMP) or
 - ☐ Environmental and Social Management System (ESMS)
 - ☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.)
- ☒ Annex 7 Summary of consultations and stakeholder engagement plan
- ☒ Annex 8 Gender assessment and project/programme-level action plan [\(template provided\)](#)
- ☒ Annex 9 Legal due diligence (regulation, taxation and insurance)
- ☒ Annex 10 Procurement plan [\(template provided\)](#)
- ☒ Annex 11 Monitoring and evaluation plan [\(template provided\)](#)
- ☒ Annex 12 AE fee request [\(template provided\)](#)
- ☒ Annex 13 Co-financing commitment letter, if applicable [\(template provided\)](#)
- ☒ Annex 14 Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

- ☒ Annex 15 Evidence of internal approval [\(template provided\)](#)
- ☐ Annex 16 Map(s) indicating the location of proposed interventions
- ☐ Annex 17 Multi-country project/programme information [\(template provided\)](#)
- ☒ Annex 18 Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
- ☐ Annex 19 Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
- ☐ Annex 20 First level AML/CFT (KYC) assessment
- ☒ Annex 21 Operations manual (Operations and maintenance)
- ☐ Annex 22 Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects)⁸⁵
- ☒ Annex 23 Beneficiary estimation methodology

* Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.

⁸⁵ Annex 22 is mandatory for mitigation and cross-cutting projects.

No-objection letter issued by the national designated authority(ies) or focal point(s)

République Islamique de
Mauritanie

Honneur - Fraternité – Justice

Ministère de l'Environnement



الجمهورية الإسلامية الموريتانية

شرف - إخاء - عدل

وزارة البيئة

Direction Climat et Economie Verte

مديرية المناخ والإقتصاد الأخضر

N° : 022/23

أنواكشوت، 19/09/2023

Le Directeur المدير

To: The Green Climate Fund ("GCF")

Re : **Funding proposal for the GCF by UN Environment programme (UNEP) regarding *Strengthening the resilience of ecosystems and populations in four regional hubs in northern Mauritania*.**

Dear Madam, Sir,

We refer to the programme titled ***Strengthening the resilience of ecosystems and populations in four regional hubs in northern Mauritania*** as included in the funding proposal submitted by United Nations Environment Programme (UNEP) to us on 01 September 2023.

The undersigned is the duly authorized representative of Sidi Mohamed El wavi, the National Designated Authority of Mauritania.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Mauritania has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Mauritania;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Sidi Mohamed El Wavi



Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Strengthening the resilience of ecosystems and populations in four regional hubs in northern Mauritania
Existence of subproject(s) to be identified after GCF Board approval	No
Sector (public or private)	Public
Accredited entity	United Nations Environment Programme (UNEP)
Environmental and social safeguards (ESS) category	Category B
Location – specific location(s) of project or target country or location(s) of programme	Islamic Republic of Mauritania: • Aoujeft Hub (Wilaya of Adrar), communes of Ain Savra, El Medah, Maeden, Aoujeft • Rachid Hub (Wilaya of Tagant), communes of Tidjikja and El Wahat • Tamcheket Hub (Wilaya of Hodh El Gharbi), communes of El Mabrouk and Gueate Teidoume • Nema Hub (Wilaya of Hodh El Chargui), communes of Oualata, N'Beiket Lahwach, Noual and Jreif
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	An ESMP consistent with the requirements for a Category B project is contained in Annex 6 "Environmental and Social Management Plan (ESMP)" of the Funding Proposal. Risk screening of funded activities rated from low to moderate necessitated the development of an ESMP and not an ESIA.
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	Friday, May 16, 2025
Language(s) of disclosure	English, Arabic, and French (for the full document and Executive Summary)
Explanation on language	Arabic is the official language of Mauritania, as well as the written language of the affected populations. French is the working language of the administration.
Link to disclosure	English: https://open.unep.org/docs/gcf/UNEPFP_Mauritania_Annex%206_ESMP_English.docx Arabic: https://open.unep.org/docs/gcf/UNEPFP_Mauritania_Annex%206_ESMP_Arabic.docx French:

	https://open.unep.org/docs/gcf/UNEPFP_Mauritania_Annex%206_ESMP_French.docx
Other link(s)	UNEP Open Data: https://open.unep.org/project/GCF-GCF-MAUR-XXX Ministry of Environment and Sustainable Development, Mauritania: (Executive Summary in French) http://www.environnement.gov.mr/fr/index.php/accueil/actualite/636-resume-du-pges-du-projet-4-poles-partage-pour-information-publique-et-remarques
Remarks	An ESMP consistent with the requirements for a Category B project is contained in Annex 6 “Environmental and Social Management Plan (ESMP)” of the Funding Proposal
Environmental and Social Management System (ESMS) (if applicable)	
Date of disclosure on accredited entity’s website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), Indigenous Peoples Planning Framework (IPPF) (if applicable)	
Description of report	Annex 7: Stakeholder Consultations and Engagement Plan; Annex 8: Gender Assessment and Gender Action Plan
Date of disclosure on accredited entity’s website	Tuesday, May 20, 2025
Language(s) of disclosure	English
Explanation on language	These documents are available in English.
Link to disclosure	Stakeholder Consultations and Engagement Plan: https://open.unep.org/docs/gcf/UNEPFP_Mauritania_Annex_7_SEP.pdf Gender Assessment and Gender Action Plan: https://open.unep.org/docs/gcf/UNEPFP_Mauritania_Annex_8_GAP.pdf
Other link(s)	https://open.unep.org/project/GCF-GCF-MAUR-XXX
Remarks	A Stakeholder Engagement Plan (SEP), and a Gender Assessment and Gender Action Plan (GAAP) are included in Funding Proposal Annexes 7 and 8, respectively
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Monday, May 26, 2025
Place	Physical copies of the Executive Summary document of the ESMP in Arabic and French have been displayed at the locations listed below, on dates between 15th and 26th of May, 2025. Wilaya of Adrar: - Office of the Regional Delegation for Environment and Sustainable Development (DREDD), Atar

	- Hakem's office, Moughataa (Department) of Aoujeft - Mayors' offices, Communes of Ain Savra, Aoujeft, El Medah and Maeden Wilaya of Tagant: - Office of the Regional Delegation for Environment and Sustainable Development (DREDD), Tidjikja - District Chief's office, District of Rachid - Mayors' offices, Communes of Rachid, Tidjikja and El Wahat Wilaya of Hodh El Gharbi: - Office of the Regional Delegation for Environment and Sustainable Development (DREDD), Aouin - Hakem's office, Moughataa (Department) of Tamcheket - Mayors' offices, Communes of El Mabrouk, Tamcheket and Gueate Teidoume Wilaya of Hodh El Chargui: - Office of the Regional Delegation for Environment and Sustainable Development (DREDD), Nema - Hakem's office, Moughataa (Department) of Nema - Mayors' offices, Communes of Nema, N'Beiket Lahwach, Jreif, Oualata and Noual
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	N/A
Date of GCF's Board meeting	Monday, June 30, 2025

Note: This form was prepared by the accredited entity stated above.

Secretariat's assessment of FP266

Proposal name:	Strengthening the resilience of ecosystems and populations in four regional hubs in northern Mauritania
Accredited entity:	United Nations Environmental Programme (UNEP)
Country:	Mauritania
Project size:	Small

I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
Ecosystem-based adaptation approach to restore ecosystems and stabilize dunes, utilizing green-grey infrastructure. This includes 2,123 hectares of dune fixation and 120 kilometres of protective fencing, benefiting approximately 230,237 people in total.	The four regions where the project activities will be implemented are not immediately adjacent but are within a few hundred kilometres of each other. If the distances between sites make implementation challenging, the project's impact could be compromised.
The project promotes locally-led adaptation by actively involving communities in decision-making, ensuring context-specific, sustainable climate resilience measures that address local needs.	The accredited entity does not have in-country presence in Mauritania, which could limit executing entity oversight. The accredited entity has budgeted for a full-time national Chief Technical Advisor to mitigate this risk. In addition, the United Nations Environment Programme Task Manager will have biannual supervision missions.
The project will establish a Small Grants Facility to support the continued investment in upscaling successful ecosystem-based adaptation activities and climate-resilient livelihoods, providing financial support to local communities and organizations to help them adapt to climate change risks.	

2. The Board may wish to consider approving this funding proposal in accordance with the terms listed in the term sheet agreed between the Secretariat and the accredited entity (AE), and, if considered appropriate, subject to the conditions set out in annex II to document GCF/B.42/02.

II. Summary of the Secretariat's assessment

2.1 Project background

3. Climate change is profoundly impacting Mauritanian communities, particularly those living along the Sahel-Saharan boundary, which extends from the north-west to the south-east of the country. A noticeable impact of climate change-induced desertification is the accelerated advance of sand dunes. Arable land, rangeland and pastures are being inundated, severely compromising the agricultural and livestock practices in the oases. In addition, sand inundation threatens critical socioeconomic infrastructure, including, inter alia, roads, school buildings and hospitals. In extreme cases, sand dunes have displaced entire villages, in some cases more than once. Additionally, the declining water table has dried up many pastoral wells, disrupting the traditional nomadic lifestyles of pastoralists and forcing them to adopt sedentary lifestyles around rural oases. This shift increases pressure on limited natural resources and heightens the risk of conflicts over resources.

4. This proposed project focuses on addressing these climate-induced challenges by adopting an integrated ecosystem-based adaptation (EbA) approach to restore ecosystems, prevent desertification, and enhance water and food security. Targeted interventions will be implemented across four regional hubs – Aoujeft, Rachid, Tamcheket and Nema – located in the country's most climate-impacted regions. This approach includes establishing 2,123 hectares (ha) of green-grey dune-fixation infrastructure and 120 kilometres of protective fence line to facilitate the rehabilitation and maintenance of degraded landscapes and enhance ecosystem services related to dune stabilization and the supply of natural resources. Additionally, the project will develop or strengthen sustainable agriculture-based livelihoods and enhance water security.

5. The project is expected to directly benefit 85,229 people (40,877 men and 44,352 women) and indirectly benefit 145,008 people (66,519 men and 78,489 women), reaching a total of 230,237 beneficiaries. The total project financing is USD 33,560,345, of which a GCF grant of USD 30,902,775 is requested. The Government of Mauritania will provide in-kind co-financing of USD 2,657,570. The project is submitted under the environmental and social safeguards Category B, and the total duration of the project is six years.

2.2 Component-by-component analysis

2.2.1. Component 1: Governance and Capacity-building

Outcome 1: Increased implementation and upscaling of ecosystem-based adaptation measures across the Sahara-Sahel boundary in Mauritania (total cost: USD 2.2 million; GCF cost: USD 2.1 million)

6. Component 1 will be crucial in laying the foundation for long-term climate resilience and sustainable land-use practices in the Sahara-Sahel region of Mauritania. The component aims to integrate EbA approaches into governance and land-use planning, with a particular focus on strengthening the capacity of regional and local institutions to support climate-resilient interventions. It will work towards mainstreaming climate adaptation and EbA measures into government policies and budgets, particularly in vulnerable areas.

7. A key feature of Component 1 is the establishment of 12 commune-level coordination and technical committees (CTCs). These CTCs will facilitate knowledge-sharing, natural resource management, and the implementation of EbA activities across the four target hubs. Each CTC will be composed of local stakeholders, including community representatives, technical experts and government officials. These committees will be trained to manage and upscale climate adaptation measures at the local level, ensuring that decision-making processes reflect the needs and priorities of local communities.

8. Additionally, Component 1 will focus on the creation and dissemination of knowledge products such as training manuals, best practices and policy briefs. These products will aid in

informed decision-making and the scaling of successful EbA practices, thus enhancing the resilience of ecosystems and communities. The integration of climate change considerations into subnational planning processes will be central to ensuring that climate adaptation becomes a key aspect of local governance, fostering long-term sustainability beyond the project's lifetime.

9. The component also recognizes the importance of gender-responsive planning, aiming to ensure that women and vulnerable groups are actively involved in the planning and implementation of adaptation measures. The participatory approach will help to secure broad-based community ownership of the project interventions, which is essential for the project's success and continuity.

2.2.2. **Component 2: Strengthening Ecosystem Resilience and Climate-Adaptive Livelihoods**

Outcome 2: Communities in four rural-urban hubs along the Sahara-Sahel boundary in Mauritania have increased livelihood and water security as a result of investment in EbA measures, water access, and sustainable land and natural resources management practices (total cost: USD 28 million; GCF cost: USD 25.6 million)

10. Component 2 of the proposed project aims to significantly enhance the resilience of communities and ecosystems in four rural-urban hubs along the Sahara-Sahel boundary of Mauritania. It focuses on EbA measures and other sustainable natural resource management practices to address desertification and improve livelihoods in vulnerable communities.

11. One of the core outputs under this component is establishing ~2,123 ha of green-grey dune-stabilization infrastructure and 120 kilometres of protective fencing around dune-fixation sites. These interventions will stabilize sand dunes and restore critical ecosystem services like soil stabilization and water infiltration, which will significantly reduce the impacts of desertification and enhance local biodiversity. These efforts will help mitigate sand encroachment on infrastructure and agriculture, leading to increased resilience and economic opportunities for local populations.

12. In addition to dune stabilization, the component focuses on improving water access for agricultural and land rehabilitation activities. This will be achieved by installing improved water management infrastructure, including rainwater harvesting systems and cisterns, enhancing water availability for both agricultural use and household consumption. This infrastructure will directly benefit communities by improving food and water security, which is crucial for their resilience against climate change.

13. The component will also support climate-resilient agricultural livelihoods by introducing sustainable land and natural resource-use practices. This includes training local farmers on climate-smart agriculture and alternative livelihoods such as beekeeping and poultry farming. The project will also assist in market access and develop value chains to help farmers diversify income sources, reducing their dependency on fragile ecosystems.

14. A Small Grants Facility (SGF) will be established to provide continued financial support for scaling up successful EbA interventions and sustainable livelihoods, fostering community-led initiatives and enhancing long-term resilience in the target hubs.

Project management (total cost: USD 1.7 million; GCF cost: USD 1.5 million)

15. The project will be executed by the Ministry of Environment and Sustainable Development (MEDD) in Mauritania, which will be supported by the United Nations Environment Programme (UNEP) as the accredited entity (AE). UNEP will oversee the entire project, ensuring that it aligns with GCF standards. As part of this arrangement, UNEP will also be responsible for managing the project's financial oversight, reporting and ensuring compliance with both GCF and UNEP policies.

16. A Project Steering Committee (PSC) will provide strategic direction, consisting of representatives from MEDD, key ministries, and local authorities. A project management unit (PMU) within MEDD, led by a Project Coordinator, will handle daily operations and reporting. The PMU will include specialists in financial management, procurement, gender, safeguards and monitoring. A Technical Committee, chaired by MEDD, will provide technical guidance.

17. In addition, implementing partners, such as local non-governmental organizations like Naforé, will be engaged for on-the-ground interventions in target hubs. Naforé will assist with ecosystem restoration, livelihood improvement, and water infrastructure activities. Other partners may be onboarded as the project progresses to enhance local implementation.

18. The GCF portion of this project management cost is less than 5 per cent of the total requested GCF funding and is compliant with the GCF Policy on Fees.

Monitoring and evaluation component (total cost: USD 1.8 million; GCF cost: USD 1.8 million)

19. The project's monitoring, reporting, and evaluation arrangements will be coordinated by a full-time Monitoring Officer, responsible for designing a performance monitoring framework. This framework will track progress against project indicators and ensure regular reporting to the PSC and the PMU. Gender-disaggregated indicators will also be monitored to ensure gender targets are met.

20. Key evaluation activities include an interim and a final evaluation. The interim evaluation, managed by UNEP, will assess the project's performance at its midpoint, addressing challenges and recommending corrective actions. The final evaluation, conducted by an independent consultant overseen by the UNEP Evaluation Office, will assess the overall project impact based on established evaluation criteria, including relevance, effectiveness and sustainability. These evaluations will be shared with stakeholders for feedback before public disclosure.

21. Furthermore, the project will produce annual progress reports and semi-annual financial reports. UNEP will manage these reports, ensuring they meet GCF standards. Field visits and supervision will be conducted by UNEP to ensure continuous monitoring and address any implementation challenges.

III. Assessment of performance against investment criteria

3.1 Impact potential

Scale: N/A

22. The adaptation impact potential of the project is considerable, focusing on enhancing the resilience of communities and ecosystems along the Sahara-Sahel boundary in Mauritania. The project is expected to directly benefit 85,229 individuals by providing improved protection against sand inundation, enhancing access to markets, income-generating opportunities, and clean water. Additionally, it will indirectly benefit 145,008 people through ecosystem restoration, better water management infrastructure, and strengthened climate adaptation capacity. The project aims to rehabilitate 2,123 ha of degraded land, enhance water security through improved infrastructure, and promote climate-resilient agriculture, with positive spillover effects across surrounding regions. This comprehensive approach will help increase resilience for the entire population in the four target hubs, benefiting approximately 230,237 people in total.

3.2 Paradigm shift potential

Scale: N/A

23. The project's paradigm shift potential is considerable as it aims to create systemic changes in local climate adaptation practices through the establishment of coordination platforms and the integration of EbA into national and local policies. By creating these platforms and promoting knowledge-sharing, the project enhances the capacity of local and regional stakeholders to manage climate risks and implement climate-resilient land and water management practices. This can lead to the institutionalization of EbA approaches at the local level and create a foundation for long-term climate resilience.

24. However, the scalability of these solutions beyond the four target regions is largely dependent on broader institutional uptake and the availability of funding. While the project can serve as a demonstration model for future initiatives, its success in catalysing systemic change across Mauritania depends on the ability to secure continued financial support and the political will to integrate EbA into national policies. The wider replication of these practices will require strong engagement from national and regional institutions to embed the project's outcomes into broader policy frameworks and funding mechanisms.

3.3 Sustainable development potential

Scale: N/A

25. The project demonstrates strong sustainable development potential, providing multiple co-benefits that align with the GCF Investment Criteria and contribute to several Sustainable Development Goals (SDGs). The project will enhance biodiversity through ecosystem restoration activities and will also help to conserve biodiversity by mitigating desertification and preventing further ecosystem degradation. This aligns with SDG 15 (life on land), which focuses on protecting, restoring and promoting the sustainable use of terrestrial ecosystems.

26. The project will also have a direct impact on food security by promoting climate-resilient agriculture and livelihood diversification. The project will strengthen local food production by training farmers in climate-smart agricultural practices and providing improved irrigation systems, thereby reducing vulnerability to climate shocks and enhancing SDG 2 (zero hunger). Additionally, the establishment of alternative livelihoods such as beekeeping and poultry farming will provide communities with diversified income sources, ensuring food availability even during challenging climatic conditions.

27. Furthermore, improved community health is a key outcome of the project. The restoration of ecosystems and improved water management infrastructure will enhance water quality and availability, reducing waterborne diseases and improving overall sanitation. Access to clean water, especially through communal water points and rainwater harvesting systems, will directly support SDG 6 (clean water and sanitation). The project also seeks to reduce health risks through better infrastructure protection from sand encroachment, which will safeguard vulnerable communities from respiratory diseases caused by dust storms and improve the general well-being of the population.

28. Additionally, the project emphasizes gender equality by ensuring that women are actively engaged in the planning, decision-making and implementation of climate adaptation interventions. This supports SDG 5 (gender equality), as women's participation in leadership roles and sustainable livelihood initiatives helps to empower them economically and socially, while also contributing to broader community development.

3.4 Needs of the recipient

Scale: N/A

29. The project effectively responds to the most pressing challenges for Mauritania by addressing the country's high vulnerability to climate change impacts and the challenges related to desertification, water scarcity and economic development. As one of the most vulnerable countries globally (twelfth on the Notre Dame Global Adaptation Initiative Vulnerability Index)

and with relatively low readiness for climate action (118th on the Readiness Index), Mauritania faces significant barriers to climate adaptation.

30. The project tackles desertification directly by implementing EbA measures, such as dune fixation and ecosystem restoration, across 2,123 ha. This addresses one of the country's most pressing environmental challenges, aiming to slow the spread of the desert into fertile agricultural areas and protecting livelihoods from sand encroachment.

31. Water scarcity is another critical issue in Mauritania. The project improves water security by installing water management infrastructure, including groundwater recharge dams, solar-powered boreholes, and rainwater harvesting systems. Additionally, the project responds to the country's low economic development by promoting climate-resilient livelihoods through sustainable agriculture, livelihood diversification (such as beekeeping and poultry farming), and strengthening food security.

3.5 Country ownership

Scale: N/A

32. The project demonstrates strong country ownership by aligning closely with the national adaptation priorities of Mauritania and has been designed in collaboration with stakeholders at all levels. It directly addresses several key objectives from first nationally determined contribution of Mauritania, particularly in areas such as restoring natural grasslands, conducting hydrological assessments, strengthening ecosystem resilience, and enhancing the resilience of vulnerable populations, especially in rural regions. The project also contributes to the country's national adaptation programme of action (NAPA), which encourages a cross-sectoral approach to climate adaptation. It directly addresses 19 of the 27 priority adaptation measures outlined in the NAPA, including EbA interventions, water management techniques, and capacity-building for communities affected by climate change.

33. The project also supports the ongoing national adaptation plan development process in Mauritania, which began in 2018, by enhancing the capacity of both policymakers and vulnerable groups. This will include activities such as awareness-raising campaigns targeting local authorities, and training programmes for relevant government staff. Additionally, the project aligns with the country's spatial planning strategy, which uses an oasis-centred administrative structure to enhance resilience to desertification and mitigate rural exodus.

34. Extensive consultations with a wide range of stakeholders – including government representatives, civil society organizations, the private sector and local communities – have been conducted to ensure the project meets the country's adaptation needs. As the national designated authority, MEDD has been actively involved since the inception of the project, contributing to the design process and validating the proposed interventions. Over 330 local community members from the target hubs participated in consultations, ensuring the project reflects the needs and priorities of the people of Mauritania.

3.6 Efficiency and effectiveness

Scale: N/A

35. The proposal presents a robust and context-specific economic and financial analysis (EFA), with clearly delineated methodologies for monetizing ecosystem services and estimating household-level returns. The EFA captures both market and non-market benefits across agricultural production, livestock management, water access and health system cost savings. The economic analysis utilizes an ecosystem service indexing approach, combined with macroeconomic contribution analysis (including multiplier effects and health-cost savings). For the financial analysis, a household-level analysis simulates with/without project scenarios, integrating productivity gains expected through GCF-financed interventions. The structure is consistent with GCF EFA guidance, and sensitivity analysis is included.

36. Long-term economic viability is supported by positive economic indicators. The economic analysis estimates a net present value of USD 151.5 million over a 20-year horizon, against project costs of USD 31.2 million. This yields a benefit-to-cost ratio of 5.86 and an economic internal rate of return (IRR) of 29 per cent. The analysis uses a 2 per cent social discount rate, lower than the GCF standard benchmark of 5–10 per cent, but justified given the non-market nature of ecosystem services and the strong resilience co-benefits delivered. The benefits encompass improved productivity in crop and livestock systems, enhanced access to water, economy-wide multipliers, and monetized health-cost savings from reduced malnutrition.

37. The financial analysis presents beneficiary-level cash flows and returns under “without” and “with” GCF support scenarios for oasis-based agriculture and livestock systems. With GCF interventions, the financial IRR improves from 12 per cent to 22 per cent (with a financial net present value of USD 42.6 million), indicating enhanced viability at household level.¹ However, the analysis shows that most financial value accrues via subsistence consumption, limiting direct revenue generation. The EFA also incorporates a qualitative justification for 100 per cent grant support, citing multidimensional poverty, limited surplus income, low private sector participation, and fiscal constraints. The SGF is mentioned in the analysis, but it was not modelled in the financial analysis. Further development of activity-specific financial models for SGF-supported interventions would enhance the robustness and credibility of the analysis, particularly in demonstrating the financial viability and sustainability of these subprojects from a beneficiary perspective.

38. The adaptation cost per direct beneficiary is estimated at USD 362.6, and USD 134.2 per total beneficiary. These figures are above the GCF portfolio averages for Africa, which are usually under USD 200 per direct beneficiary. While the cost per beneficiary is relatively high, this is explained by the project’s focus on foundational EbA measures, and the inclusion of public goods such as water access and land restoration. The EFA demonstrates strong value for money by linking these cost levels to substantial economic benefits – with an estimated benefit-to-cost ratio of 5.86 and an economic IRR of 29 per cent. Overall, the economic and financial metrics reinforce the justification for full grant financing in the context of a least developed country.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

39. The Environmental and Social Management Plan (ESMP) shared by the AE including a Stakeholder Engagement Plan (SEP) aligns with GCF’s revised Environmental and Social Policy (ESP). The project has been classified as Category B, based the outcome of risk screening against the ESS standards that showed moderate to low risks. The environmental and social risk categorization is consistent with that of the revised ESP. Based on the on the environmental and social risk screening undertaken by the AE, potential risks are likely to be associated with;

40. **Biodiversity/Ecological:** While the project interventions and associated outcomes have been designed to reduce degradation and improve the provisioning of ecosystem services,

¹ The financial discount rate of 12 per cent provided in the assumptions sheet serves as a benchmark for evaluating financial viability but is not directly applied in the financial IRR calculation. Financial IRR is computed independently of any discount rate, based solely on projected cash flows. Furthermore, the baseline cost structure is reverse-engineered using a goal-seek approach that back-calculates costs from an assumed IRR. This method does not align with standard financial analysis practice, which requires costs to be independently estimated from actual or benchmarked inputs. As a result, both baseline and project IRR figures should be interpreted with caution.

there are ecological risks associated with the implementation of project activities. Poorly designed and implemented interventions are likely to baseline conditions and result in downstream impacts. This may lead to incidents of soil erosion, increased sand inundation, siltation of water resources, proliferation of alien invasives or affect normal hydrological functioning and the provision of ecosystem services for downstream communities. Further the inclusion of *Prosopis Juliflora* as a potential specie for biological dune fixation as it establishes faster than indigenous species increases the potential of invasive species. The major threat to biodiversity and ecosystems remains, dune stabilisation and the establishment of water infrastructure. It is expected adopting mitigation measures proposed in the ESMP will reduce and eliminate potential risks.

41. **Labour and Working Conditions:** All activities requiring physical labour hold an implicit occupational health and safety risk, for project employees and implementing entities. Civil works such as construction activities, dune fixation and water resource desilting interventions pose labour and working condition risks. All physical activities or civil works including construction activities will be required to the proposed mitigation measures and acceptable best practices. This will require continuous training interventions, issuance and maintenance of suitable personal protective equipment (PPE), constant supervision and compliance monitoring by the employees themselves, their site management and the PMU.

42. **Community Conflict:** During project implementation incidents and issues related to financial mismanagement, lack of trust and lack of buy in of project can also lead to potential inter-and-intra-community conflict if perceptions of preferential or differential benefits stemming from project activities becomes an issue, or an observable reality. Stakeholder engagement is critical in to enhance transparency, accountability, regular feedback and information dissemination on project performance to beneficiary communities. This will reduce or eliminate potential risks associated with nepotism, fraudulent or corrupt practices as it can exacerbate community conflicts that have the potential to derail project success. Any potential social conflicts that may arise will need to be assessed and managed through the appropriate project consultation forums that will be constituted by relevant stakeholders (residents, commune council representatives and leadership etc.). Additionally, the GRM will serve as an important pathway for community members to voice concerns in an anonymous manner without fear of retaliation. Furthermore, existing tensions over water resources are well documented in the target hubs and the project will need to ensure that these realities and constraints are suitably identified and addressed during implementation. Effective stakeholder engagement and an operational GRM will place an emphasis on the early identification and resolution of avoidable conflict situations or scenarios.

43. **Involuntary Resettlement:** Mauritanian law recognizes traditional property rights except when it interferes with the collective interest. During project implementation, it is expected that decisions, including the selection of sites for project interventions, will be undertaken through consultative and participatory processes to ensure they do not lead the involuntary resettlement of any communities or community members. No involuntary resettlement is anticipated prior or during project implementation

44. **GCF Indigenous Peoples Policy and Environmental and Social Standard 7 (Indigenous Peoples).** The AE has undertaken an assessment against the Indigenous Peoples Policy and engaged with authorities and civil society organizations to determine the presence of Indigenous Peoples within the project area, and confirms that none have been identified. The project is committed to inclusive and culturally appropriate engagement with all vulnerable and mobile groups through broader environmental and social safeguards and will establish a community-based coordination mechanism to facilitate management of resources. The project further indicates that, should Indigenous Peoples be identified during project implementation, an Indigenous Peoples Plan or an Indigenous Peoples Planning Framework will be developed.

45. **Sexual exploitation, sexual abuse, and sexual harassment.** The revised GCF Environmental and Social Policy adopted by decision B.BM-2021/18 requires safeguarding from sexual exploitation, abuse and harassment (SEAH) in GCF-financed activities. The AE provided SEAH safeguarding in its submission to this funding proposal. The project ESMP identifies SEAH as one of the social risks and highlights key measures to address SEAH and gender-based violence (GBV) in alignment with the AE's approach to SEAH and/or its equivalence. The risks in relation to SEAH have been identified from: (i) the outcomes of the UNEP risk screening; (ii) consideration of local legal requirements; and (iii) general good practice principles in the context of the proposed project interventions. Both annexes 6 and 8 have ensured that the approach to SEAH is survivor-centred. The AE noted that because of the absence of a legal framework for GBV in the country, several limitations affecting the GBV and SEAH grievance redress mechanisms have been identified. These include: (i) weak protection and accountability structures; (ii) limited standardized reporting mechanisms; and (iii) limited coordination with law enforcement. The AE has therefore put in place a SEAH management process (SEAH 9-step procedure) to ensure the safety of the survivors and continued psychosocial support services. In addition to this, the project will allocate SEAH focal points as well as the development of indicators to track the effectiveness of SEAH/GBV prevention and reporting systems. The PMU will comprise a gender officer to oversee any grievances related to GBV or SEAH. In addition, gender-related training will be provided to the PMU and include topics such as understanding and preventing SEAH and GBV.

4.2 Gender policy

46. The UNEP gender assessment for this project highlights the various challenges women and marginalized groups face, especially in the four regional hubs of northern Mauritania. The assessment describes the limiting factors for women including technical skills and access to means of production, such as land and credit, as well as gender norms and perceptions. The assessment describes potential risks for the participation of women in the project. Key gender project management outputs have been categorized as per each project outcome area. The project will ensure that there is a gender balance when establishing the PMU and management of its human resources. A gender officer will also be recruited in the PMU to support and monitor the effective implementation of the Gender Action Plan and oversee grievances related to SEAH/GBV.

47. Activities to be implemented in this project include: gender training for the PMU; facilitating the participation of women in the project monitoring and learning; building women's leadership and agency; ensuring that national policies, plans and strategies on natural resource management and land-use planning are gender-responsive; developing knowledge products on project lessons learned and best practices through a participatory process that engages the communities; and demonstrating gender-responsive best practices in adaptation planning. The project will further document case studies featuring women's leadership in EbA interventions; provide gender-sensitive technical training on dune-fixation techniques; ensure at least 30 per cent female representation in water user groups; develop gender-sensitive water regulation plans that consider women's water needs for domestic and agricultural use; ensure women's participation in selecting locations for rainwater-harvesting systems and communal cisterns; and prioritize access to harvested rainwater for women-led smallholder farms and household use. It has been recommended that the AE should make any relevant mid-course corrections as and when applicable during implementation of the Gender Action Plan and leverage other opportunities where the gender targets, especially for women, could be increased given the social and cultural context. The Gender Action Plan consists of activities, performance indicators, baselines, targets, means of verification, budget, timelines and responsibilities.

4.3 Risks

4.3.1. Overall project assessment (medium risk)

48. The Secretariat assesses overall programme risk as medium.

4.3.2. Accredited entity/executing entity capability to execute the current programme (medium risk)

49. UNEP will serve as the accredited entity for this project. UNEP has experience in implementing environmental projects of this nature; it has 10 GCF-approved funded projects with a budget of USD 249 million in its portfolio. MEDD will act as the executing entity (EE). MEDD has been assessed for its capacity to manage the project effectively, with adequate financial management and procurement systems in place. However, the AE does not have an in-country presence in Mauritania, which could limit oversight of the EE. The AE has budgeted for a full-time national Chief Technical Advisor to mitigate this risk. In addition, the UNEP Task Manager will have biannual supervision missions.

4.3.3. Project-specific execution risks (medium risk)

50. Implementation challenges: The four target regions where the project will be implemented – Aoujeft, Rachid, Tamcheket and Nema – are not immediately adjacent but are within a few hundred kilometres of each other. The distance between these regions may present logistical challenges that could affect the efficiency and timeliness of implementation. To mitigate this, the project budget accounts for the logistical demands of coordinating activities across multiple sites, including transportation and communication.

51. Lack of buy-in: There is a risk of resistance to the project activities due to a lack of understanding of the benefits or vested interests conflicting with the project goals. While consultations have been conducted to encourage community buy-in, ongoing engagement through awareness campaigns is necessary to address this challenge. A participatory approach will be integral to overcoming such challenges, ensuring that local communities take ownership of the project and are committed to its long-term success.

52. Co-finance: Although co-financing from the government accounts for only 8 per cent of the total project financing and is in-kind in the form of staff time as well as office facilities to support the operations of the PMU, the government has committed USD 1.5 million for specific project activities, specifically to provide inputs required to cover 905 ha (42 per cent) of the total 2123 ha of dune stabilization to be undertaken. If the co-financing does not materialize fully, it could lead to delays and/or reduced scope of activities. The AE has secured co-financing commitment letters from the government, and it will be required to report to GCF on co-financing applied to the implementation of the funded activity.

4.3.4. Compliance risk (medium risk)

53. The money laundering/terrorist financing (ML/TF) risks associated with certain proposed project activities, such as green and grey infrastructure enhancements and the creation of an on-granting mechanism, are relatively elevated.

54. The AE has assessed ML/TF risk as low based on its internal controls and governance mechanisms, as well as mitigation measures that will be implemented based on identified risks. For example, the AE's due diligence of downstream implementing partners includes assessments of their governance and financial management capacities and a risk assessment that covers financial, operational, and reputational risks. The AE conducted a micro assessment of the EE/national designated authority under its Harmonized Approach to Cash Transfers

framework, which did not flag any material deficiencies relating to the EE's anti-money laundering/countering the financing of terrorism (AML/CFT) or prohibited practices controls. Moreover, the AE will provide AML/CFT training to the EE as part of project preparation. In addition, financial management and procurement, including funding allocation mechanisms such as eligibility criteria, will be subject to the UN's established rules and procedures as detailed in paragraphs five and six in the Fiduciary section.

55. On the basis of the elevated inherent ML/TF risks presented by certain project activities and the AE's existing controls and mitigation measures to be implemented to address any project-specific risks, the compliance risk is determined to be medium.

4.3.5. GCF portfolio concentration risk

56. In the case of approval, the impact of this proposal on the GCF portfolio concentration in terms of the accredited entity, the results area and a single proposal is within monitoring threshold.

4.3.6. Recommendation

57. It is recommended that the Board consider the above factors in its decision.

Summary risk assessment	
Overall project	Medium
Accredited entity/executing entity capability	Medium
Project-specific execution	Medium
Compliance	Medium
GCF portfolio concentration	Within monitoring threshold

4.4 Fiduciary

58. The project will be implemented by the United Nations Environment Programme (UNEP), as the accredited entity (AE), in collaboration with the Ministry of Environment and Sustainable Development (MEDD) of Mauritania, which serves as the executing entity (EE). UNEP will retain overall responsibility for project supervision, fiduciary compliance, financial oversight, and reporting to GCF. MEDD will be responsible for executing day-to-day project activities and managing country-level implementation in accordance with agreed plans. A dedicated Project Management Unit (PMU) will be established to coordinate project execution.

59. Oversight and strategic direction will be provided by a PSC composed of relevant government institutions, including the Ministries of Finance, Health, and Agriculture. A representative of UNEP will also form part of the PSC. This committee will have a decision-making capacity and will primarily serve to provide project oversight and advisory support, including: a) overseeing project implementation; and b) reviewing annual budget and workplans.

60. All financial management will be governed by the financial regulations and rules of UNEP. UNEP will manage GCF proceeds through a dedicated grant within its trust fund structure, and expenditures will be tracked through the UNEP enterprise resource planning system using a work breakdown structure. MEDD will submit quarterly financial requests for the activities under its responsibility, which will be reviewed and approved by UNEP. Financial reporting will follow international standards and UNEP policies, and will be supported by

annual compliance audits conducted by a government auditor or an internationally certified firm.

61. Procurement for the project will follow United Nations Financial Regulations and the procurement rules and procedures of UNEP. All procurement will initially be managed by UNEP. However, during the project inception phase, UNEP will conduct a comprehensive procurement capacity assessment of MEDD. Based on this assessment, a procurement monitoring plan will be developed to determine the threshold and scope of procurement that may be delegated to MEDD. All procurement undertaken by MEDD, if authorized, must comply with United Nations procurement standards and will be subject to regular monitoring and review.

62. The following condition precedents are recommended to be included in the term sheet and funded activity agreement.

- (a) Prior to the first disbursement, UNEP should submit, to the satisfaction of GCF, i) a capacity assessment of MEDD, specifically in relation to its procurement management and ability to manage an on-granting mechanism in line with GCF fiduciary standards; and ii) based on the findings of this assessment, UNEP should also submit an action plan that outlines specific measures to address identified capacity gaps.
- (b) Prior to the disbursement of funds under the SGF (activity 2.3.2), UNEP should submit to the GCF Secretariat, to the satisfaction of the Secretariat, the final institutional and governance structure of the SGF.

4.5 Results monitoring and reporting

63. The project will contribute to all adaptation results areas largely towards increased resilience of most vulnerable people and communities (35 per cent) and of ecosystems and ecosystem services (35 per cent), and partly towards health and well-being, and food and water security (15 per cent) and infrastructure and built environment (15 per cent). These results areas are represented in the theory of change where appropriate assumptions were also articulated.

64. A logical framework was developed to include indicators that relate to these results areas and as consistent with the GCF Integrated Results Framework and draft Results Handbook. Specifically, the proposed project will monitor and report progress against: Core Indicator 2: Direct and Indirect Beneficiaries Reached; Supplementary 2.1: Beneficiaries (female/male) adopting improved and/or new climate-resilient livelihood options; Supplementary 2.5: Beneficiaries (female/male) adopting innovations that strengthen climate change resilience; Supplementary 2.3: Beneficiaries (female/male) with more climate-resilient water security; Core 4: Hectares of natural resources brought under improved low-emission and/or climate-resilient management practice; Supplementary 4.1: Hectares of terrestrial forest, terrestrial non-forest, fresh-water and coastal marine areas brought under restoration and/or improved ecosystems; and Supplementary 4.2: Number of livestock brought under sustainable management practices.

65. Baseline data and mid-line and endline targets have been specified for all the above indicators. The means of verification combines primary and secondary data sources including those collected through household surveys, geographic information systems mapping, government data, interviews and focus group discussions. With a budget of USD 1.8 million for monitoring and evaluation, the proposal provides confidence that data collection for monitoring and reporting purposes would be sufficient and robust. Thus, based on the assessments, the proposed project has provided for the required resources to ensure compliance with the requirements of the monitoring and accountability framework, Integrated Results Management Framework and the Evaluation Policy.

4.6 Legal assessment

66. The Accreditation Master Agreement was signed with the Accredited Entity on 15 December 2016, and became effective on 20 February 2017, which was amended and restated pursuant to a first amendment and restatement agreement dated 13 July 2023, and which became effective on 2 August 2023 (the “AMA”)

67. The Accredited Entity has provided a certificate confirming that it has obtained all internal approvals and it has the capacity and authority to implement the project.

68. The proposed project will be implemented in the Islamic Republic of Mauritania, country in which GCF is not provided with privileges and immunities. This means that, amongst other things, GCF is not protected against litigation or expropriation in this country, which risks need to be further assessed. Moreover, the ability of GCF to undertake redress activities and/or investigations in the country may be hindered due to the absence of privileges and immunities for relevant GCF personnel.

69. Therefore, it is recommended that the Board considers whether disbursements of GCF proceeds should only be made after GCF has obtained satisfactory protection against litigation and expropriation in the country, or has been provided with appropriate privileges and immunities for GCF and its personnel.

70. To address the matters raised in this section, it is recommended that any approval by the Board is made subject to the following conditions:

- (a) Signature of the funded activity agreement in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval; and
- (b) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

Independent Technical Advisory Panel's assessment of FP266

Proposal name:	Strengthening the resilience of ecosystems and populations in four regional hubs in northern Mauritania
Accredited entity:	United Nations Environmental Programme (UNEP)
Country:	Mauritania
Project size:	Small

I. Assessment of the independent Technical Advisory Panel

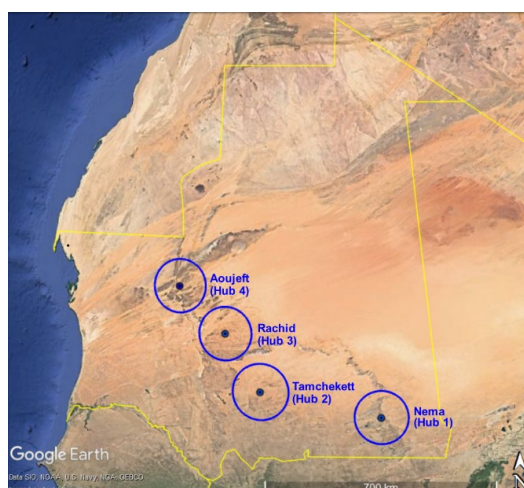
1.1 Overview

1. This funding proposal¹ for a small-sized adaptation project in environmental and social safeguards category B is submitted by the Government of Mauritania (GoM) through the United Nations Environment Programme (UNEP) as the proposed project's accredited entity (AE). The executing entity (EE) is the Ministry of Environment and Sustainable Development (MEDD) of Mauritania. The total cost of the funding proposal is USD 33.56 million, of which USD 30.90 million is proposed as GCF finance and USD 2.66 million as in-kind co-finance from the GoM. The project implementation period is six years, while benefits are calculated over a lifespan of 20 years.

2. The proposed project's objective is to enhance the resilience of communities and ecosystems against climate change-induced desertification in the area of the Sahara-Sahel boundary within Mauritania. The project will focus on four rural-urban hubs (see the figure): Aoujeft (in the Adrar wilayah, or region), Rachid (in the Tagant wilayah), Tamchekett (in the Hodh El Gharbi wilayah) and Nema (in the Hodh Chargui wilayah). The project is designed to help vulnerable communities and their settlements and livelihoods to adapt to climate change-intensified desert encroachment, sandstorms, heat, drought and flood, with a particular focus on combating sand inundation through the construction of "ecological infrastructure".

Figure 1: Satellite image showing locations of four project hubs (Source: figure 3 in annex 2, feasibility study)

¹ This assessment is based on the full and final funding proposal package received by the independent Technical Advisory Panel on 17 April 2025.



3. The project will involve implementation of the components of work shown in the table.

Component	Outputs	GCF funding (USD)	Co-finance (USD)
Component 1: Governance and capacity building	<u>Output 1.1.</u> Governance structures are strengthened, sustainable financial mechanisms are established and policy briefs are prepared to support the integration of climate change considerations and ecosystem-based adaptation into plans, policies and budgets	2,149,000	9,600
	<u>Output 1.2.</u> Knowledge products are developed and disseminated to support decision-making and upscaling	393,000	24,000
Component 2: Strengthening ecosystem resilience and climate-adaptive livelihoods	<u>Output 2.1.</u> Grey-green dune fixation infrastructure is established to control sand encroachment, enhance the provision of ecosystem services and slow the rate of desertification within the four target hubs	8,453,000	2,119,000
	<u>Output 2.2.</u> Access to water is improved for agricultural and land rehabilitation activities	6,685,000	152,800
	<u>Output 2.3.</u> Climate-resilient agricultural livelihoods based on sustainable land use and natural resource use are developed and/or strengthened to reduce land degradation and support climate-resilient income generation by community members within the target regions	10,522,000	142,000
Monitoring and evaluation component		1,444,000	-

1.2 Impact potential

Scale: N/A

4. The project targets all four of the GCF results areas for adaptation, aiming to enhance the resilience of the most vulnerable people and communities; health and well-being, and food and water security; infrastructure and the built environment; and ecosystems and ecosystem services.

5. **Climate change risk and impacts.** Observations of historical climate trends show that the mean annual temperature in Mauritania has increased by 0.3 °C per decade over the past 70 years – from 26.10 °C in 1950 to 28.46 °C in 2020². In addition to shifting temperatures, climate change in Mauritania has led to more frequent extreme rainfall events over short time periods (with associated flash floods, and a decline in the kind of low-intensity, long-duration rainfall that infiltrates into the ground), increased rainfall irregularity and more prolonged dry spells.³ Precipitation has decreased during the wet season and over the entire year in Rachid, Tamchekett and Nema. In contrast, the Aoujeft hub has experienced a small but statistically significant increase in its overall precipitation. All four hubs have seen an increase in extreme daily rainfall, which often results in flash floods. The observed potential evapotranspiration (ETP) values are high in all four hubs (1,700–2,000 mm/year), with negative differences between precipitation and ETP, indicating that water loss via ETP exceeds water gained via precipitation. Sand movements are driven by the prevalence of strong winds from the north of Mauritania, including both the north-eastern trade winds and the *harmattan* – a dry wind strongest from late November to mid-March. Together, these winds carry large amounts of dust and sand across the country.

6. In future, all four target hubs are projected to experience increases in both minimum and maximum temperatures under future Representative Concentration Pathway (RCP) 4.5 (intermediate) and RCP 8.5 (high) greenhouse gas emissions scenarios, with frequency of heatwaves predicted to increase across all four hubs by 2050 under both scenarios.⁴ The projected evolution of precipitation trends varies between the hubs. No significant change in precipitation is projected for Aoujeft by 2050 under either scenario, but annual precipitation is expected to significantly decrease in Rachid, Tamchekett and Nema, relative to the reference period (1981–2010), under both scenarios. In line with increasing temperature trends and decreasing precipitation trends, potential ETP is projected to increase across all four hubs by 2050 under both scenarios, and the water balance deficit in all four hubs is also projected to increase significantly ($p < 0.1$) by 2050.⁵

7. The combination of projected temperature rise, ETP and declining rainfall will increase the extent and severity of droughts at the national and local levels. By 2040, the median number of consecutive dry days for Mauritania is predicted to be between 219 and 230, compared to the historical value of 216 days in 1995. The Standardised Precipitation Evapotranspiration Index (SPEI)⁶ displays a clear trend towards worsening drought conditions, with values below zero by 2040.⁷ By 2100, the ninetieth-percentile values are expected to fall below –2, signifying an increase in extremely dry years. In its analysis, the AE has applied an assumption that wind conditions will remain constant under future climate scenarios in all four hubs. The combination of constant windy conditions and a generally warmer and drier climate is likely to facilitate the southward expansion of the Sahara into the Sahelian region of the country, further impacting the communities of the project's focal hubs.

8. Prolonged droughts across the country have also accelerated vegetation loss in desert environments, reducing ecosystem services related to soil stabilization and water infiltration. As a result, the rate of dune advancement along the Sahara-Sahel boundary has increased and greater sand encroachment has been observed in oases, with subsequent impacts on water resources, agricultural productivity and income generation across northern Mauritania. Arable land, rangeland and pastures are being inundated, as is infrastructure such as roads, schools

² Unless stated otherwise, all sources for statistics are shown in the funding proposal.

³ See table 32 in annex 2 of the funding proposal for observed precipitation trends in the four focal hubs.

⁴ See annex 2 of the funding proposal, appendix 2: "Trend analysis report for Mauritania".

⁵ See annex 2 of the funding proposal, appendix 4: "Climate change risk assessment for Mauritania".

⁶ The SPEI is an indicator of drought severity that accounts for precipitation and potential evapotranspiration. Values of +2 or higher indicate extremely wet years, while values –2 or less indicate extremely dry years.

⁷ Sixth Assessment Report of the Intergovernmental Panel on Climate Change.

and hospitals, with sand sometimes covering entire villages. Siltation of dams, boreholes, wells and rivers has reduced the accessibility and availability of already scarce water resources in the region. This is combined with reduced rainfall and water infiltration resulting from vegetation loss increasing the depth of the water table. This has led to many pastoral wells drying up, impacting pastoralists' abilities to maintain nomadic lifestyles. Consequently, many nomadic pastoralists are adopting sedentary lifestyles around rural oases, increasing the pressure on limited natural resources, and heightening the risk of conflicts over resources.

9. **Adaptation impact.** The adaptation impact potential of the project is considerable, focusing on enhancing the resilience of communities living in and around oases in the vulnerable interface between the arid Sahel zone and the hyperarid Sahara Desert to drought, flood and desert encroachment. Through Output 2.1, the exposure of urban and rural communities in the target hubs to sand inundation, escalated by climate change, will be reduced. This will be achieved through enhanced ecosystem services generated by physical dune fixation using a wattling technique to create a stabilized grid, fenced in to allow full establishment, and then mulching and vegetating these dunes with grasses, shrubs and trees to create permanent buffers against sand. These measures will reduce the vulnerability of water sources, agricultural production and human respiratory health to sand encroachment and dust storms. A set of grey-green adaptation measures will be used to reduce vulnerability to flood and drought, installing (a) groundwater dams, to improve aquifer recharge; (b) gabions, dykes and stone bunds, to reduce flood risk; and (c) reservoirs, boreholes, well points and rainwater-harvesting systems, to increase rainwater capture and storage. This will be complemented through Output 2.2, which will provide (a) training and inputs for climate-resilient crop agriculture and natural resource-based livelihoods; (b) training on climate-resilient and sustainable livestock management practices; (c) communal boreholes equipped with solar pumps in *graras*⁸ and along historical transhumance routes, as well as rainwater-harvesting tanks; and (d) support to community-managed water user groups in the form of commune-level (see paragraph 10 below) water management plans.

10. The project is expected to directly benefit 85,229 people (40,877 men and 44,352 women) living within 12 priority communes across the four target hubs, who will directly experience improved protection against sand inundation, enhanced access to markets and income-generating opportunities, and increased access to clean water. Additionally, the AE estimates that there will be about 145,008 indirect beneficiaries (66,519 men and 78,489 women), representing individuals in surrounding or non-priority communes who will experience knowledge-sharing benefits, improved coordination at the regional level, and access to the small grants facility for scaling climate change adaptation (CCA) interventions (around 3 per cent of the country's total population and 12 per cent of the target wilayahs). The project aims to rehabilitate 2,123 hectares of degraded land, enhance water security through improved infrastructure and promote climate-resilient agriculture. This comprehensive approach will help to increase the resilience of the entire population in the four target hubs, benefiting approximately 230,237 people in total.

11. **Risks to new ecological infrastructure.** In an exchange of written questions and responses,⁹ the independent Technical Advisory Panel (iTAP) asked the AE to comment on two questions. The first related to the short-term prospects of the success of revegetation efforts in the face of the high winds, extreme temperatures and erratic rainfall likely to be experienced while the new ecological infrastructure is being established, as well as the potential scarcity of water for irrigation in the early stages. The second related to the long-term prospects of success of the artificial dune structures and green belts in preventing inundation of settlements.

⁸ According to the feasibility study, *graras* are low-lying basins where water collects during rainfall events.

⁹ Written questions from the iTAP with International Fund for Agricultural Development (IFAD) responses, 11 April 2025.

12. In response to the question on the shorter-term risks to successful impact, the AE stated that “planting of greenbelts will be informed by best practices and lessons learnt from previous greenbelt initiatives in Mauritania, including several GEF [Global Environment Facility] projects and the FAO working paper (*Fighting sand encroachment: lessons from Mauritania*¹⁰). This will include using native species adapted to low soil moisture and with deep taproot systems...and implementing rainwater harvesting infrastructure to supply additional water to greenbelts. Greenbelt trees will be...grown to more survivable sizes in nurseries and planted immediately following rains during the rainy season...Additionally, tree mortality during establishment is anticipated, and dead trees will be restocked.” UNEP highlighted that proximity to a water source was one criterion for dune stabilization site selection and that the project will establish boreholes for this purpose where necessary, as well as promoting measures for aquifer recharge. “The initial watering phase will employ gravel-filled bottomless cylinders to direct water to the roots and minimise evaporation, and the surface layer immediately surrounding the plant will be covered in mulch or other water-retaining products to further increase moisture retention.”¹¹

13. On the longer-term question, the AE stated: “Assuming that trees and perennial vegetation are successfully established by the end of the 6-year implementation period, the prognosis for the ecological infrastructure over the 20-year project lifespan is generally positive. If favourable ecological, institutional and social conditions are maintained, restored dunes can remain stable and functionally intact for decades. Furthermore, the areas will be protected by law under the Forest Code, as areas revegetated by the State.” In addition to outlining lessons learned from similar initiatives in the Sahel¹² about the need for water availability and effective protection of new infrastructure from grazing, the AE highlighted the need for careful species selection – for example, non-native *Prosopis juliflora* is one of the species with the best chance of surviving harsh desert conditions, but if poorly managed can become invasive.

14. **Use of SWAT model.** The iTAP notes that the AE used a soil and water assessment tool (SWAT+) for hydrological modelling to evaluate the potential impacts of water resource management strategies by comparing a business-as-usual scenario with a scenario with interventions (involving installation of physical structures to manage and increase water infiltration). The analysis focused on identifying spatial patterns of run-off generation, infiltration potential and water balance components, to generate for each potential site the expected difference between scenarios, indicating where investment would be most effective.

15. Chapter 7 of the feasibility study (annex 2) outlines the SWAT+ model set-up (inputs, baseline and intervention runs) but fails to include calibration of the model and its validation against measured streamflow or observed flood events. In response to a question about this, the AE acknowledged this gap, which was a result of limited availability of continuous and reliable discharge records at appropriate spatial resolution across the study domain. As a result, the AE relied on spatial differentiation in run-off and evaporation regimes as proxies for identifying hydrologically responsive sub-basins suitable for engineered or ecosystem-based interventions. Because it was not possible to calibrate and validate the SWAT+ model against observed flows, its outputs were not used for site selection; instead, the AE conducted a multi-criteria risk assessment, which was then validated with stakeholders.

16. The iTAP assesses the impact potential of the project as high.

¹⁰ *Fighting Sand Encroachment: Lessons from Mauritania*. Available at www.fao.org/3/i1488e/i1488e.pdf.

¹¹ Further details may be found in the funding proposal. See section 5 of annex 2 (“Feasibility study”) and annex 21 (“Operations manual”).

¹² Including tree lines established in Tamchekett under a previous Adaptation Fund project.

1.3 Paradigm shift potential

Scale: N/A

17. The proposed project aspires to have “four paradigm-shifting pathways, as identified in the GCF sectoral guidance, aimed at enhancing the climate resilience of ecosystems, and the agricultural and water sectors. These pathways include: i) transformational planning and programming; ii) catalysing climate innovation; iii) mobilising finance at scale; and iv) establishing coalitions and enhancing knowledge to scale up successful outcomes.”¹³ However, these approaches will be applied at sub-national level, in the communes of the four target hubs and the four wilayahs within which they fall. Activity 1.1.1 (“Establish coordination platforms to facilitate knowledge-sharing”) is limited to wilayah-, *moughataa*-¹⁴ and commune-level development plans, policies and budgets in the target hubs. Subactivity 1.2.2.3 does cover the dissemination of knowledge products in non-target communes across the four project wilayahs, but does not go beyond this to the country’s other eight wilayahs or the national policy level.

18. **Knowledge-sharing and learning.** Numerous project studies and assessments will be conducted within the target hubs to improve the country’s climate change knowledge base, including assessments of value chains, surface and groundwater quality and quantity, and community vulnerability (disaggregated by gender). All the assessments will contribute to a CCA knowledge management platform, which will be used as a centralized repository accessible to stakeholders across different administrative levels. Best practices on CCA will be collated, tailored and incorporated into knowledge products, informing the design and implementation of subsequent adaptation interventions. This approach to knowledge management is intended to support both vertical scaling (enhancing the impact within the target hubs) and horizontal scaling (supporting CCA in other hubs, countries or regions) of successful adaptation measures. These knowledge products will also be integrated into the country’s national adaptation plan (NAP) knowledge management platform, which manages knowledge related to desertification and climate change in Mauritania.

19. **Enabling environment.** At commune level, processes will be put in place to achieve a paradigm shift in the way resources are managed – including establishing commune-level technical committees and water user groups, strengthening local institutional and regulatory frameworks for monitoring water use and availability, training farmer organizations (including women) on climate-smart agriculture, and establishing a small grants facility to recycle some of the proceeds from sustainable livelihood activities back into adaptation beyond the project implementation period. This new approach is not reflected at the national level, however. In the project’s theory of change, it is suggested that climate change resilience will be enhanced “if supported by transformational change in policies, planning, knowledge management and capacity-building”, but the project focus only at the local and regional levels limits such change.

20. The AE acknowledges that the project “takes a bottom-up approach to achieving transformational change and the integration of EbA [ecosystem-based adaptation] approaches into national and sub-national policies” and “lays the foundation for the integration of EbA approaches into wilayah-, *moughataa*- and commune-level development plans”.¹⁵ This is to be achieved through the development and dissemination of knowledge products, including policy briefs, but there are no planned points of formal engagement with national government and other regional government agencies that would allow for long-term changes to policy and financing frameworks for Mauritania as a whole, particularly for other oases and other areas vulnerable to sand encroachment. This seems a missed opportunity for the GCF project to contribute to transformational change by fostering long-term policy reforms in natural resource

¹³ Section D.2 of the funding proposal.

¹⁴ Departments (second-level administrative subdivisions).

¹⁵ Written questions from the iTAP with IFAD responses, 11 April 2025.

management that include EbA approaches, at national and subnational levels across the country, enabling replication and scaling.

21. Based on the above, the paradigm shift potential of the project is assessed as medium.

1.4 Sustainable development potential

Scale: N/A

22. The proposed project is aligned with the United Nations 2030 Agenda for Sustainable Development, contributing to a wide range of the Sustainable Development Goals (SDGs), namely: SDG 2 – Zero hunger; SDG 3 – Good health and well-being; SDG 5 – Gender equality; SDG 6 – Clean water and sanitation; SDG 8 – Decent work and economic growth; SDG 10 – Reduced inequalities; SDG 11 – Sustainable cities and communities; SDG 12 – Responsible consumption and production; SDG 13 – Climate action; SDG 15 – Life on land; and SDG 16 – Peace, justice and strong institutions. Specific co-benefits anticipated from project interventions are discussed below.

23. **Environmental co-benefits.** Capturing rainwater and allowing infiltration through on-the-ground water-harvesting structures, and better rangeland management, may allow for natural regeneration of seedbanks and root systems in degraded grazing lands, enabling native species to flourish again. With mostly native species to be used for revegetation, this will also have benefits for biodiversity. Dune stabilization and revegetation will reduce erosion of productive soils and land degradation, and will enhance soil organic carbon, as above- and below-ground biomass increase.

24. **Socioeconomic benefits.** Support of climate-resilient agriculture, forest products and agriprocessing is expected to improve households' nutritional status, reducing dependence on imports and, in some cases, food aid. Dune stabilization efforts should reduce the prevalence of respiratory diseases (such as lung cancer) and respiratory infection resulting from exposure to airborne dust, decreasing healthcare costs for households and government. Improved access to clean drinking water in the target hubs should lower the number of epidemics linked to waterborne diseases. Overall, there should be reduced losses from damage to property, infrastructure and crops, as well as increased lifespan and capacity of water infrastructure (such as weirs, gabions and wells). Improving access to water may have positive social effects, including reduced conflict over water resources and reduced risk of gender-based violence. It may also free up women's time, as women will no longer have to travel far from their households to collect water, and improve school attendance for girls, who frequently leave school to fulfil the task of collecting water.

25. **Gender and social inclusion.** The gender assessment and action plan (annex 8 to the funding proposal) highlights that although women in the rural areas of Mauritania are involved in subsistence farming, food processing and livestock management, they have limited access to land, financial resources, training and agricultural inputs. Less than 10 per cent of women have access to formal agricultural extension services, even though up to 50 per cent of households are headed by women in many areas. This limits women's capacity to improve productivity and resilience to climate impacts. Section 1.13 of annex 8 also highlights the ongoing challenges of social inclusion in a country with a complex past characterized by enslavement of parts of the population. The assessment notes that "women are responsible for a considerable amount of household work, including agricultural production, post-harvest activities, collecting fuelwood and water, food preparation and childcare" and acknowledges the need to avoid additional burdens on women through new climate-smart agricultural practices, sustainable livelihoods and water management techniques.

26. The proposed project interventions will address gender equality through several pathways. One will be to incorporate a gender action plan into new development plans and community climate action plans, based on the specific gaps identified within each community.

Another will be to integrate women into the project-specific institutional structures and involve them in decision-making. Women will be included in all CCA training, ensuring equal opportunity and access for on-the-ground adaptation interventions. Women will form at least 50% of the beneficiaries of technical assistance on new climate-resilient livelihoods, while awareness is maintained of the need to balance such new tasks by freeing up other time – for example, by reducing how far women need to walk to collect water (see paragraph 24 above).

27. **Risk of overextraction of groundwater.** The project's lengthy and detailed feasibility study (annex 2) notes that groundwater extraction in oases is hindered by low levels of aquifer recharge, with a continually falling water table; that enforcement of regulations to control water extraction is limited; that overuse adversely affects marginalized groups, particularly women; and that baseline assessments conducted during project development highlight the risk of groundwater depletion when the use of solar-powered pumps is not strictly monitored. The funding proposal includes hydrogeological studies and maps in subactivity 2.2.1.3 as the basis for the appropriate siting of four new boreholes to be drilled per community, and the iTAP welcomes these measures. But there is no discussion of ongoing water-level monitoring and community governance of pumping in order to avoid unsustainable extraction, which is nowhere treated as a project risk. The iTAP believes this is an oversight. It recommends that this issue be included in the environmental and social management plan and that a subactivity be added to establish a governance system to determine and enforce pumping times and volumes, to monitor changes in the water table, and to take action to further restrict pumping if water levels fall beneath a predetermined threshold.

28. The iTAP assesses the sustainable development potential of the project as medium to high.

1.5 Needs of the recipient

Scale: N/A

29. Mauritania is classified as a least developed country, with a gross domestic product (GDP) of about USD 7.9 billion and a GDP per capita of about USD 1,700.¹⁶ As a result of high poverty rates, particularly within rural areas, the population is heavily dependent on agriculture for subsistence, with most households' capacities to adopt new climate-resilient livelihoods being limited. The country's Human Capital Index score of 38, which ranks it 150th out of 157 countries, indicates that a child born in Mauritania can only expect to attain 38 per cent of maximum potential human capital by the age of 18, even if they receive complete health and education, with the country still in the "low human development" stage despite significant progress over the past 30 years. In 2019, Mauritania was ranked the country twentieth most vulnerable to climate change and other global challenges, according to the Notre Dame Global Adaptation Initiative country index. With the highest global sensitivity score (0.602), Mauritania faces substantial insecurity in both the food sector and the water sector.

30. Between 1900 and 2018, natural hazards such as droughts, disease epidemics, insect infestations and storms (strong winds and heavy rainfall) impacted more than 11.5 million people in Mauritania, according to the Emergency Events Database. In 2019, an estimated 33,600 Mauritania were impacted by floods, while in 2020 approximately 609,000 people were affected by droughts. Flooding alone caused an estimated USD 5.5–11.4 billion in damages and losses between 2000 and 2008. Under future climate change scenarios, Mauritania is expected to face an overall decrease in water availability of 10–15 per cent; an increase in land degradation and erosion; continued loss of arable land through southward expansion of the Sahara Desert; a decrease in rangeland availability for livestock; and a reduction in food availability.

¹⁶ As mentioned previously, sources for all statistics are shown by the AE in the funding proposal.

31. The country's high public and external debt – estimated at 79 per cent of GDP and 68 per cent of GDP, respectively – constrain the ability of the GoM to allocate funding from the national budget for CCA. Moreover, limited opportunities exist for public and private sector stakeholders to access financial resources for adaptation, which is expected to cost around USD 9.3 billion between 2015 and 2030. Given that this cost is likely to increase considerably under future climate change conditions, external investment is crucial to enhance the resilience of the population of Mauritania to the impacts of climate change – which are projected to intensify in the future – on food and water security, economic growth, and human health. The project will effectively respond to the most pressing challenges Mauritania is facing, addressing the country's high vulnerability to climate change impacts and its challenges related to desertification, water scarcity and economic development.

32. The vulnerability of the population of Mauritania is high because of elevated levels of poverty and a strong dependence on climate change-sensitive sectors, including agriculture, fisheries, mining and livestock. In addition, access to basic services and facilities is unequal between rural and urban areas, and those attempting to transition out of poverty encounter major obstacles. As a result, approximately 6 per cent of the population of Mauritania live below the international poverty line (USD 1.9 per day in 2011 with purchasing power parity) and about 28 per cent live below the national poverty line. Moreover, 70 per cent of the country's impoverished households live in rural areas and are mainly engaged in agriculture and livestock-rearing. This high prevalence of poverty and limited access to basic resources means that rural communities living within the four target hubs lack the adaptive and financial capacity to invest in climate-resilient sustainable development and are almost entirely dependent on external support from both government and development partners.

33. Overall, based on the above, the needs of the recipients are rated as high.

1.6 Country ownership

Scale: N/A

34. The proposed project has been designed following extensive consultation with the GoM and a range of stakeholders, and is aligned with the policies of Mauritania. The roles of the AE and EE, as well as those of technical partners who will support implementation, are clearly set out in the funding proposal.

35. **Policy and strategy alignment.** The proposed project aligns with the Mauritania national spatial planning strategy, which aims to strengthen resilience using an oasis-centred administrative structure to prevent rural exodus and enhance resilience to the impacts of climate change-induced desertification. The project is well aligned with the national adaptation priorities of Mauritania, as expressed in the country's nationally determined contribution (NDC) to the Paris Agreement, updated in 2021. The project directly addresses five of the objectives of the NDC, namely: (a) restoring natural grasslands; (b) conducting hydrological assessments and water resource availability analyses; (c) strengthening ecosystems' resilience against climate change; (d) enhancing vulnerable populations' resilience to climate change, particularly in rural areas; and (e) strengthening institutional and technical capacities at local, regional and national levels to enable more effective planning, financing and implementation of adaptation measures.

36. The project's objectives are also consistent with the cross-sectoral approach set out in the country's 2004 National Adaptation Programme of Action (NAPA), which includes the following priority adaptation measures: (a) enhancement of knowledge of surface water's hydrological cycle; (b) construction of flooding breakdown dykes in pluvial and oasis zones; (c) promotion of water-saving techniques in oasis zones; (d) participatory reforestation for agroforestry in agricultural zones; and (e) reorganization and capacity-building in communities adversely affected by climate change. The project is complementary to the country's ongoing GCF-supported NAP development process, initiated in 2018. As such, it will build on efforts to enhance the capacity of policymakers, GoM staff and vulnerable groups, and will conduct

awareness-raising initiatives with local authorities in each wilayah as well as with civil society organizations and local communities.

37. **Engagement with stakeholders, including civil society.** As the AE, UNEP has facilitated dialogue between national consultants, the MEDD, other relevant institutions and community stakeholders, ensuring that the project responds to the adaptation priorities of Mauritania. Consultations have been held with GoM representatives, national consultants, public authorities, civil societies, local community members and the private sector. The EE, the MEDD (which is also the Mauritania national designated authority for GCF), has been actively involved in the project's development since inception by contributing to the consultation process and providing inputs during field missions. UNEP organized community consultations through four field missions (each covering all or most of the Aoujeft, Rachid, Tamchekett and Nema hubs¹⁷) over a two-year period, consulting 330 local community members across the target hubs.

38. **Roles of key entities.** As the AE, UNEP will be responsible for oversight of the project's implementation, monitoring and evaluation – in coordination with the project steering committee (representing a range of government and civil society stakeholders) and a project management unit embedded in the MEDD. UNEP has a broad global portfolio of climate change projects funded via GCF, the Adaptation Fund, the Least Developed Countries Fund, the Special Climate Change Fund, and bilateral funds such as the European Commission. UNEP also has a long-standing relationship with Mauritania, having supported the country in the development of the NAPA, the GCF NAP Readiness and Preparatory Support Programme and several adaptation projects including two GEF-funded projects.

39. The proposed project will be executed by the MEDD. This Ministry has the mandate to develop and implement the National Sustainable Development Plan and the National Environmental Action Plan, and has created sector-specific plans and programmes addressing desertification and climate change, biodiversity, and wetland conservation. The MEDD hosts a coordination cell for the National Climate Change Programme; this cell handles reporting to the United Nations Framework Convention on Climate Change and coordinates and implements climate change initiatives. Since no mention is made in the funding proposal of the MEDD having executed a project of this nature previously, or of a specific capacity assessment in this regard, it is welcomed that the exit strategy (section B.6) mentions that “the project will work closely with the designated government entity to build its capacity in fund management, monitoring and evaluation, as well as in the strategic allocation of resources towards high-impact EbA interventions. This collaboration will include the development of clear guidelines for fund disbursement, criteria for project selection, and mechanisms for tracking and reporting on the impact of funded projects.”

40. A local non-governmental organization in Mauritania, Association Naforé,¹⁸ has been identified as an eligible implementing partner to assist the MEDD to implement on-the-ground interventions on ecosystem restoration, livelihood and water infrastructure activities (Outputs 2.1 and 2.2) within the proposed project's four target hubs. Since its inception in 2009, Naforé has implemented several projects in the domains of environmental protection, health, hygiene and sanitation, showcasing its capacity through diverse projects and a wide range of expertise among its members.

41. Country ownership of the project is thus seen to be medium to high.

1.7 Efficiency and effectiveness

Scale: N/A

¹⁷ See details in annex 7 to the funding proposal, “Stakeholder consultations and engagement plan”.

¹⁸ Association Naforé: *A propos de nous*. Available at <https://naforeorg.wordpress.com>.

42. The GoM is requesting USD 30,902,000 in grant finance from GCF, to be supported by in-kind co-financing of USD 2,657,000 from the MEDD and regional authorities. This co-financing will be used to cover baseline costs, including the operations and maintenance necessary to ensure ongoing sustainable land and water resource management. The GCF funds form 92 per cent of the total project costs, while the co-finance forms 8 per cent. This co-finance figure is low but reflects severe budgetary constraints faced by the GoM. The request for 100 per cent concessionality is justified in section B.5 of the funding proposal, in relation to the high level of government indebtedness and the public goods nature of the ecological infrastructure investment at the heart of the project.

43. Beyond the direct co-financing, the proposed project is strategically positioned to benefit from parallel financing totalling around USD 41.3 million, drawn from complementary initiatives within the target regions. These include the Integrated Management of Protected Areas in the Arid Regions of Mauritania (IMPADRA) project (USD 20 million), the Sustainable Management of Natural Resources (PROGRES) project (USD 6.6 million), and the Africa Integrated Climate Risk Management Programme (AICRMP) (USD 14.7 million).¹⁹ The combined investment will be used to support local, regional and national institutions to improve access to climate-risk information for strategic land-use planning and natural resource management, and continued investment in ecosystem restoration, climate-resilient livelihoods and sustainable natural resource management, reducing pressure from the impacts of climate change-induced desertification.

44. The feasibility study (annex 2) provides a thorough analysis of recent and current related initiatives, lessons learned and proposed synergies. Although the project will support commune-level coordination platforms, it is not clear how national-level coordination with the many related initiatives will be undertaken to maximize effectiveness. As noted in the feasibility study, the GoM is actively engaged in the Great Green Wall (GGW) effort through its National Agency for the Great Green Wall, and there are significant opportunities for synergies and exchange between this proposed project and ongoing and planned GGW-related initiatives in Mauritania. A coordination mechanism for all these related projects, through the MEDD coordination cell, is recommended.

45. **Return on investment.** Both an economic analysis and a financial analysis (see annex 3 to the funding proposal) were conducted to inform the development of the project. The economic analysis determined the costs and benefits associated with implementing EbA interventions, including dune fixation and land rehabilitation activities; facilitating sustainable natural resource management practices and the adoption of new climate-resilient livelihoods by developing value chains; and supporting the GoM in developing local and regional frameworks for mainstreaming climate change and gender-responsive adaptation solutions into plans and policies. The economic analysis estimated the net present value (NPV) of the net benefits (economic benefits less project costs) to be USD 152 million (discounted at a social discount rate of 2 per cent over 20 years), with an economic internal rate of return (IRR) of 29 per cent and a benefit-to-cost ratio of 5.86 (USD 5.86 of benefits generated per USD 1 spent), reflecting a cost-effective investment for GCF and the GoM. A financial analysis of income potential through the adoption of climate-smart agriculture, climate-resilient livestock management and new climate-resilient livelihoods was carried out using the best available data and research. This generated an NPV of USD 31.16 million with a financial IRR on these activities of 22 per cent.

46. The Secretariat's assessment of the funding proposal package indicates that the adaptation cost per direct beneficiary, estimated at USD 363 (with USD 134 per total beneficiary), is "above the GCF portfolio averages for Africa, which are usually under USD 200 per direct beneficiary. While the cost per beneficiary is relatively high, this is explained by the project's focus on foundational EbA measures, and the inclusion of public goods such as water

¹⁹ See table 50 in annex 2 ("Feasibility study") for a detailed list of parallel finance opportunities.

access and land restoration. The economic and financial analysis demonstrates strong value for money by linking these cost levels to substantial economic benefits.”

47. **Financial sustainability.** With regard to financial sustainability post-project, the AE’s exit strategy (section B.6 of the funding proposal) indicates that several financing opportunities at the government and community levels will enable the continued operation and maintenance of interventions beyond the project’s lifespan: “At the government level, the prioritisation of CCA, gender considerations and natural resource management into policies, plans and strategies will promote the allocation of sufficient funds to CCA initiatives through national, regional and local government budgets.” This seems optimistic, since the AE has simultaneously noted in section B.5 that “the high level of public debt (99% of GDP in 2016) constrains public financing for CCA, making it difficult for the GoM to secure financing for adaptation projects” and that “further investments to address climate change are almost entirely reliant on assistance from external multilateral development partners, including the GCF”.

48. **Sustainability of small grants.** According to the funding proposal, a critical component of achieving financial sustainability will be the project’s extension of an existing MEDD-based small grants facility (SGF) under Activity 2.3.2, designed to provide financial stability and facilitate continued investment in upscaling successful EbA activities and sustainable livelihoods. In response to a question from iTAP about how realistic it would be for the GoM to replenish this fund, the AE stated that “a modest level of replenishment of the SGF funding from government would be realistic to expect. However, additional funding sources will also be sought to support the continuation of the SGF after the project ends, including through linkages with other financial mechanisms and instruments. This will be in line with the long-term financing strategy for the SGF, to be developed under project sub-activity 2.3.2.2.”²⁰ This area will need more attention during project implementation, in order to ensure that the fund is replenished, ideally with a long-term sustainable revenue source.

49. The project’s overall efficiency and effectiveness is rated as medium.

II. Overall remarks from the independent Technical Advisory Panel

50. The iTAP believes that the project on “Strengthening the resilience of ecosystems and populations in four regional hubs in northern Mauritania” applies a best practice ecosystem-based adaptation approach to building the resilience of oasis communities threatened by the impacts of climate change-accelerated drought, flood and desert encroachment upon their lives and livelihoods.

51. In order to minimize any risk of maladaptation, given the harsh conditions in which the project is operating and uncertainty around how climate hazards will evolve, the iTAP recommends that the AE undertake the following before project inception:

- (a) Create a coordination mechanism to share lessons being learned and adaptive management practices being undertaken in response to evolving climate hazards – both in the three projects with parallel co-finance (IMPADRA, PROGRES and AICRMP) and in the other projects of the Great Green Wall being financed by GCF and other technical and financial partners of the Government of Mauritania; and
- (b) Include in the environmental and social management plan the risk of potential overextraction of groundwater through new boreholes with solar photovoltaic systems, with an added project subactivity to establish a governance system to determine and enforce pumping times and volumes, to monitor changes in the water table, and to take

²⁰ Written questions from the iTAP with IFAD responses, 11 April 2025.



action to further restrict pumping if water levels fall beneath a predetermined threshold.

52. The iTAP recommends that the Board approve this funding proposal.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP266)

Proposal name:	Strengthening the resilience of ecosystems and populations in four regional hubs in northern Mauritania
Accredited entity:	United Nations Environmental Programme (UNEP)
Country:	Mauritania
Project size:	Small

Impact potential
We acknowledge iTAP's assessment of the project's impact potential as "high". We note that no specific recommendations are made in this regard.
Paradigm shift potential
We acknowledge iTAP's conclusion that the project's paradigm shift potential is "medium" owing to the project's focus being at the sub-national level, potentially reducing its ability to effect transformational change in national policies and planning. As noted in UNEP's written responses to iTAP comments, it is envisaged that by providing tangible bottom-up examples of EbA interventions and their integration into sub-national policies and plans, the project will create an enabling environment and catalyze their consideration also in national-level processes. The primary mechanism for this will be the NAP knowledge management platform, which is currently under development in Mauritania, where project reports and best practices guidance will be stored and dissemination. This centralised repository will enhance access to critical adaptation information for policymakers, practitioners and community-level stakeholders.
Sustainable development potential
We acknowledge iTAP's assessment of the project's sustainable development potential as "medium to high", and agree with iTAP's concern regarding the potential risk of unsustainable extraction of groundwater. Please see our response to the associated iTAP recommendation under the "overall remarks" section.
Needs of the recipient
We acknowledge iTAP's assessment of the needs of the recipient as "high". We note that no specific recommendations are made in this regard.
Country ownership
We acknowledge iTAP's assessment of country ownership as "medium to high". Regarding MEDD's experience in executing similar projects, we would like to clarify that MEDD has successfully executed a number of UNEP projects, including ones on EbA approaches in arid areas. A micro-HACT assessment of MEDD confirmed its capacity to act as the project Executing Entity. The risk mitigation plan measures to address identified gaps and

weaknesses will further strengthen MEDD's project execution capacity, as will the specific capacity building included in the project to strengthen MEDD's capacity for the design, execution and oversight of the Small Grants Facility.

Efficiency and effectiveness

We acknowledge iTAP's assessment of the project's efficiency and effectiveness as "medium", and agree on the need for a coordination mechanism for related initiatives. On this recommendation, please see our response under the "overall remarks" section.

Regarding the government's capacity to fund adaptation, although this currently low, with the increased priority allocated to addressing climate change and its improved integration in policies, plans and budgets, this is likely to improve. As for post-project sustainability, government investment in the maintenance of interventions is a realistic expectation in the Mauritanian context.

Finally, we agree with the importance of identifying long-term financing sources for replenishing the SGF, to ensure its continued operation beyond the project timeframe.

Overall remarks from the independent Technical Advisory Panel:

We have taken note of iTAP's recommendations, and commit to addressing them as follows:

Recommendation 1: Create a coordination mechanism to share lessons being learned and adaptive management practices being undertaken in response to evolving climate hazards

Response: Under the National Adaptation Plan (NAP) process, in which UNEP is a Delivery Partner, a multi-stakeholder coordination mechanism for climate change adaptation in Mauritania will be established. To ensure that coordination between initiatives isn't limited to the duration of this project, a working group under the NAP coordination platform will be established by UNEP and MEDD to this effect. This will be used to share lessons learnt between the parallel co-finance projects, other projects in the Great Green Wall financed by GCF, and other technical and financial partners of Government of Mauritania.

Recommendation 2: Include in the environmental and social management plan the risk of potential overextraction of groundwater (...), with an added project subactivity to establish a governance system (...)

Response: Before project inception, the ESMP will be revised to include risk mitigation activities focused on determining and enforcing pumping times and volumes, monitoring changes in water tables and restricting pumping if needed, as well as on the training of Water User Groups (WUGs) on these specific aspects. The description of the WUGs roles will be updated in the FP and Annex 2, to specifically highlight these responsibilities, and they will also be included in the TOR for WUGs. Budget for these additional mitigation measures (including for the necessary technical expertise) will be allocated through a budget revision.

Strengthening the resilience of ecosystems and populations in four regional hubs in Northern Mauritania

ANNEX 8

GENDER ASSESSMENT AND GENDER ACTION PLAN



Annex 8: Gender Action Plan

Acronyms and abbreviations

CEDAW	Convention on the Elimination of All Forms of Discrimination Against Women
EbA	Ecosystem-based adaptation
FGM	Female genital mutilation
GBV	Gender-based violence
PMU	Project Management Unit

1. Part 1: Gender Analysis

1.1. Introduction

The overall objective of the 'Strengthening the resilience of ecosystems and populations' project is to contribute to improved resilience of local and vulnerable communities in four regional hubs in northern Mauritania. To fulfil the project's design requirements, this gender analysis was undertaken in accordance with GCF guidelines to inform the project design on gender barriers, constraints and opportunities for women to benefit from the project's activities. In addition, the Gender Action Plan (GAP) will ensure women and other marginalised groups are included in the project's activities. The GAP is based on the proposed project's logical framework and is focused on avoiding and minimising the negative impacts of the project's activities on local communities and maximising the positive impacts and social benefits, particularly for women and marginalised groups.

1.2. Methodology

This gender analysis primarily used qualitative data collection methods, including informant interviews with multiple stakeholders and partners (See Annex 7: stakeholder engagement plan) as well as mixed community consultations. The method involving open community meetings proved particularly effective as it allowed for the provision of information by individuals through open discussions, accompanied by immediate expressions of approval or disapproval and corroborated by testimonies or cases from other group members. This facilitated an accurate representation of the gender-related challenges faced by their respective communities. The gender analysis has also made use of: i) a desktop review of available literature; ii) gender equality studies conducted by research institutions and donor agencies; and iii) details made available by the government of Mauritania. The questions posed to the community members focused on the lived experiences of individuals and communities regarding the effects of climate change and their gendered responses to these challenges. This data collection method was preceded and supplemented by a review of existing literature on gender dynamics and equality in Mauritania — particularly within the rural and agricultural sectors — to establish a framework aligned with the project's objectives and anticipated impact. The findings of the gender analysis (Part 1) form the basis of the GAP (part 2).

1.3. Women's rights in Mauritania

Mauritania has ratified the Convention on the Elimination of All Forms of Discrimination Against Women (CEDAW), and its legal system includes four areas of law that impact women differently: Islamic law; Customary African law; Customary Arabic-Bedouin and Berber law; and (modern) Civil law. Women's rights in marriage remain uncertain, particularly under Islamic and customary African law.

In 2005, Mauritania ratified the Protocol to the African Charter on Human and Peoples' Rights on the Rights of Women in Africa (Maputo Protocol)¹, pledging to promote women's rights and eliminate violence against them. However, at the national level, no comprehensive law addressing violence against women exists to date. Specific laws address particular forms of violence, such as rape and female genital mutilation, but offer no legal protection or recourse for women experiencing other types violence, like spousal abuse and sexual harassment. The absence of a comprehensive legal framework

¹ United Nations. 2003. Protocol to the African Charter on Human and Peoples' Rights on the Rights of Women in Africa (The Maputo Protocol). Available at: <https://www.un.org/shestandsforspeace/sites/www.un.org/shestandsforspeace/files/protocolontherightsofwomen.pdf>

to safeguard women and combat violence, coupled with cultural acceptance of harmful practices against women, has perpetuated violence against women in the country².

The high illiteracy rate among women often leaves them unaware of their rights, and the distribution of information concerning women's rights remains inadequate.

1.4. General situation of women in households

Globally, progress has been made regarding women's rights; however, substantial effort is needed to achieve gender equality in Mauritania. In 2021, the Gender Development Index for Mauritania was 0.890 which categorises Mauritania in group five, the lowest group that comprises countries with low equality in HDI achievements between women and men. In addition, the Gender Inequality Index was 0.632 which ranks Mauritania 161st out of 170 countries³.

The legal age for marriage is 18 for both women and men (Personal Code, Art. 6)⁴, with child marriage being prohibited and punishable by imprisonment and fines (Juvenile Justice Code, Art. 41)⁵. However, child marriage is widespread in the country, especially among rural, economically disadvantaged, and less educated girls who are married for their families' social and financial benefit. Nationally, there has been a trend of women marrying later. Some women in the age bracket 40-49 married at the average age of 16, whereas women in the 25-29-year age bracket married at 19. There is also an increase in age for the birth of the first child⁶. These changes are closely linked to the increase in the level of women's education. However, women of reproductive age (15-49 years) often face barriers with respect to their sexual and reproductive health and rights: from 2019-2021, only 29% of women had their family planning needs met with modern methods, and 37% of women aged 20-24 years were married or in a union before age 18⁷. The adolescent birth rate is 78 per 1,000 women aged 15-19 as of 2021⁸, up from 71 per 1,000 in 2011⁹.

Under the Personal Status Code, marriage is defined as a union between a man and a woman and requires the consent of both parties (Art. 1)¹⁰. However, consent is loosely defined, with "the silence of a young girl" considered as consent (Art. 9)¹¹. According to the law, a legal marriage comprises the bride and groom, the guardian or the *weli* of the bride (who serves to represent her in concluding the marriage contract), the dowry or the *sadaq*, and the consent of both parties (Art. 5)¹². Polygamy is legal, with men allowed up to four wives with the consent of existing spouses (Art. 28)¹³. Polygamy is rarely practiced among Moor communities but is prevalent among other ethnic groups in the country. Under customary law, which is not recognised by the state, a woman's first marriage requires parental consent, and Muslim women are expected to marry within their faith, whereas Muslim men do not face the same restriction. Cultural taboos against the mixing of social classes or inter-caste unions result in marriages among members within the same clan¹⁴. The husband is considered the head of the household

² Committee on the Elimination of Discrimination against Women, *Concluding observations on the combined second and third periodic reports of Mauritania*, 2014

³ UNDP. 2022. Human Development Report 2021/2022, Available at: https://hdr.undp.org/system/files/documents/global-report-document/hdr2021-22pdf_1.pdf

⁴ Government of Mauritania. 2001. Personal Status Code. Available at:

<https://www.ilo.org/dyn/natlex/docs/MONOGRAPH/96689/114337/F-2050531859/MRT-96689.pdf>

⁵ Government of Mauritania. 2001. Juvenile Justice Code. Available at: <https://features.hrw.org/features/african-union/files/Mauritania%20-%20Ordonnance%20n%202005-015%20portant%20protection%20pe%CC%81nale%20de%20%E2%80%99enfant.pdf>

⁶ Study on Demographics and Health in Mauritania (ESDM)

⁷ Republic of Mauritania. 2022. Demographic and Health Survey 2019-2021. Available at:

<https://www.dhsprogram.com/pubs/pdf/SR274/SR274.pdf>

⁸ World Bank. 2021. Adolescent birth rate. Available at:

<https://data.worldbank.org/indicator/SP.ADO.TFRT?locations=MR>

⁹ <https://data.unwomen.org/country/mauritania>

¹⁰ Government of Mauritania. 2001. Personal Status Code. Available at:

<https://www.ilo.org/dyn/natlex/docs/MONOGRAPH/96689/114337/F-2050531859/MRT-96689.pdf>

¹¹ Government of Mauritania. 2001. Personal Status Code. Available at:

<https://www.ilo.org/dyn/natlex/docs/MONOGRAPH/96689/114337/F-2050531859/MRT-96689.pdf>

¹² Government of Mauritania. 2001. Personal Status Code. Available at:

<https://www.ilo.org/dyn/natlex/docs/MONOGRAPH/96689/114337/F-2050531859/MRT-96689.pdf>

¹³ Government of Mauritania. 2001. Personal Status Code. Available at:

<https://www.ilo.org/dyn/natlex/docs/MONOGRAPH/96689/114337/F-2050531859/MRT-96689.pdf>

¹⁴ Emery, Robert, *Cultural Sociology of Divorce: An Encyclopaedia*, SAGE Publications, Virginia, 2013.

(Personal Status Code, Art. 1)¹⁵ and is the legal authority of the household who can make decisions on behalf of family without the wife's consent¹⁶.

Although women and men both have inheritance rights, women's entitlements differs considerable from men's. Under Sharia law, a woman inherits only half of what a man does (Personal Status Code, Art. 253-259)¹⁷. Accordingly, a daughter may inherit only half of the property entitled to her male sibling¹⁸. When a family has only daughters and no sons, the inheritance of the deceased father is first distributed among his brothers and other members of the family and the remainder to his daughters and his wife¹⁹. Women face additional challenges in retaining property (both land and non-land assets) in divorce and widowhood, where their right to ownership of assets is often limited to objects related to women's work and daily life such as cooking and cleaning utensils (Personal Code, Art. 73-74)²⁰. In some cases, the family of the deceased father is granted full custody of the child (rather than to the living mother), allowing his family sole rights over the child's inheritance.

1.5. *Division of labour*

Rural women play an essential role in agriculture. Although only 25.4% of women in the country are classified as employed, many more participate in unpaid work in the informal sector and agriculture. Women are responsible for vital tasks related to food provision and household maintenance. They are involved in all phases of the agricultural cycle, from planting and weeding to the harvesting, conservation and maintenance of fields, storage, processing and marketing of produce. Women play an important role in market gardening and in milk product processing. In the fishery sector, women also contribute to the processing and marketing of fish and fish products. Women participate in almost all aspects of agricultural and livestock activities, while men primarily focus on land clearing and cultivating specific crops such as wetland rice. Men also take care of the herding and watering of large animals, while the curing of skins, the processing of milk products, and the care of small ruminants and poultry is the task of women. In addition, women undertake household tasks such as food processing and preparation, fetching water and collecting wood. Women are also the practitioners of traditional medicine. Although women in certain areas were traditionally responsible for rice cultivation, wetland rice cultivation is now predominantly handled by men, with women mainly focusing on irrigation, storage and marketing. In some areas, both men and women participate in the construction of small dams. Major decisions on land transfer and agricultural investments are generally made by the men in the extended family. While women generally do not own land or livestock, they usually make decisions regarding their work responsibilities and the income, derived from their activities.

1.6. *Impact of Climate Change*

As stated in Mauritania's National Adaptation Programme of Action to Climate Change (2004)²¹ and, more recently, in the country's first Nationally Determined Contribution (2021)²², climate change impacts men and women differently, with women often disproportionately affected by adverse impacts. For example, the increasing frequency of drought in Mauritania results in women having to walk longer distances for water and firewood or develop new income-generating activities, such as weaving and tie-dyeing. Men and women have different access to financial, social, or political resources and generally hold distinct roles and responsibilities within households. Women tend to balance domestic tasks such as cooking, cleaning, and childrearing in addition to agricultural responsibilities. It is common for women to maintain kitchen gardens for household consumption, while men maintain cash crops. Finally, because of gender inequalities with regards to financial assets, women tend to be responsible for poultry

¹⁵ Government of Mauritania. 2001. Personal Status Code. Available at:

<https://www.ilo.org/dyn/natlex/docs/MONOGRAPH/96689/114337/F-2050531859/MRT-96689.pdf>

¹⁶ OECD. 2019. Gender, Institutions and Development Database. Available at: <https://www.genderindex.org/wp-content/uploads/files/datasheets/2019/MR.pdf>

¹⁷ Government of Mauritania. 2001. Personal Status Code. Available at:

<https://www.ilo.org/dyn/natlex/docs/MONOGRAPH/96689/114337/F-2050531859/MRT-96689.pdf>

¹⁸ Committee on the Elimination of Discrimination against Women, *Concluding observations on the combined second and third periodic reports of Mauritania*, 2014

¹⁹ Anwar, Zainah, *A Global Report on Equality in the Muslim Family: Mauritania*, 2010

²⁰ Government of Mauritania. 2001. Personal Status Code. Available at:

<https://www.ilo.org/dyn/natlex/docs/MONOGRAPH/96689/114337/F-2050531859/MRT-96689.pdf>

²¹ Republic of Mauritania. 2004. National Adaptation Programme of Action to Climate Change. Available at: <https://unfccc.int/resource/docs/napa/mau01e.pdf>

²² MEDD. 2021. Contribution Determinee au Niveau National (CDN): Mauritania. Available at:

https://unfccc.int/sites/default/files/NDC/2022-06/CDN-actualis%C3%A9%202021_%20Mauritania.pdf

and small ruminants, whereas men care for larger and more financially advantageous livestock such as cattle. Consequently, men and women tend to experience the effects of climate change differently.

The West African Sahel has experienced periods of long-term decline in rainfall that was intensified by shorter periods of severe drought. In addition to the high human toll, this series of droughts contributed to a decrease in species diversification, tree density and plant coverage throughout the region. Climate variability challenges dependence on natural resources for sustenance, particularly when alternative livelihood options are limited in rural areas. Insufficient water sources force women and girls to travel farther for water, exposing them to potential harm, violence and the social repercussions resulting from sexual assault. The health concerns that arise from carrying heavy water containers over long distances include dehydration, back and spinal injuries, fatigue, dehydration, and reproductive complications. This extra work also detracts from time that women would otherwise spend on productive activities, which sometimes results in decreased agricultural productivity, household food security, and overall household income. This problem is termed 'time poverty' and highlights the ways women experience less free time available than men as they are often involved in very time-consuming non-remunerated household work on top of outside work to supplement their household income. This means that they may have less flexibility to diversify their activities far from the home or participate in projects designed to improve their resilience. To compensate for this time allocation problem with regards to water collection, some households assign girl children the responsibility of collecting water, forcing them to leave school and contribute to household labour, thereby perpetuating social inequity. In addition, women are often excluded as climate change resources prioritise information dissemination methods accommodate men's abilities and needs. However, emerging literature on women's adaptation strategies, such as new technologies and climate-smart agriculture practices and collective action and information sharing strategies, emphasises the importance of recognising women's knowledge and preferred ways of learning across diverse contexts. Development agencies have started incorporating these considerations into their programme initiatives²³.

1.7. Access to land

Land rights are a particular concern in Mauritania, as few women own land. In the traditional economy, women, youth, and the servant class do not possess land. Decisions on and-use are generally determined by men and the elderly. Land governance in Mauritania represents a major component of sustainable development and poverty reduction. Implementing effective land policies is crucial for reducing poverty by stimulating economic prosperity and fostering self-sufficiency. Viable and inclusive land governance stimulates the consolidation of social peace and the inclusion of women and the poor in an open discussion about access to land security. Although many Mauritanian women desire land ownership, only 2,146 of 27,000 title deeds registered at the national level, or slightly less than 8%, are held by women (Table 1). Despite women's limited access to permits and land titles, authorities have not yet adopted an affirmative action policy that promotes increasing women's access to land. Investigations conducted with the help of officials and an analysis of the Land Tenure Office database showed that most of the data had not yet been disaggregated by gender. However, in 2015, the Land Tenure Office made efforts to allow data disaggregation by identifying names by gender, notably creating a database that includes gender as a key variable²⁴.

²³ McOmber, C. (2020). "Women and Climate Change in the Sahel", West African Papers, No. 27, OECD Publishing, Paris

²⁴ World Bank. 2015. Women's access to land in Mauritania: A case study in preparation for the COP. Available at: <https://documents1.worldbank.org/curated/pt/224621467991907919/pdf/100049-WP-PUBLIC-Box3932168-Women-s-Access-to-Land-in-Mauritania.pdf>

Table 1: A breakdown of access to land ownership in Mauritania, disaggregated by gender.

Titles	Total	Disaggregated by Gender		Percentage by Gender	
Permis d'Occuper (PO; occupying permit)	38,574	Women	3,864	Women	10%
		Men	34,710	Men	90%
Official Deeds	27,075	Women	2,146	Women	7.9%
		Men	24,876	Men	91.9%
Urban	27,003	Women	2,143	Women	7.9%
		Men	24,860	Men	92.1%
Rural	72	Women	3	Women	4.2%
		Men	16	Men	22.2%

Source: Directorate of Land and State Assets

1.8. Access to and control over resources

In Mauritania, considerable gender disparities exist in men's and women's access to and control over resources, largely as a result of entrenched socio-cultural norms and legal frameworks. Women in rural areas have restricted access to land, water and credit, despite legal provisions recognising their rights. For example, less than 15% of women in Mauritania own land, and even fewer have formal land titles, which limits their ability to access agricultural inputs and loans²⁵. This gender gap is compounded by traditional customs that prioritise male inheritance of land and resources, further marginalising women's participation in agricultural productivity and economic decision-making²⁶.

A case study from the southern region of Mauritania highlights these challenges, where men largely dominate decision-making processes related to natural resources, including water management and grazing rights. This control affects women's ability to engage in productive activities such as farming or livestock rearing, which are necessary for household income and food security. However, gender-sensitive initiatives, such as the World Bank-funded Community-Based Watershed Management Project²⁷, have demonstrated some progress in empowering women by improving their access to water for irrigation and livestock²⁸.

1.9. Gender and agricultural livelihoods

Agriculture in Mauritania is characterised by gender-specific roles, with men and women participating in different tasks within the sector. Women in rural areas are involved in subsistence farming, food processing and livestock management, all of which contribute to household food security²⁹. However, they face obstacles in fully participating in and benefiting from the agricultural economy. Women make up nearly 43% of the agricultural labour force, yet they have limited access to land, financial resources, training and agricultural inputs. Less than 10% of women have access to formal agricultural extension services, which limits their capacity to improve productivity and resilience to climate impacts. These

²⁵ World Bank. 2021. Gender Equality: A Necessary Condition for Promoting Inclusive Growth in Mauritania Available at: <https://www.worldbank.org/en/news/press-release/2021/06/01/gender-equality-a-necessary-condition-for-promoting-inclusive-growth-in-mauritania>

²⁶ World Bank. 2015. Women's access to land in Mauritania: a case study in preparation for the COP. Available at: <https://documents.worldbank.org/en/publication/documents-reports/documentdetail/224621467991907919/women-s-access-to-land-in-mauritania-a-case-study-in-preparation-for-the-cop>

²⁷ World Bank. 2005. Community Based Watershed Management Project. Available at: <https://projects.worldbank.org/en/projects-operations/project-detail/P087670>

²⁸ World Bank. 2021. Gender Equality: A Necessary Condition for Promoting Inclusive Growth in Mauritania Available at: <https://www.worldbank.org/en/news/press-release/2021/06/01/gender-equality-a-necessary-condition-for-promoting-inclusive-growth-in-mauritania>

²⁹ FAO. n/d. Role of women in agriculture. Available at: <https://www.fao.org/4/v9324e/v9324e01.htm>

disparities create a gap in the agricultural sector, reducing women's opportunities in commercial farming and agribusiness³⁰.

Instances in the Brakna region demonstrates the challenges and opportunities for women in agriculture. Women's cooperatives engaged in rice production have benefited from increased access to irrigation and farming technologies, resulting in higher yields and income³¹. However, limited access to land tenure and markets remains a considerable challenge for these women. Further interventions are required to ensure equitable access to resources, training, and decision-making in the agricultural sector, which is necessary for improving food security and reducing poverty in Mauritania.

1.10. *Participation of women in politics*

Under the Constitution, women and men have equal rights to hold public office. The government has instituted several measures to ensure and increase the participation of women in the public and political space. One such measure is the electoral law introduced in 2006 that requires a minimum of 20% of all municipal council seats to be reserved for women. In 2012, the country reformed the electoral law to add 51 new seats to the National Assembly, of which 20 are reserved for women. In 2013, the government introduced financial incentives for political parties that nominate more women than required by the quota for national assembly seats. As of February 2021, only 20.3% of seats in parliament were held by women.

In 2015, women members of parliament (MPs) in Mauritania created an action plan to advance gender equality. This plan followed a training session in Nouakchott on strategic planning on the work of the women's caucus in parliament involving approximately 40 female MPs, civil society organisations and government officials. The workshop, jointly organised by the parliamentary women's group REFPAM, IPU and the National Assembly, included training on frameworks for protecting women's rights. Priority areas for 2016 include promoting new laws to combat violence against women and girls, and strengthening female MPs through study visits, training activities and other projects.

Despite these measures, women remain underrepresented in public service, particularly in ministerial, judiciary and foreign service roles. Limited access to education and employment, the lack of financial support for female political candidates, and the traditional stereotypes of women's role in society compared to men contribute to the low level of female political participation.³² Women from Haratine communities, in particular, have limited or no access to public and political life³³.

1.11. *Access to justice*

The justice system in Mauritania consists of courts that apply the principles of both civil and Sharia law. Although the Personal Status Code outlines provisions for the application of these two legal systems, judicial officials do not always abide by it. Under Sharia law, a woman's testimony holds half of the evidentiary weight of that of a man. Moreover, in cases of rape, judges can accuse the victim of fornication outside marriage (*zina*), and theoretically, imprison the victim on these grounds³⁴.

Regarding legal proceedings, many courts conduct their sessions in Arabic, and interpreters are not always available. In addition, there is inadequate legal aid and counselling available to women, a lack of training among legal and judicial officers on women's rights and gender sensitivity, and a reliance on conciliators (*mouslihines*) to settle cases outside of the formal justice system³⁵. Consequently, certain

³⁰ World Bank. 2015. Women's access to land in Mauritania: a case study in preparation for the COP. Available at: <https://documents.worldbank.org/en/publication/documents-reports/documentdetail/224621467991907919/women-s-access-to-land-in-mauritania-a-case-study-in-preparation-for-the-cop>

³¹ World Bank. 2015. Women's access to land in Mauritania: a case study in preparation for the COP. Available at: <https://documents.worldbank.org/en/publication/documents-reports/documentdetail/224621467991907919/women-s-access-to-land-in-mauritania-a-case-study-in-preparation-for-the-cop>

³² Mauritanian For Women's Rights, *Mauritanienne Pour les Droits de la Femme (MPDF) Parallel Report to the Government's Report*, 2014.

³³ Anti - Slavery International and Minority Rights Group International, SOS Slaves, *Observations on the second and third reports periodic submitted by MAURITANIA, Committee on the Elimination of Discrimination against women 58th Session*, 2014.

³⁴ Human Rights Council Working Group, Universal Periodic Review Twenty-third session 2-13 - Mauritania, 2015.

³⁵ Committee on the Elimination of Discrimination against Women, *Concluding observations on the combined second and third periodic reports of Mauritania*, 2014

crimes against women, including rape and domestic violence, have been largely unreported due to the stigmatisation of victims, the potential for imprisonment, and the low conviction rate of offenders³⁶. Wealthy perpetrators often avoid prosecution, or if prosecuted, avoid prison by settling outside of formal courts.

1.12. Gender-based violence (GBV)

Although there are no specific laws criminalising domestic violence in Mauritania, the government has instituted measures to prevent it. The government has developed a National Action Plan on Gender Based Violence (2014–2018) and a National Strategy for the Promotion of Women (2015–2019), both of which address violence against women. In addition, the government has organised community-based awareness-raising campaigns in cooperation with civil society and established family dispute units in all regional capitals of the country. However, in the absence of a specific law, women rarely seek legal redress for domestic violence, and there is limited police intervention in reported cases. Instead, women rely on family members, community leaders and local NGOs to resolve domestic violence cases³⁷.

Domestic violence is a pervasive problem throughout the country³⁸. Under the Penal Code, rape is a criminal offense, with perpetrators subject to penalties ranging from forced labour to death penalty (Art. 309–310)³⁹. However, the code fails to provide an explicit definition of rape, allowing for subjective interpretation of the law and the omission of marital rape⁴⁰. In practice, rape cases often go unreported in the country due to the social stigma attached to rape victims. Moreover, if the rape victim is married, a judge may, under Sharia law, accuse the woman of adultery or sexual intercourse outside of marriage (*zina*), and if found guilty the woman may face imprisonment. To address this issue, the government has enacted various awareness-raising campaigns to encourage women victims of rape to file a formal complaint and provided in-service training for judges and law officers to ensure a clear distinction between rape victims and those who have committed *zina*. Nevertheless, impunity for rape continues to be a problem, with girls held in slavery often being victims of sexual violence and assault, and rape offenders often avoid prosecution through monetary settlements outside of court.

To date, there is no law prohibiting sexual harassment, which occurs often, especially in the workplace and schools, where girls are reportedly at risk of sexual harassment and abuse by teachers. Regarding female genital mutilation (FGM), there is no specific legislation prohibiting this practice; however, the government has instituted efforts, including legal measures to combat it. Under the Juvenile Justice Code, FGM is prohibited by law and is punishable by imprisonment as well as fines ranging from 120,000 to 300,000 ouguiyas (Art. 12)⁴¹. The Penal Code also bans child mutilation (Art. 328)⁴². Moreover, the government developed a National Strategy against Female Genital Mutilation in 2008 and a draft law criminalising FGM in 2010. In 2011, religious leaders in the country issued an opinion on Islamic Law (*fatwa*) forbidding FGM. At the local level, the government continues to work with international organisations and local NGOs to conduct awareness and outreach campaigns to shift cultural attitudes surrounding the practice. However, authorities seldom apply the laws pertaining to FGM, and there has been no progress on the approval of the draft bill criminalising FGM. FGM remains widely practiced in Mauritania, where it is prevalent among all ethnic groups to some degree and performed on young girls, sometimes as early as a week after birth. The practice is deeply rooted in cultural tradition where it is seen as a rite of passage or religious requirement, providing social acceptance or higher social recognition of the girl and her family, and reducing female sexual desire⁴³.

³⁶ Committee on the Elimination of Discrimination against Women, *Concluding observations on the combined second and third periodic reports of Mauritania*, 2014

³⁷ United States Department of State, *2016 Country Reports on Human Rights Practices - Mauritania*, 2017.

³⁸ United States Department of State, *2016 Country Reports on Human Rights Practices - Mauritania*, 2017.

³⁹ Government of Mauritania. 2001. Penal Code. Available at:

<file:///C:/Users/mccon/Downloads/Mauritania%20Penal%20Code%201983.pdf>

⁴⁰ Committee on the Elimination of Discrimination against Women, *Concluding observations on the combined second and third periodic reports of Mauritania*, 2014

⁴¹ Government of Mauritania. 2001. Juvenile Justice Code. Available at: <https://features.hrw.org/features/african-union/files/Mauritania%20-%20Ordonnance%20n%202005-015%20portant%20protection%20pe%CC%81nale%20de%20l%E2%80%99enfant.pdf>

⁴² Government of Mauritania. 2001. Penal Code. Available at:

<file:///C:/Users/mccon/Downloads/Mauritania%20Penal%20Code%201983.pdf>

⁴³ Ouldzeidoune, Nacerdine, Keating, Joseph, Bertrand, Jane, Rice, Janet, *A Description of Female Genital Mutilation and Force-Feeding Practices in Mauritania: Implications for the Protection of Child Rights and Health*, 2013.

1.13. Gender and Minorities

Minority groups in Mauritania include sub-Saharan ethnic groups such as Peulh and Toucouleur, Soninké, and Wolof, which together make up 25% of the population, and Haratine who constitute 45% of the population. It is difficult to provide transparent data on the ethnic composition of the population since the Mauritanian government systematically refuses to disaggregate the data in terms of ethnicity. In addition, the government considers that Haratines are part of the wider Moorish society, forming part of the majority (Haratines and Beydans share the same language and cultural practices, although it is the result of decades of enslavement and assimilation). The Haratine are an ethnic minority found in Mauritania, whose population is essentially composed of Moors (or Beidanes) who live in the northern part of the country. The term Haratine refers to slaves (freed slaves) and persons of slave descent. Haratine are descendants of Black ethnic groups that were captured by the majority white Maures or Berber-Arabs and have been discriminated against and have lived under slavery for years. Slavery, and in particular slavery of Haratine women, underpinned by deep-seated discrimination as well as marginalisation, remains a common practice in Mauritania regardless of its prohibition under the law. Mauritanian adults and children continue to be sold and bought as slaves, children are enslaved by their parents' masters, and women often do not choose their marital partners. Some Haratine are also dispossessed of their ancestral property by their masters or masters of their parents upon their parents' deaths.⁴⁴

Although Arabic is the official language, some minorities speak Hassaniyya – a dialect of Arabic (83%), Peulh or Pulaar and Toucouleur (5 %), Soninké (1%), and Wolof (0.3%). All minority groups in Mauritania share the same religion, Sunni Islam.

2. Part 2: Gender and Social Inclusion Action Plan

The following Gender Action Plan has been developed on the basis of consultations led and is in line with the GCF Updated Gender Policy and Gender Action Plan 2020–2023 (GCF/B.24/15).

2.1. Project Management

Project Management Unit (PMU) staff will be required to take the UN course on gender and environment⁴⁵ and the UN Women course on introduction to gender found in the UN Women training Centre portal⁴⁶. Participation in the Introduction to Gender course will be mandatory for all PMU staff as part of their onboarding and capacity-building process. In addition to this mandatory course, staff will be strongly encouraged to enrol in at least one other relevant course offered through the portal to further enhance their understanding of gender, sex, and gender-mainstreaming.

To ensure compliance, completion of the mandatory courses will be monitored and documented as part of the staff performance and development plan. Certificates of completion will be required for inclusion in individual performance reviews. As an additional incentive, participation in supplementary courses will be factored into professional development assessments and opportunities for recognition or career advancement within the project.

The outcomes of this training will be measured through ongoing monitoring and evaluation (M&E) processes. PMU staff will be expected to demonstrate an applied understanding of gender mainstreaming, which will be reflected in project activities, reports, and decision-making processes. Progress will be assessed using gender statistics and qualitative analysis, beyond just gender-disaggregated indicators, and these metrics will be integrated into the project's logical framework for regular tracking and evaluation.

Throughout the project, members of the PMU will regularly engage with the project Gender Specialist — an expert with extensive experience in gender mainstreaming — to ensure compliance with gender-mainstreaming requirements. Women-based organisations with appropriate mandates and capacities will be carefully selected and engaged in project implementation to facilitate greater women engagement in consultation processes. The project will engage both men and women proportionally while being cautious not to increase their workloads unequally (as women typically also carry household

⁴⁴ <https://minorityrights.org/law-and-legal-cases/mauritania-challenging-slavery-of-haratine-women/>

⁴⁵ <https://www.unclearn.org/courses/open-online-course-on-gender-and-environment/>

⁴⁶ <https://portal.trainingcentre.unwomen.org/unw-catalog-mobile/>

responsibilities that may not be easily handed over to men). In addition, care will be taken to respond to the fact that men and women may have different educational levels and interests.

The project will promote the quality of participation and the actual impact of women on decision-making processes. The project will ensure that participation moves beyond nominal membership and provides women access to decision-making spaces and processes alongside men, but also the ability to actively influence and lead those processes. This will be done through trainings for both women and men on women leadership and the importance of their participation, in order to help advocate for the inclusion and participation of women in decision-making platforms of the community.

2.2. *Project Monitoring*

Monitoring and evaluation methods have been designed to be participatory, and to engage with women in the process. Gender-response data collection will enable the progress measurement, gender issues assessment, and making appropriate adjustments where needed, constituting gender statistics. This data will be shared with project teams and communities in a timely and understandable manner.

Gender indicators highlight differences and inequalities in all aspects of the lives of women and men⁴⁷. By revealing the gender equalities and gender gaps in key areas, gender inequalities are made visible, which can in turn inform policy-making to address identified gender gaps. The indicators help identify data that is relevant to women's and men's lives and to key areas of policy-making. The project will generate a gender-sensitive database on socio-economic characteristics to monitor gender advancement throughout the project.

It is important to distinguish between sex-disaggregated data and gender statistics. While sex-disaggregated data is separated by sex, gender statistics considers broader gender inequalities and bias in data collection methods and tools. In addition, gender statistics have the potential to reflect different groups of women and men, taking into consideration that gender intersects with age, education, family composition and parenthood, country of birth and disability. This means that gender statistics can reflect a deeper understanding of women's situations and needs, and thereby support the analysis of intersecting inequalities.

For indicators to be gender-sensitive, they must encompass the following characteristics:

- data are collected and presented in disaggregated form by sex as a primary and overall classification;
- data reflect gender issues;
- data adequately reflect the diversity of women and men and capture all aspects of their lives; and
- data-collection methods consider stereotypes and socio-economic and cultural factors that may induce gender biases.

Two data collection methods will be used for gender statistics and indicators. Quantitative methods provide quantifiable results focusing on countable issues such as percentages of women and men in the labour market, male and female wage rates, or school enrolment rates for girls and boys. These statistics are relevant to the project as they are important for assessing capacity needs for women wanting to take up sustainable, alternative livelihoods. These statistics are relevant to the project as they are important for assessing capacity needs for women wanting to take up sustainable, alternative livelihoods. Quantitative data can also show the magnitude of changes in gender equality over time — for example, the percentage of women married before the age of 15 or the gender pay gap over time. Qualitative methods capture people's experiences, opinions, attitudes, and feelings — for example, women's experiences the constraints or advantages of working in the informal sector, or men's and women's views on the causes and consequences of underrepresentation of women in senior positions in the economy or in politics. This data will inform the development of guidelines and policies for the membership structure of coordination committees established under the proposed project — for example, commune-level coordination and technical committees (Activity 1.1.1) and water user groups (Activity 2.2.2). Additionally, gathering information about women's experiences in the informal sector will inform the design of training workshops focused on the uptake of alternative livelihoods in the target region. Participatory methods, such as focus group discussions and social mapping tools, are often

⁴⁷ United Nations, Integrating a gender perspective into statistics, New York, 2006, cited in United Nations Statistical Division. 2016, p. 1. <https://unstats.un.org/unsd/demographic-social/Standards-and-Methods/files/Handbooks/gender/Integrating-a-Gender-Perspective-into-Statistics-E.pdf>

used for qualitative indicators. Qualitative data can also be collected through in-depth surveys measuring perceptions and opinions.

2.3. Gender considerations in the project design

2.3.1. Approach and problem analysis

If the project is not adequately informed on the specific roles, responsibilities, and opportunities assigned to men and women in Mauritania, it may inadvertently exacerbate ‘time poverty’ among women. ‘Time poverty’ refers to the unpaid household and care duties that women perform instead of engaging in economically lucrative activities. Mauritanian women are responsible for a considerable amount of household work, including agricultural production, post-harvest activities, collecting fuelwood and water, food preparation and childcare — both productive and reproductive in nature. By incorporating climate-smart agricultural practices, sustainable livelihoods and water management techniques into the project, women could modify and even increase their use of time.

Evaluating the gender barriers prevalent across the project’s four hubs in Mauritania and considering gender-specific concerns in agriculture, land, and natural resource use helps identify the primary issues tied to gender. Table 22 provides a problem analysis of the main gender challenges and the subsequent sections describe how these are mainstreamed into the project design.

Table 22. Project-related gender challenges in Mauritania.

Main gender challenges	Limited property rights and access to land ownership	Gender-blind policies and non-inclusive social and cultural norms related to land and natural resource use	Gender gaps in education and skills; limited access to information and knowledge	Access to finance and labour force participation
Causes	Women’s access to land is limited, particularly in rural areas as governance structures are generally weaker. Customary law reinforces this limitation	Limited representation of women in decision-making processes related to land, agriculture and water.	Women’s limited literacy rates result in their exclusion from awareness-raising campaigns and information dissemination from knowledge management platforms. This limits their ability to make climate-smart decisions.	Restricted participation in certain economic activities and greater demand on their time to undertake time-consuming and non-remunerating household work (travelling long distances to collect firewood and water)
Immediate effects	Land degradation and compromised agricultural yield	Limited entitlements to assets outside of those related to their household work	Limited adaptive capacity	Women have limited time to engage in income generating activities and diversify their livelihoods
Effects	Reduced productivity and income generated from natural resources	Limited access to resources reduces their coping capacity in the face of climate change	Individual and household resilience to climate change is reduced	Women have less opportunity to earn an income, are economically disadvantaged and subsequently have limited resilience and adaptive capacity to extreme climatic events

Climate change affects men and women differently, with cultural practices making women more vulnerable than men. Mainstreaming gender into the ‘Strengthening the resilience of ecosystems and populations’ project design – including conceptualisation, implementation, monitoring and evaluation –

is therefore necessary to build the adaptive capacity of households and communities in the four hubs of northern Mauritania. The results from this gender assessment will guide the design of project activities, ensuring that women, the youth, elderly and other vulnerable groups are not only included in the project, but that the relationships between the cultural roles of men and women are considered. The proposed project interventions will, therefore, address gender equality through several pathways, including:

- incorporating a gender action plan into the development of the development plans and community climate action plans, based on the specific gaps identified within each community (see **Error! Reference source not found.**);
- integrating women into the project specific institutional structures and involving them in the project's decision-making process;
- ensuring women's participation in all climate change adaptation training;
- ensuring equal opportunity and access to women farmers for on-the-ground adaptation interventions;
- targeting women for specific adaptation interventions, with a particular focus on climate-resilient additional livelihoods;
- raising awareness of the importance of equal opportunity for all people and the role women play in climate change adaptation across a community;
- establishing long-term monitoring and evaluation programmes to assess the effectiveness of interventions for achieving gender equality; and
- sharing lessons learned through project interventions to help create a paradigm shift in the approach to gender equality in climate change adaptation projects.

2.3.2. Gender mainstreaming in the project design

Gender mainstreaming ensures visibility and support for the participation and benefit of both women and men, with the main objective of achieving gender equality. Considering gender challenges in this project is necessary given the distinct roles men and women hold within a community, their differing access to and control over natural resources, and their unique needs. It is important to consider gender-specific challenges when designing ecosystem-based adaptation (EbA) approaches for sustainable natural resource management and strengthening ecosystem and community resilience to climate change impacts while enhancing ecosystem service provision. To ensure that the project employs a gender-response approach — such that men and women are given equal opportunities for capacity-building, improved access to finance and strengthened climate resilience — the specific adaptation needs of men and women, and the roles they might play in implementing EbA interventions, have been carefully considered in the project development process.

It is also essential to avoid exposing women to greater risks of sexual exploitation and harassment (SEAH), GBV and intimate partner violence as a result of their empowerment through project activities that increase their income accessibility. Project interventions should be sensitive to these dynamics, ensuring that women can exercise their agency and empowerment, while also engaging male community members in women empowerment.

The gender assessment and problem analysis identified gaps in access and control over resources and opportunities. Mainstreaming gender in the project design is achieved by using sex-disaggregated data, facilitating awareness raising on grievance redress mechanisms (GRM) for sexual exploitation and harassment (SEAH) and gender-based violence (GBV), developing strategies and actions to close those gaps, allocating resources and expertise for implementing equality strategies, and monitoring results using gender-sensitive indicators.

Project component description

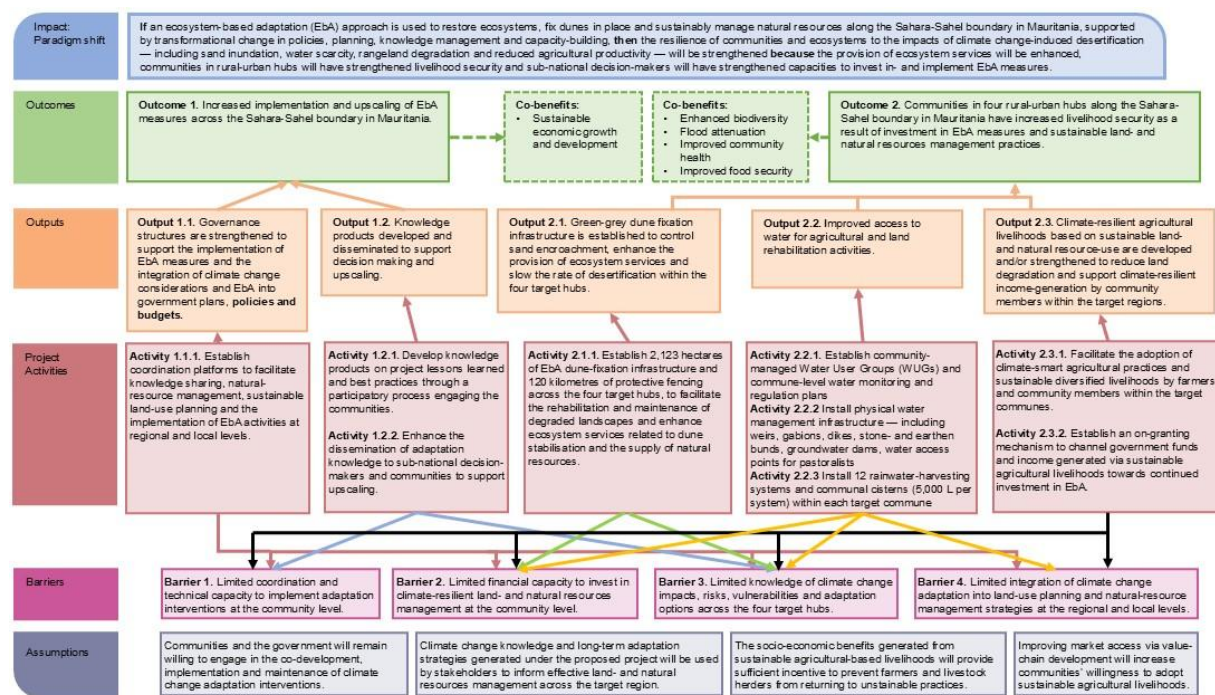


Figure 1: A diagram of the Theory of Change detailing expected Outcomes, Outputs and Activities.

Activities under Output 1.1 (Figure 1) will be participatory, ensuring that while strengthening governance structures, women's inclusion in decision-making processes and their placement in positions of authority will allow them to participate in sustainable land and natural resource management. Mainstreaming gender considerations in land and natural resource use activities can be done by integrating gender-responsiveness into various legislative documents. Under Output 1.1, Activity 1.1.1 involves reviewing existing plans, policies and budgets at the *wilayah*-, *moughataa*- and commune-level (Figure 1), thereafter, having policy briefs prepared for the integration of gender-responsiveness into these various documents. These policy briefs will also be presented at a workshop with relevant government ministry and department representatives. Additionally, Output 1.1 ensures that staff who provide training workshops, such as extension and technical staff, also receive gender training, to avoid following and reinforcing socio-cultural norms that could negatively impact women's participation in the project activities. Moreover, Output 1.1 will make sure that the training workshops carried out under Activity 1.1.1 will include modules on gender roles and mainstreaming gender in sustainable natural resource management and strategic land-use planning. Under Activity 1.2.2, gender-responsive best practices will be captured in the knowledge products generated and disseminated by the project. In addition,

Gender mainstreaming will consider capacity-building activities provided under Output 2.1, Output 2.2 and Output 2.3, including involving women in identifying degraded zones for rehabilitation and ensuring that this process is sensitive to women's rights to land. This will also include integrating women's perspectives into land tenure security policies to ensure equitable access to rehabilitated land. Capacity-building activities proposed by the project will also increase women's access to inputs and resources delivered under the project — for example, by increasing access to agricultural inputs and improved agricultural technologies, including drip irrigation kits, integrated pest management strategies, solar-powered pumps and tools and training for the establishment of half-moons and zai pits. Specific targets for improved access to inputs and resources for women are detailed in Sub-section 2.3.3. In addition, specific targets for women's participation in training and technology adoption will be incorporated to track gender-responsive outcomes.

In addition to increasing women's access to and use of improved agricultural technologies (stated under Output 2.3), project agricultural support will prioritise climate-resilient crop agriculture and will focus on increasing their productivity to enhance sustainable agricultural livelihoods. Women will actively participate in the sub-activity interventions listed under Activity 2.3.1, namely: i) active and ongoing participation in the establishment of nurseries and seed banks in the target communes; ii) assist in the collection and creation of stock material for these nurseries and seed banks; and iii) engage in

sustainable livelihood practices, such as market-gardening, apiculture, poultry farming, livestock feed production and collecting and selling non-timber products.

Moreover, under Activity 2.2.1, it is necessary to include women in the Water User Groups (WUGs) of each target commune so that they are involved in all water resource management sub-activities. During the identification of sites for water management infrastructure and rainwater-harvesting systems, women will provide input and guidance given their local knowledge and experience having to access water sources. This will ensure that the identified sites will also be appropriately situated according to women's views and needs — addressing the long distances women and women children typically travel to access water.

Project interventions will include raising awareness and developing skills for extension officers to advocate for gender equality in all aspects of training and agricultural support activities. Both men and women extension officers will be equipped with participatory tools and skills to manage a variety of cultural settings and power relations. The project recognises the intersectional systems of advantage and accessibility in the target locations, which could result in women's exclusion from advisory and decision-making processes and spaces. Women's needs, concerns, and expertise frequently become marginalised during the planning and execution of adaptation interventions. To address this inequality, the project will actively involve women in activities, ensuring that their needs, insights, and interests are represented, while creating gender-sensitive communication materials in terms of content, design, and dissemination.

2.3.3. Project expected benefits promoting gender equality

The project will provide tangible benefits for women and promote a shift in gender dynamics by adopting a gender-sensitive and responsive strategy. Involving women in various decision-making processes is necessary for ensuring that they benefit from the project. The gender action plan outlines the specific steps and activities to ensure women's participation in decision-making processes and project activities. By building their capacity, skills, knowledge, and access to resources, they will be better equipped to make decisions at both household and community levels. This impact is anticipated to continue beyond the project's timeline.

Women depend on natural resources to fulfil their numerous economic and domestic responsibilities. Both men and women engage in agricultural activities; by ensuring women have access to agricultural training, resources, climate-smart crops and water-efficient irrigation investments, their resilience to present and future climate change impacts will be strengthened. Moreover, by targeting 30%–50% of women in the project's alternative livelihoods activities and enabling their full participation, their economic empowerment and adaptive capacity will be enhanced. Land use planning in the target hub communities and climate-resilient agriculture related interventions are expected to lead to better-managed natural resources which will substantially reduce women's work burden as less time will be spent to collect these resources. Additionally, for women in the four hubs, improved environmental management — including water source conservation and the establishment of water management infrastructure at strategic sites within target communes, to ensure water is stored closer to the household (Activity 2.2.2) — will improve access to water for domestic and productive purposes, easing the load on women responsible for fetching water. The exposure to the threat of GBV or other forms of violence will also be reduced as they will no longer be compelled to travel far from their households in search of natural resources such as water and wood.

Project activities	Gender action	Baseline, indicators and targets	Timeline	Responsible entity	Budget
Impact Statement: If an ecosystem-based adaptation (EbA) approach is used to restore ecosystems, fix dunes in place and sustainably manage natural resources along the Sahara-Sahel boundary in Mauritania, supported by transformational change in policies, planning, knowledge management and capacity-building, then the resilience of communities and ecosystems to the impacts of climate change-induced desertification — including sand inundation, water scarcity, rangeland degradation and reduced agricultural productivity — will be strengthened because the provision of ecosystem services will be enhanced, communities in rural-urban hubs will have strengthened livelihood security and sub-national decision-makers will have strengthened capacities to invest in- and implement EbA measures. Research and anecdotal evidence indicate that women, children and youth are among the groups most vulnerable to the impacts of climate change. In addition, a large number of vulnerable people in communities that are highly dependent on local natural resources for their survival are women and children. Therefore, the Government of the Islamic Republic of Mauritania (GoM) has established that gender-responsive planning, resource-allocation and implementation in the context of this proposed project will be a priority. Supporting transformational change both in the lives of women beneficiaries and in prompting gender mainstreaming in national climate change policy-making and implementation was therefore seen as a priority in stakeholder consultations. Expected total numbers of direct and indirect beneficiaries (reduced vulnerability or increased resilience) are: • 85,229 people directly benefit from project interventions (40,877 men and 44,352 women); and • 145,008 people indirectly benefit from the project interventions (66,519 men and 78,489 women). The project will also: • ensure that all reviewed policies, recommended policy revisions, and economic development planning frameworks and guidelines promote gender inclusion and responsiveness.; • ensure sustainability and long-term impact of gender-responsive outcomes, outputs and actions; • build capacity for government institutions, traditional authorities and faith-based institutions in acknowledging gender differentials in responding to climate risks; • ensure that women and youth take leadership with regards to project components related to them, however, without the marginalisation of men with the intent to avoid their resistance and negative influence; and • increase social accountability.					
	Project Management				

	1. Ensure a gender balance in establishing the Project Management Unit and in managing its human resources.	<u>Baseline:</u> No PMU established <u>Indicator:</u> Documentation of contracting process (vacancy announcements, shortlisting, interview notes) + proportion of women employed in unskilled / technical / management / supervisory roles <u>Target:</u> Demonstrable efforts to recruit at least 50% women, including in positions with project responsibilities	Year 1	UNEP task manager	/
	2. Recruit a gender officer ⁴⁸ in the Project Management Unit to support and monitor the effective implementation of the Gender Action Plan (50%) and oversee any grievances related to gender-based violence (GBV) or sexual exploitation and harassment (SEAH).	<u>Baseline:</u> No TORs <u>Indicator:</u> TORs to recruit for this position <u>Target:</u> Fifty percent of his/her time is dedicated to support the gender-integrated implementation of the project. 15% of his/her time is allocated to support grievances for SEAH or GBV.	Year 1	UNEP task manager	/

⁴⁸ This gender officer will have decision-making power in relation to the implementation of the GAP. The gender officer will be accountable for ensuring consistent and timeous implementation of the GAP activities and report to the Project Coordinator.

	3. Gender Training for the PMU. Training materials will include topics of, <i>inter alia</i> : i) SEAH and GBV; understanding how to analyse the differential impacts of projects on men and women; iii) strategies to promote equitable access to project benefits for all social groups; and iv) familiarisation with national, regional and international gender policies and conventions.	<u>Baseline</u>: No training; limited gender expertise available among PMU staff (level to be assessed at start of training) <u>Indicator</u>: Online and in-situ training provided to team members (certificates, modules), knowledge assessed through interviews <u>Target</u>: PMU team members are knowledgeable on gender challenges and able to report on gender-responsive indicators of the project such as certificates, modules and knowledge assessments.	Year 1	PMU gender officer	/
	4. Facilitate participation of women in project monitoring, evaluation, and learning (MEL) activities.	<u>Baseline</u>: No MEL system in plan <u>Indicator</u>: Number and % of men and women actively participating in the MEL process <u>Target</u>: Each MEL activity includes at least 40% women	Year 1–6	PMU gender officer	/
	5. Enable women to actively participate in decision-making processes affecting their lands, resources, activities, families, and livelihoods, and to overcome access	<u>Baseline</u>: No qualitative measures to ensure gender-mainstreaming <u>Indicator</u>: Level of engagement of women in project consultation (% of women participating in discussions⁴⁹ disaggregated by ethnic group, age, and marital status +	Year 1–6	PMU gender officer	/

⁴⁹ Participation of sub-groups to be measured from onset; decrease may indicate an issue related to limited interest in the project and should trigger further consultations.

	constraints to natural resources.	opinions expressed + decisions impacted) <u>Target:</u> Women are actively participating in project consultation, accounting for 50% of representatives, including representatives of different communities, women of all ages, and marital status; women express their opinions, and these are taken into account in influencing each decision-making process related to the project			
Outcome 1 Statement: Increased implementation and upscaling of EbA measures across the Sahara-Sahel boundary in Mauritania.					
Project activities	Gender action	Baseline, indicators and targets	Responsible entity	Timeline	Budget
Activity 1.1.1. Establish coordination platforms to facilitate knowledge-sharing, natural resource management, sustainable land-use planning and the implementation of EbA activities at the regional and local levels.	6. Ensure at least 30% female representation in coordination and technical committees (CTCs).	<u>Baseline:</u> Limited representation of women in local and regional governance structures (CTCs). <u>Indicator:</u> % of women in CTCs. <u>Target:</u> At least 30% of CTC members are women.	PMU: Gender Officer ensures gender considerations are integrated Monitoring & Evaluations (M&E) Officer tracks recruitment metrics	Year 1	The cost of implementing the gender activities for Activity 1.1.1 will be covered by the PMU Gender Officer's salary.
	7. Ensure training materials include gender-responsive approaches to land-use and natural resource management. The module should also include information on GBV and	<u>Baseline:</u> No training provided yet under 1.1.1.2 and 1.1.1.3. <u>Indicator:</u> % of training modules that incorporate gender-responsive adaptation strategies. <u>Target:</u> 100% of training sessions integrate gender considerations. One module in	PMU: Gender Officer leads training development and delivery. Chief Technical Advisor supports integration with project goals. Regional Project Coordinators ensures training is regionally contextual	Years 1–2	

	grievance redress mechanisms (GRM). 8. Ensure trainings and conferences take place in the mornings on weekdays to account for women's household responsibilities.	each training is also specifically on GBV grievance mechanisms, gender roles and gender mainstreaming options in sustainable natural resource management and strategic land-use planning.			
	9. Build women's leadership capacity through trainings for both men and women. Ensuring mixed-gender group discussions encourages inclusivity and knowledge-sharing. 10. Ensure trainings and conferences take place in the mornings on weekdays to account for women's household responsibilities.	<u>Baseline:</u> No structured training provided yet under 1.1.1.2 and 1.1.1.3. <u>Indicator:</u> % of persons who receive trainings, by type of training, disaggregated by gender. <u>Target:</u> At least 30% of training and skill building activities participants are women.	PMU: Gender Officer leads training development and delivery. Chief Technical Advisor supports integration with project goals. Regional Project Coordinators ensures training is regionally contextual	Years 1–2	

	11. National policies, plans and strategies on natural resource management and land-use planning are gender-responsive. These policies should also promote awareness raising on SEAH and GBV. 12. Consult with gender experts, relevant policy makers and women's groups on how to incorporate actionable gender activities into the national policies.	<u>Baseline:</u> No policy brief written. <u>Indicator:</u> Number of gender-responsive measures proposed. <u>Target:</u> Each policy briefs is gender-responsive and one is focused on gender in natural resource management and land-use planning.	PMU facilitates overall governance structure Gender Officer ensures gender considerations are integrated MEDD & Local Governments lead policy integration.	Years 2–4 (policy development) Years 5–6 (Evaluations and reinforcement)	
Activity 1.2.1: Develop knowledge products on project lessons learned and best practices through a participatory process engaging the communities.	13. Ensure all knowledge products collect and incorporate gender-disaggregated data.	<u>Baseline:</u> No gender-specific knowledge products exist. <u>Indicator:</u> % of knowledge products with gender-disaggregated data. <u>Target:</u> 100% of knowledge products include gender-disaggregated data.	PMU: M&E Officer leads documentation and knowledge-sharing efforts. Gender Officer ensures gender considerations are integrated	Years 1–3	The cost of implementing the gender activities for Activity 1.2.1 will be covered by the PMU Gender Officer's salary.

	14. Demonstrate gender-responsive best practices in adaptation planning. Women often use different adaptation strategies such as water conservation techniques and drought-resistant farming methods. Capturing these practices enhances inclusivity.	<u>Baseline:</u> Gender-specific adaptation strategies are not systematically documented. <u>Indicator:</u> % of knowledge products featuring gender-sensitive adaptation strategies. <u>Target:</u> At least one gender-sensitive case study per knowledge product.	PMU: Gender Officer ensures gender considerations are integrated	Years 1–3	
	15. Document case studies featuring women's leadership in EbA interventions.	<u>Baseline:</u> Women's perspectives are underrepresented in community engagement activities. <u>Indicator:</u> % of women participating in knowledge-gathering consultations. <u>Target:</u> At least 30% of participants in consultations are women.		Years 1–3	

	16. Identify barriers to women's participation in EbA interventions and propose solutions. Women may face challenges such as land tenure restrictions, exclusion from decision-making, or time constraints. These barriers should be identified and addressed in policy recommendations.	<u>Baseline:</u> Barriers to women's participation in EbA activities are not well-documented. <u>Indicator:</u> Existence of a gender barrier analysis in knowledge products. <u>Target:</u> At least one policy recommendation per product addressing gender barriers.		Years 1–3	
	17. Develop gender-sensitive communication formats for knowledge products. Women may have lower literacy rates or limited access to digital information sources. Using visual storytelling, radio broadcasts, or local women's networks ensures inclusive dissemination.	<u>Baseline:</u> Knowledge dissemination is not always accessible to women due to literacy and technology gaps. <u>Indicator:</u> % of knowledge products available in gender-inclusive formats such as radio and visual content. <u>Target:</u> At least one alternative dissemination format per product.		Years 1–3	5,000

Activity 1.2.2: Enhance the dissemination of adaptation knowledge to sub-national decision-makers and communities to support upscaling.	18. Ensure gender-disaggregated data and gender-sensitive content are included in all disseminated knowledge products.	<u>Baseline:</u> Knowledge products do not consistently integrate gender-disaggregated data or gender-sensitive insights. <u>Indicator:</u> % of disseminated knowledge products with gender-disaggregated data. <u>Target:</u> 100% of knowledge products include gender-disaggregated data.		Years 3–6	The cost of implementing the gender activities for Activity 1.2.2 will be covered by the PMU Gender Officer's salary.
	19. Knowledge dissemination must reflect how climate change impacts women and men differently and highlight gender-responsive adaptation approaches.				
	20. Use gender-inclusive dissemination formats such as radio broadcasts, visual materials and local women's networks.	<u>Baseline:</u> Women have less access to formal knowledge-sharing platforms than men. <u>Indicator:</u> % of knowledge-sharing events with at least 30% female participants. <u>Target:</u> At least 30% of participants in knowledge-sharing events are women.		Years 3–6	
	21. Ensure women's organisations, gender focal points and female leaders are engaged in dissemination efforts.	<u>Baseline:</u> Knowledge dissemination is not always accessible to women due to literacy and technology gaps. <u>Indicator:</u> % of knowledge products available in gender-inclusive formats such as radio and visual content. <u>Target:</u> At least one alternative dissemination format per product.		Years 3–6	
	22. Women's networks can help amplify knowledge products, ensuring that women benefit from and apply the shared information.				

	23. Include case studies or policy briefs focusing on gender-specific climate adaptation challenges.	<u>Baseline:</u> Women's perspectives are not systematically integrated into international platforms. <u>Indicator:</u> % of international knowledge-sharing events featuring gender-sensitive discussions or case studies. <u>Target:</u> At least one gender-focused discussion or case study in every international dissemination event.		Years 3–6	
Outcome 2 Statement: Communities in four rural-urban hubs along the Sahara-Sahel boundary in Mauritania have increased livelihood security as a result of investment in EbA measures and sustainable land- and natural resources management practices.					

Project activities	Gender action	Baseline, indicators and targets	Responsible entity	Timeline	Budget
Activity 2.1.1. Establish 2,123 hectares of EbA dune-fixation infrastructure and 120 kilometres of protective fencing across the four target hubs, to facilitate the rehabilitation and maintenance of degraded landscapes and enhance ecosystem services related to dune stabilisation and the supply of natural resources.	24. Include women in processes related to the identification of sites for: i) restoration of degraded land ; and ii) managed aquifer recharge infrastructure within key recharge areas. This will ensure that women's adaptation needs — and the distance women may be expected to travel when accessing natural resources — are considered when identifying target sites. Additionally, this will ensure that women are given equal opportunities to men when accessing communal land/acquiring land for project interventions.	<u>Baseline:</u> Areas dedicated to women's activities have not been identified nor assessed (Activity 2.2.1) <u>Indicator:</u> ha of lands available for women's activities and location information (distance from the village, type of terrain) <u>Target:</u> The rehabilitated lands do not encroach women's activities and allows for appropriate areas to be preserved for women's activities	PMU: Regional Project Coordinators identify target areas. Gender Officer ensures gender balance.	Years 1–3	The cost of implementing the gender activities for Activity 2.1.1 will be covered by the PMU Gender Officer's salary.
	25. Guarantee at least 30% female employment in dune fixation and fencing projects.	<u>Baseline:</u> Dune fixation and fencing projects are typically male-dominated. <u>Indicator:</u> % of women employed in dune fixation and fencing works. <u>Target:</u> At least 30% of employees in these projects are women.	PMU: Regional Project Coordinators identify target areas. Gender Officer ensures gender balance.	Years 1–3	

	26. Provide gender-sensitive technical training on dune fixation techniques. Women may not traditionally engage in land rehabilitation work, but training enables equal participation in implementation and future maintenance.	<u>Baseline:</u> Women often lack technical skills for dune fixation and land restoration. <u>Indicator:</u> % of women trained in dune fixation techniques. <u>Target:</u> At least 30% of training participants are women.	PMU: Gender Officer ensures gender-sensitive training.	Years 1–3	US\$ 4,000.00 ⁵⁰
	27. Implement safety measures to prevent GBV risks in project sites. Women working in outdoor environments may face harassment or exploitation. GBV protocols help create a safe and inclusive work environment.	<u>Baseline:</u> No standardised GBV safety measures in outdoor work environments. <u>Indicator:</u> Existence of GBV prevention protocols in project sites. <u>Target:</u> GBV prevention measures implemented at all project sites.	PMU: Gender Officer ensures gender considerations are integrated	Continuous for the duration of the project	The cost of implementing the gender activities for Activity 2.1.1 will be covered by the PMU Gender Officer's salary.
	28. Conduct assessments to identify the concerns, adaptation needs, and roles of women in target communities and incorporate findings into the design of awareness-raising campaigns focused on climate change, enhanced resilience	<u>Baseline:</u> gender-responsive awareness campaigns have not been designed <u>Indicator:</u> Messages tailored for women included in the campaign (ToRs, comms material) <u>Target:</u> Each campaign includes women protagonists, tackles a subject related to women's activities, includes	PMU extension offices of relevant ministries located within the urban areas in the four target hubs	Continuous for the duration of the project	

⁵⁰ Cost of procuring a national consultant or NGO (environmental management consultant) to prepare standardised training material for annual one-day workshops on: i) EbA solutions and physical measures for dune fixation and rehabilitation of degraded ecosystems; ii) climate-smart agricultural and livestock practices; iii) diversified and natural resource-based livelihoods; iv) climate-resilient water resource management; and v) channelling funds generated from sustainable livelihoods and ecosystem services towards reinvestment in the operation, maintenance and upscaling of successful project interventions. National Consultant is to spend 10 days at US\$400,00 per day developing training material remotely. Training material will be passed on to the PMU and extension officers from the MEDD, for use in training CTC members on an annual basis.

	to desertification and SEAH and GBV GRMs.	awareness raising campaigns on GBV and SEAH and includes at least one message regarding women's rights			
Activity 2.2.1. Establish community-managed Water User Groups (WUGs) and commune-level water monitoring and regulation plans	29. Ensure at least 30% female representation in Water User Groups (WUGs). Women are primary water users for domestic and agricultural purposes but are often excluded from formal water governance structures. Ensuring representation enhances decision-making inclusivity.	<u>Baseline:</u> No women in Water User Groups (WUGs). <u>Indicator:</u> % of WUG members who are women. <u>Target:</u> At least 30% of WUG members are women.	PMU: Regional Project Coordinators identify target areas. Gender Officer ensures gender balance.	Years 1–3	The cost of implementing the gender activities for Activity 2.2.1 will be covered by the PMU Gender Officer's salary.
	30. Provide targeted training for women on water governance, monitoring, and technical aspects of water management. Women's participation is often limited by lack of technical knowledge. Training ensures they can effectively engage in WUGs and water resource decision-making.	<u>Baseline:</u> Women's participation in water governance training is limited. <u>Indicator:</u> % of training participants who are women. <u>Target:</u> At least 30% of training participants are women.	PMU: Gender Officer ensures gender balance.	Years 1–3	US\$ 4,000.00 ⁵¹

51 Cost of procuring a national consultant or NGO (environmental management consultant) to prepare standardised training material for annual one-day workshops on: i) EbA solutions and physical measures for dune fixation and rehabilitation of degraded ecosystems; ii) climate-smart agricultural and livestock practices; iii) diversified and natural resource-based livelihoods; iv) climate-resilient water resource management; and v) channelling funds generated from sustainable livelihoods and ecosystem services towards reinvestment in the operation, maintenance and upscaling of successful project interventions. National Consultant is to spend 10 days at US\$400,00 per day developing training material remotely. Training material will be passed on to the PMU and extension officers from the MEDD, for use in training CTC members on an annual basis.

	31. Ensure women's access to leadership roles within WUGs. Women in WUGs often hold lower-status roles. Encouraging leadership strengthens gender-responsive water governance.	<u>Baseline:</u> Women hold few leadership roles in WUGs. <u>Indicator:</u> Number of WUG leadership positions held by women. <u>Target:</u> At least 1 WUG leadership positions are held by a woman.			The cost of implementing the gender activities for Activity 2.2.1 will be covered by the PMU Gender Officer's salary.
	32. Develop gender-sensitive water regulation plans that consider women's water needs for domestic and agricultural use. Women often prioritise household water use, while men may focus on agricultural or livestock needs. Integrating women's perspectives ensures balanced water allocation.	<u>Baseline:</u> Water regulation plans do not consider gendered water needs. <u>Indicator:</u> % of water regulation plans that include gender-responsive considerations. <u>Target:</u> 100% of water regulation plans integrate gender-specific needs.	MEDD ensures alignment with national frameworks. Water Ministry leads technical development. PMU: Gender officer ensures gender considerations are considered.	Years 1–3	The cost of implementing the gender activities for Activity 2.2.1 will be covered by the PMU Gender Officer's salary.
Activity 2.2.2: Install physical water management infrastructure — including weirs, gabions, dikes, stone- and earthen bunds, groundwater dams, water access points for pastoralists.	33. Ensure women's participation in site selection for water infrastructure. Women and men use water differently (e.g., domestic use vs. agriculture/livestock). Ensuring women's voices in site selection prevents	<u>Baseline:</u> Women's voices are not systematically included in site selection for water infrastructure. <u>Indicator:</u> % of site selection consultations that include women. <u>Target:</u> At least 30% of site selection participants are women.	PMU: Regional Project Coordinators identify target sites. Gender Officer ensures gender balance.	Years 1–3	The cost of implementing the gender activities for Activity 2.2.2 will be covered by the PMU Gender Officer's salary.

	marginalisation of their needs.				
	34. Ensure water infrastructure is designed with gender-sensitive considerations (e.g., safety, proximity, and accessibility for women and children). Water points located far from homes increase women's workloads and expose them to safety risks (e.g., GBV while collecting water). Locating infrastructure closer to communities reduces burdens.	<u>Baseline:</u> Water points are not always safely or conveniently located for women. <u>Indicator:</u> % of newly installed water points located based on gender-sensitive considerations. <u>Target:</u> 100% of new water infrastructure incorporates gender-sensitive site selection.	PMU: Gender officer ensures gender considerations are considered.	Years 1–3	
Activity 2.2.3 Install 12 rainwater-harvesting systems and communal cisterns (5,000 L per system) within each target commune	35. Ensure women's participation in selecting locations for rainwater-harvesting systems and communal cisterns. Women are often responsible for collecting water, and improper site selection could increase their workload or limit accessibility.	<u>Baseline:</u> Women's voices are not systematically included in selecting rainwater-harvesting sites. <u>Indicator:</u> % of site selection consultations that include women. <u>Target:</u> At least 30% of site selection participants are women.	PMU: Regional Project Coordinators identify target sites. Gender Officer ensures gender balance.	Years 1–3	The cost of implementing the gender activities for Activity 2.2.3 will be covered by the PMU Gender Officer's salary.
	36. Prioritise access to harvested rainwater for women-led smallholder farms and household use. Women often	<u>Baseline:</u> Women's water needs are not explicitly prioritised in allocation.			

	manage household gardens and small-scale agriculture, which require reliable water access.	<u>Indicator:</u> % of water allocation plans that include gender-responsive provisions. <u>Target:</u> 100% of allocation plans consider women's household and agricultural needs.			
Activity 2.3.1. Facilitate the adoption of climate-smart agricultural practices and sustainable diversified livelihoods by farmers and community members within the target communes.	37. Capture local knowledge of men and women within the target region and facilitate knowledge-sharing via information platforms hosted by the Government of Mauritania. Using this existing knowledge, together with training material developed under Activity 1.1.1proposed project, train both women and men in climate smart crop agriculture, nurseries and sustainable livestock management practices.	<u>Baseline:</u> No training / skill-building activities have taken place <u>Indicator:</u> % of persons who receive trainings, by type of training, disaggregated by gender, ethnic group, age, and marital status <u>Target:</u> At least 50% of training and skill building activities target women + exclusive modules for women are developed	PMU: Gender officer ensures gender considerations are considered.	Continuous for the duration of the project	The cost of implementing the gender activities for Activity 2.3.1 will be covered by the PMU Gender Officer's salary.
	38. Enable women to actively participate in decision-making processes affecting their lands, resources, activities, families, and livelihoods, and to overcome access	<u>Baseline:</u> Existing community-level decision-making groups in the target communes (within the four hubs) targeted by the project demonstrate little quantitative gender parity and no qualitative measures to ensure gender-mainstreaming; community-level water-user	PMU: Gender officer ensures gender considerations are considered.	Continuous for the duration of the project	The cost of implementing the gender activities for Activity 2.3.1 will be covered by the PMU Gender Officer's salary.

	constraints to natural resources.	groups (WUGs) have not been set up yet. <u>Indicator:</u> Level of engagement of women structures (% of women participating in discussions⁵² disaggregated by ethnic group, age, and marital status + opinions expressed + decisions impacted). <u>Target:</u> Women are actively participating in local governance structures, accounting for 50% of the members, including representatives of all communities, women of all ages, and marital status; women express their opinions, and these are taken into account in influencing each decision-making process related to the project			
	39. Ensure all women, including indigenous and other marginalised women, are equitably consulted and participate in the development of the livelihood activities and value chains to be supported by the project.	<u>Baseline:</u> No consultations. <u>Indicator:</u> Number of awareness activities providing targeted information to women on potential micro-credits for livelihood alternatives. <u>Target:</u> 50% of the livelihood activities supported by the project specifically target women.	PMU: Gender officer ensures gender considerations are considered	Continuous for the duration of the project	The cost of implementing the gender activities for Activity 2.3.1 will be covered by the PMU Gender Officer's salary.
Activity 2.3.2. Establish an on-granting mechanism	40. Ensure at least 40% of grants are awarded to women-	<u>Baseline:</u> Women receive limited access to climate adaptation financing.	PMU:	Continuous for the duration of the project	The cost of implementing the gender activities for

⁵² Participation of sub-groups to be measured from onset; decrease may indicate an issue related to limited interest in the project and should trigger further consultations.

to channel government funds and income generated via sustainable agricultural livelihoods towards continued investment in EbA.	led agricultural initiatives. Women often face barriers to accessing financing, limiting their ability to scale up climate-resilient farming.	<u>Indicator:</u> % of on-granting funds awarded to women-led agricultural initiatives. <u>Target:</u> At least 40% of grants are allocated to women-led initiatives.	Gender officer ensures gender considerations are considered Financial & Administrative Officer manages grant allocations.		Activity 2.3.2 will be covered by the PMU Gender Officer's salary.
	41. Develop tailored financial literacy training for women and men in rural areas. Many rural farmers lack experience with formal banking and financial management, limiting their ability to effectively utilise grants.	<u>Baseline:</u> Women lack financial literacy training tailored to their needs. <u>Indicator:</u> Number of training initiatives per target commune. <u>Target:</u> At least one training initiative per target commune with at least 30% financial literacy trainees being women	PMU: Gender officer ensures gender considerations are considered	Years 2–3	US\$ 400.00 ⁵³
	42. Encourage collective women-led cooperatives to access financing as groups. Women often lack collateral for individual loans but can access financing through group-based funding models.	<u>Baseline:</u> Women farmers have limited access to peer support and mentoring. <u>Indicator:</u> % of trained female farmers participating in peer mentoring networks. <u>Target:</u> At least one mentoring programme per target commune.	PMU: Gender officer ensures gender considerations are considered	Continuous for the duration of the project	The cost of implementing the gender activities for Activity 2.3.2 will be covered by the PMU Gender Officer's salary.
	Total: USD 338,400				

⁵³ Cost for a contracted financial expert or NGO to design an on-granting facility with revolving fund mechanism, performance-based milestones, repayment schemes and sub-project selection criteria at US\$400.00 per day.